

Predictable Engineering of Signal-Dependent Cis-Regulatory Elements

Jake Cornwall-Scoones¹, Dirk Benzinger¹, Tianji Yu^{2,3}, Alberto Pezzotta^{1,4}, Andreas Sagner⁵, Lina Gerontogianni¹, Shaun Bernadet¹, Elizabeth Finnie¹, Giulia L. M. Boezio¹, Hannah T. Stuart^{1,6}, Manuela Melchionda¹, Oliver C. K. Inge¹, Bianca Dumitrascu^{2,3,7,8,9}, James Briscoe^{1*}, M. Joaquina Delás^{1,10*}

1. The Francis Crick Institute, Midland Road, London, NW1 1AT, UK
2. Department of Systems Biology, Columbia University, New York City, USA
3. Department of Statistics, Columbia University, New York City, USA
4. Gatsby Computational Neuroscience Unit, University College London, London, UK
5. Institut für Biochemie, Friedrich-Alexander-Universität Erlangen-Nürnberg, Fahrstraße 17, 91054 Erlangen, Germany
6. Institute of Molecular Biotechnology of the Austrian Academy of Sciences (IMBA), Vienna BioCenter, Dr. Bohr-Gasse 3, 1030 Vienna, Austria
7. Irving Institute for Cancer Dynamics, Columbia University, New York City, USA
8. Department of Computer Science, Columbia University, New York City, USA
9. Columbia Stem Cell Initiative, Columbia University, New York City, USA
10. Laboratory for Molecular Cell Biology, University College London, Gower Street, London WC1E 6BT, UK

*Corresponding authors: j.delas@ucl.ac.uk and james.briscoe@crick.ac.uk

Abstract

Cis-regulatory elements (CREs) control how genes respond to external signals, but the principles governing their structure and function remain poorly understood. While differential transcription factor binding is known to regulate gene expression, how CREs integrate the amount and combination of inputs to secure precise spatiotemporal profiles of gene expression remains unclear. Here, we developed a high-throughput combinatorial screening strategy, that we term NeMECiS, to investigate signal-dependent synthetic CREs (synCREs) in differentiating mammalian stem cells. By concatenating fragments of functional CREs from genes that respond to Sonic Hedgehog in the developing vertebrate neural tube, we found that CRE activity follows hierarchical design rules. While individual 200-base-pair fragments showed minimal activity, their combinations generated thousands of functional signal-responsive synCREs, many exceeding the activity of natural sequences. Statistical modelling revealed CRE function can be decomposed into specific quantitative contributions in which sequence fragments combine through a multiplicative rule, tuned by their relative positioning and spacing. These findings provide a predictive framework for CRE redesign, which we used to engineer synthetic CREs that alter the pattern of motor neuron differentiation in neural tissue. These findings establish quantitative principles for engineering synthetic regulatory elements with programmable signal responses to rewire genetic circuits and control stem cell differentiation, providing a basis for understanding developmental gene regulation and designing therapeutic gene expression systems.

Introduction

During embryonic development and tissue homeostasis, stem cells continuously undergo differentiation to generate a diverse array of specialised cell types. Such cell fate decisions are driven by external signals that regulate cell type specific gene expression programmes (1, 2). This process relies on cis-regulatory elements (CREs) – non-coding regions of the genome that act as molecular control switches for gene expression. These DNA sequences function as information processing devices by recruiting signal-dependent and cell type specific transcription factors (TFs) to control the transcription of target genes.

Despite their critical role, our understanding of how CREs integrate inputs to drive cell-type specific gene expression remains incomplete and lacks unifying principles. Uncovering quantitative rules for the structure and sequence-composition of CREs would have far-reaching implications (3), advancing our understanding of how stem cells interpret signalling cues to establish and maintain complex tissues, and enabling the engineering of synthetic cis-regulatory elements (synCREs) to direct cell fates away from pathological states and toward regenerative outcomes.

Short DNA motifs within CREs bind specific TFs (4) and may be activating, repressing, or both depending on the context and signalling environment (5–7). While functional genomics has demonstrated that CRE activity depends on the presence of these motifs (8, 9), we lack a fundamental understanding of how the structure and composition of CREs facilitate signal-dependent gene expression.

Various models for the design rules of CREs have been proposed (10). These fall along a spectrum (11) that place different emphasis on the relative positioning of motifs. At one extreme, the billboard model suggests that CREs are simply “bags of motifs” (12), where the activity contributions of motifs are independent of each other. At the other extreme, the enhanceosome model, exemplified by the IFN- β locus, argues that CRE activity depends on a specific physical conformation (13), where the “grammar” – the precise positioning and spacing of motifs – determines transcription (14, 15). Indeed, there are clear examples of local rigid grammar (16–18), however for other CREs, grammar appears subtle and case-dependent (19).

Recent progress in applying deep-learning to construct novel synthetic CREs (20–23) points to a hidden structure in the configuration of CREs, which can be learned from assaying the regulatory activity of large sets of DNA sequences. These deep learning models are designed to handle potential complexities and higher-order interactions between TFs and CRE in a way

that simpler mechanistic models cannot (24). It is possible that CRE structure is inherently complex, necessitating complex high dimensional models. Alternatively, the apparent complexity of deep-learning models may conceal a simpler set of rules or principles that provide an interpretable explanation of CRE function.

Synthetic genomics offers a path to understanding CRE activity (25). By going beyond natural variation, synthetic genomics allows for the systematic exploration of ‘design space’ (26, 27), providing an understanding of how CRE activity depends on its sequence building-blocks. To realise this potential, however, novel high-throughput approaches for building and testing large cohorts of synthetic regulatory sequences are required (24).

The ability to rationally design synthetic regulatory sequences with precisely tuned activities has profound implications for therapeutic applications and tissue engineering. Understanding and exploiting the rules of CRE construction could allow the design of “super-physiological-enhancers” that drive gene expression more efficiently than their natural counterparts, or engineer tissue-specific regulatory elements that enable targeted gene therapy with minimal off-target effects (28, 29). Such synthetic enhancers could be particularly valuable for fine-tuning the expression of therapeutic transgenes, reprogramming cell fates in regenerative medicine, or creating more effective gene therapy vectors that achieve therapeutic levels of expression in specific target tissues while remaining silent in others.

Here, we developed NeMECiS (Nested Modular Expressible Cis Screening), a novel combinatorial screening strategy to build and assay synCREs. Using vertebrate neural progenitor diversification as a model, in which different concentrations of the signal Sonic Hedgehog (Shh) instruct alternative neural cell types with distinct gene expression programmes (30), we generated exhaustive three-way fusions of fragments from a set of known CREs. The comprehensive, combinatorial coverage of synCRE design-space afforded by NeMECiS enabled inference of the rules of CRE activity. Surprisingly, individual 200-base-pair fragments of CREs showed minimal activity, yet in combination generated thousands of functional signal-decoding synCREs. Statistical modelling revealed the hierarchical rules of CRE activity, where specific sequence fragments contribute to signal-dependent activity, but are tuned by spacing-dependent pairwise interactions.

The learned rules extrapolate beyond the training design-space, enabling the principled re-design of CREs controlling stem cell diversification. As a proof-of-concept, we modified a hedgehog-responsive element controlling motor neuron differentiation (31–33), generating synthetic variants with enhanced or reduced activity compared to the endogenous sequence. These engineered CREs successfully directed alternative spatial patterning outcomes in

neural tissue. This synthetic regulatory toolkit provides a versatile platform for programming cell fate decisions, offering opportunities to investigate pathological consequences of aberrant signal interpretation and to optimise therapeutic gene expression in target tissues.

Results

Complex Architectural Requirements for Hedgehog Dependent CRE Activity

To analyse CRE activity we took advantage of Shh signalling dependent gene expression in developing neural progenitors using an mouse embryonic stem cell (mESC) directed differentiation model of neural tube patterning (34, 35) (Fig. S1A). Here, neural progenitors supplied with different concentrations of SAG, a Shh signalling pathway agonist, are directed to distinct identities that correspond to the *in vivo* pattern of progenitor domains (36). To assay CRE activity, we integrated a lentivirus containing regulatory sequences controlling the expression of a fluorescent reporter (ZsGreen)(37, 38) and a constitutively expressed marker, mScarlet3, to identify transduced cells (Fig. 1A). Signal-dependent CRE activity was assayed by exposing cells to different SAG concentrations and analysing by flow cytometry at day 6 of the differentiation, when Shh-responsive ventral neural progenitors are usually present (36). When performed using a well-characterised CRE for the Shh-responsive TF Olig2, we observed an increase in ZsGreen levels as a function of SAG concentration (Fig. 1B), consistent with *in vivo* reporters (31, 32).

Previous high-throughput screens of CRE activity have been limited to DNA elements <300bp in size, typically centred on ChIP-/ATAC-seq peaks (39–43). We therefore generated a nested set of reporters comprising 200bp fragments of the Olig2 CRE. Surprisingly, none of the fragments alone showed substantial SAG-dependent fold-changes in activity (Fig. 1C). This could indicate a rigid architecture of the CRE, consistent with the enhanceosome model, in which the CRE activity is dependent on its integrity. Alternatively, unknown rules of composability may describe how modules come together to generate CRE activity that is greater than the sum of its parts.

Combinatorial assay of synCRE activity demonstrates flexible composability of regulatory information.

To discriminate between alternative hypotheses, we tested whether combining CRE sequence fragments was sufficient to recover SAG-dependent activity. To test if activity can be recovered, and to infer the hidden rules of CRE composability in an unbiased manner, we

designed an exhaustive and systematic approach. We term this approach NeMECiS (Nested Modular Expressible Cis Screening).

To this end we established a sequential, combinatorial pooled cloning strategy to fuse a library of CRE fragments in exhaustive permutations to establish synCREs (Fig. 1D). In this approach, 608bp synCREs, comprising three 200bp fragments connected by 4bp spacers, are assembled spanning the full permutation-set of CRE fragments (Fig. S1B). During the assembly a concatemer of 8bp barcodes is established, allowing for direct readout of synCRE composition by high-throughput sequencing (Materials and Methods) (Fig. S1D). Using three established CREs controlling neural progenitor genes *Nkx2-2*, *Olig2* and *Pax6* (31, 44) (Fig. S1E), we generated a library comprising $25^3=15,625$ synCREs, the exhaustive permutation-set of twenty-five 200bp fragments from these CREs (32) (Fig. S2A-D).

To measure synCRE activity across this library, we used lentiviral transduction to integrate synCREs into mESCs (41) ($\text{MOI}<1$) and differentiated them to neural progenitors, treating cells with either 0nM or 500nM SAG (Fig. S1C). To quantify SAG-dependent synCRE activity, we flow-sorted neural progenitors using the ZsGreen reporter into four equally sized bins, representing different levels of activity, and sequenced the barcodes in each bin (Fig. S2E-G). From the counts of barcodes in each activity bin we used Bayesian inference to infer the mean ZsGreen level (Materials and Methods). The measurements showed good reproducibility across replicates (PCC: 0nM SAG = 0.45; 500nM SAG = 0.60; Fig. S3A) and established a set of 13,991 synCREs (90% of theoretical max) that pass quality-control.

The assay revealed a broad spectrum of signal-dependent behaviours (Fig. 1E), ranging from activation in SAG conditions (23% of synCREs) to repression (7% of synCREs) and insensitivity (Fig. S3B-E). We validated with flow cytometry and microscopy a selection of synCREs that were either induced or repressed by SAG (Fig. 1F-I & S3F-I). This showed tight SAG dose-response relationships (Fig. 1H). Strikingly, in response to 500nM SAG, some activating synCREs displayed activities that were 187% greater than the strong *Olig2* CRE. Across the whole dataset, the majority of inducing (84%) and repressing (86%) synCREs comprised combinations of fragments from the CREs of the different genes.

We therefore conclude that despite the weak or absent activity of individual 200bp CRE fragments, a surprising proportion of fragment permutations showed a range of different activities. This argues that while CRE activity is greater than the sum of its parts, the rules of composability are not as rigid as the enhanceosome model but could be consistent with a modular mechanism in which individual fragments make a specific contribution to the overall activity of the synCRE.

CRE activity can be decomposed into modular contributions

The possibility that CRE activity can be decomposed into a series of separable modules could explain why signal-dependent synCREs can be readily assembled by concatenating CRE fragments. To test this, we established a linear modelling framework (Materials and Methods). In this framework each fragment makes a discrete, predictable contribution to overall transcriptional output both with and without Hedgehog signalling, independently of the other fragments in the synCRE (Fig. 2A & S4A-C). This revealed clear statistical signatures of signal-dependent fragment activities, which we grouped into three categories based on their response to Hedgehog signalling: inducing, repressing and neutral (Fig. S4D). To validate these predictions, we compared the mean ZsGreen expression levels between synCREs containing a specific fragment versus those without it, measuring their responses to SAG (Fig. 2B & S4D). The observed changes closely matched the predictions from our linear modelling, indicating that a major component of the overall activity of each synCRE can be explained by individual fragment contributions

Such modularity in the composition of synCREs is consistent with the extensive literature on TF-motif binding driving signal-dependent activities (44–46). To understand the sequence features underlying Hedgehog-responsive modules, we performed motif enrichment analysis on synCREs' signal-dependent activity. This identified a cluster of motifs that included the DNA binding motif of Gli transcription factors (the transcriptional effectors of the Hedgehog pathway (47)) as the strongest positive correlate of logFC SAG (Fig. 2C, S5A-C). Indeed, when we mutated the Gli motif in a functional synCRE (Fig. S5D), we found that the Hedgehog signalling-dependent fold change of activity decreased substantially (Fig. 2D), both raising the activity in the absence of SAG and lowering the activity in its presence (48). This is consistent with the Gli proteins acting as transcriptional repressors in the absence of Hedgehog signalling and activators in its presence (47). Further, across the entire NeMECiS dataset, synCREs that contain Gli binding sites have elevated responses to SAG (Fig. 2E). However, the overlap in the distribution of SAG responsiveness of fragments with or without a binding motif for Gli proteins (Fig. 2E) suggested that motif composition alone is insufficient to explain the extent of synCRE signal-dependent activity: 25% of strongly inducing fragments lacked detectable Gli motifs, while 14% of non-responsive fragments contained them. Additional sequence motifs enriched in Hedgehog-responsive fragments suggest complex combinatorial control beyond Gli binding (Fig. S5A-C). These findings indicate that while Gli motifs contribute substantially to Hedgehog responsiveness, additional sequence features shape the signal-dependent activity of synCREs.

Signal-dependent CRE activity follows a multiplicative composability rule.

We next sought to identify the modular composability rules. In principle this could follow several forms: Boolean, where activity depends on the presence of a fragment; or cumulative, where activity depends on the numbers of each fragment present in a synCRE. We grouped synCREs based on the number of inducing fragments, disregarding synCREs containing repressing fragments. This revealed the activity to be cumulative, with a substantial, monotonic increase in activity in response to 500nM SAG (Fig. 2F).

The cumulative behaviour could in principle be additive, multiplicative, or adopt an alternative non-linearity (49). Whereas for an additive model each fragment would add a fixed amount to the total activity, with a multiplicative model each additional fragment multiplies the existing activity by a factor, leading to exponential-like increases in overall activity. Consistent with a multiplicative model, we found a good fit of the mean log activities, and logFC, as a function of the number of activating fragments (Fig. 2G-H). A comparison of additive versus multiplicative models across all the dataset revealed the multiplicative model captured the behaviour better across fragment compositions (Fig. S6A-C). These results agree with the predictions of the well-established thermodynamic description of CRE activity (Materials and Methods)(50–55).

We tested the multiplicative model with an arrayed, individualised experiment, where we generated exhaustive permutations of the neutral fragment 8 (f8) and the activating fragment 16 (f16). Signal-dependent activity increases as a function of the number of activating fragments, minimal in f8-f8-f8 and maximal in f16-f16-f16 (Fig. 2I). In between these extremes, activities fit well to a multiplicative scaling rule (Fig. 2J). Repeating this approach with the activating (f2) and neutral (f20) fragments combined with either f8, f16, yielded equivalent results (Fig. S6E-G). A corollary of this rule is that a linear increase in inducing fragment frequency has an exponential effect on activity, helping explain why individual fragments show low activities yet readily combine to make functional synCREs (Fig. 2K).

Synergies among modules shape activities.

The multiplicative model assumes that transcription factors bind to fragments independently. Yet, this is likely an oversimplification, as transcription factors can bind cooperatively to DNA (49, 56–60). We therefore tested whether a model that accounted for interactions between fragments, and not just their modular action, fit the data better. NeMECiS is uniquely

positioned to statistically isolate these higher-order effects (Fig. 3A), as it assays every fragment in every combination and ordering.

Given the recent success of deep-learning in capturing regulatory sequence functionality (20, 23, 61, 62) and in building synthetic CREs (21, 22, 43, 63), we first trained several deep-learning models on our data (20, 23, 64, 65) (Fig. S7A-B). These models generally perform better than our linear modular model (PCCs: 0.61-0.66 on held-out test-data compared to 0.56 of the linear model; Fig. S7B). This suggested that the linear modular multiplicative model, while a good first approximation, does not account for the full complexity of regulatory activity across our synCRE design space.

Because the deep learning models provided superior predictive power for CRE activity, but their complexity limited mechanistic interpretation, we developed an interpretable framework. We decomposed fragment interactions into two components. First, we quantified how pairs of fragments jointly influence CRE activity beyond their individual effects, which we term 'interaction'. Second, we measured how the spacing between fragments affects CRE activity independent of these pairwise interactions, termed 'syntax' (Fig. 3A). This outperforms the simple multiplicative model across experimental replicates, revealing significant and reproducible effects for both interaction and syntax terms (Fig. S8).

Strikingly, the fit quality of our synergistic model matched those of the deep-learning models (Fig. S7B). There was a clear correspondence in predictions for synCRE activity between the synergistic model and the deep-learning models, which suggests that both classes of model are learning similar rules (Fig. S7A). This raises the possibility that sequence-to-activity deep-learning models are performing a two-step process, first identifying local regulatory modules, then applying a multiplicative rule modulated by pairwise interactions to predict the activity of a whole CRE from its collection of modules.

We compared the simple multiplicative model to the model incorporating pairwise interactions between fragments (Fig. 3B & S9A-B). Most fragment pairs were well described by the multiplicative behaviour, as shown by their distribution along the diagonal. However, there were significant instances of deviations above (positive interaction; 15% at $P < 0.05$) and below (negative interaction; 13% at $P < 0.05$) the diagonal. Analysis of all pairwise fragment combinations revealed that positive interactions occurred more frequently between pairs of inducing fragments (1.6-fold enrichment, permutation test), whereas negative interactions were more common between inducing-repressing pairs (1.3-fold enrichment; Fig. 3C, Fig. S9A-B).

To test the role of interactions in modular composition, we examined combinations of two activating fragments with different strengths: f2 (weaker) and f16 (stronger). The multiplicative model predicted that sequentially replacing f2 with f16 in the f2-f2-f2 synCRE would produce a monotonic increase in SAG-dependent activity. However, our experimental results showed that synCREs with combinations of f2 and f16 fragments produced higher activities than predicted, exceeding the f16-f16-f16 arrangement (Fig. 3D). This finding aligns with our global analysis showing interactions between inducing fragments, and with the strong positive interaction coefficient between f2-f16 predicted in our synergistic model (Fig. 3C). By combining both modular and synergistic effects, synCREs achieved activities up to 276% higher than the Olig2 CRE. These results confirm that while CREs can be broken down into modules, pairwise interactions can significantly influence their activity.

Pairwise syntax rules explain CRE grammar

We next asked whether synCRE activity is dependent on the spacing and ordering of its fragments. By analogy to linguistics, we term such contributions syntax. Our exhaustive, combinatorial synCRE library contains every pair of fragments in every combination of spacing (adjacent vs separated by a fragment) and ordering ('AB' vs 'BA'). These four classes act as components in the 'syntax' layer of our synergistic model (Fig. 3A). To identify contributions of syntax to signal-dependent CRE activity, we devised a positionality score (Materials and Methods), quantifying the relative variance of activity due to the spacing of a pair of fragments (positionality score > 1 implies spacing-dependent grammar). This revealed that, while most fragment pairs show 'syntax-free' interaction (positionality score ≈ 1), 86 fragment pairs showed significant spacing-dependent interactions (26.5%; positionality score > 2) (Fig. 3E).

Having identified a subset of fragment pairs displaying syntax effects, we sought to classify the different types of spacing-dependent activity. We transformed the measurements of spatial interactions into combinations of fundamental patterns (orthonormal bases) (Materials and Methods) (Fig. 3F). This allowed us to project the data into a 2D space, revealing examples of distance-dependent order-independent (e.g. f9-f6), and distance- and order-dependent (e.g. f4-f2), syntax (Fig. 3G-H). Similar results were found by k-means clustering of spacing-dependent synergies (Fig. S9D).

A consequence of these syntax rules is that the ordering of fragments in a synCRE should influence its activity. Specifically, permuting the order of fragments comprising a synCRE should maximise or minimise combined syntax-effects across the pairs of these fragment trimers. By building exhaustive six-way permutations of a series of fragment trimers (Fig. 3I,

Fig. S10A), we found repeated examples of order-dependence, with the levels of ZsGreen induced by 500nM varying between 2.3 and 4.6-fold (6 triplet sets x 6 permutations). In some instances, these ordering effects are sufficiently influential that some orderings show strongly SAG-dependent activities, whereas other orderings are insensitive to SAG, despite being comprised of the same modules.

This order-dependence is explained by our measured pairwise syntax rules. For example, the ordering in f4-f2-f21 maximises spacing-dependent synergy for each comprising pair (Fig. 3H). Signal-dependent activity is maximised when: 1) f4 is immediately before f2; 2) when f2 and f21 are adjacent in either ordering; and (3) when f4 is 408bp upstream of f21 (Fig. 3I-J). Indeed, the learned spacing-dependence predicts well the relative signal-dependent activities when validated individually (Fig. 3K & S10B-C). This argues that the ordering-dependent differences in synCRE activity can be explained by the linear combination of comprising pairwise spacing rules, with deviations from this trend (e.g. f10-f11-f9, f11-f10-f9; Fig. S10) potentially indicating three-way interactions that defy pairwise reduction.

Principled re-engineering of motor neuron progenitor patterning

The widespread examples of spacing-dependent synergies prompted us to ask whether endogenous Shh-responsive CREs have a fragment ordering that maximises expression. Analysis of 600bp windows from the Nkx2-2 and Olig2 CREs, divided into 200bp fragments and permuted exhaustively, showed that non-natural fragment orderings could achieve higher signal-dependent activity than natural orderings (e.g. f4-f2-f3 > f2-f3-f4; f9-f11-f10 > f9-f10-f11; Fig. S10). We then extended our prediction framework to evaluate signal-dependent activity across arbitrary-length fragment combinations (Materials and Methods; Fig. 4A). Applying this to the 1400bp Olig2 CRE sequence, we analysed systematically all possible arrangements of its seven 200bp fragments (5,040 permutations) to predict signal-dependent activity across the complete full-length synCRE (FL-synCRE) design-space (Fig. 4B-C & S11A). This analysis revealed that while the natural Olig2 CRE fragment ordering performs above average (73rd percentile; Fig. 4C & S11A), it does not achieve maximal signal-dependent activity.

Surprised by this observation, we selected a set of candidate FL-synCREs from the top and bottom 10% of the design space to test the predictions (Fig. 4D-E & S11B). Remarkably, we found that re-ordering the fragments of the Olig2 CRE can result in up to a 61% increase in SAG-dependent activity (Fig. 4D). Conversely, we found that candidates in the bottom 10% of activity predictions were able to reduce signal-dependent activity almost completely (Fig. 4E).

Principled re-engineering of motor neuron progenitor patterning

We investigated whether such ‘optimised’ re-orderings of fragments of the Olig2 CRE could be used to modify Hedgehog signalling responses and reprogramme signal-dependent cell fate transitions. Specifically, natural CREs are embedded within signal-decoding gene regulatory networks (GRNs), hence modifying CRE design enables the creation of synthetic GRNs (synGRNs). As a proof of concept, we asked whether we could predictably rewire the Shh-decoding GRN regulating dorsoventral patterning of spinal cord neural progenitors by placing Olig2, the initiator of motor-neuron development, under FL-synCRE control (Fig. 4F).

We exploited a well-established Ordinary Differential Equation (ODE) model of dorsoventral neural patterning to predict the consequences of GRN rewiring with FL-synCREs (33, 66, 67). This model integrates both signal strength and spatial distribution to predict how modified CREs alter gene expression patterns across the tissue. We first calibrated the model using fluorescence measurements from FL-synCRE reporters (Fig. S11C), then used the fit parameters to predict the spatial distribution of Olig2 expression upon GRN re-wiring (Fig. S11D). This revealed that replacing Olig2 regulation with a diminished FL-synCREs should reduce the Olig2 expression, shrinking or eliminating the Olig2+ motor neuron progenitor domain, whereas a boosted FL-synCREs should enhance Olig2 expression and expand the domain of Olig2 (Fig. 4G-H & S11E-F). These predictions capture not just changes in expression level, but crucially, how synthetic CREs reshape the tissue-wide organisation of cell types through altered interpretation of the morphogen gradient (Fig. 4G-H & S11D-F).

To test these predictions, we devised a synthetic reconstitution strategy of the cis-regulation of Olig2 to generate synGRNs (Fig. 4I). We made an Olig2-null mutant mESC line, then integrated, with PiggyBac, a cassette driving an Olig2 coding region under the control of a minimal promoter and either WT or FL-synCREs. Differentiating cells under varying SAG concentrations and performing flow cytometry with intracellular staining revealed a rescue of Hedgehog-dependent Olig2 expression (Fig. 4J). Strikingly, under maximum SAG, Olig2 was tuned to different levels depending on the FL-synCRE, with the diminished FL-synCRE showing lower and the boosted FL-synCRE showing higher expression than their wild-type counterpart (Fig. 4K).

These results suggest that neural patterning can be predictably controlled by re-engineering TF control under synCREs. To test this, we generated neural tube organoids (NTOs) (68) from synGRN-containing mESCs. Here, single mESCs embedded in Matrigel and cultured appropriately (Materials and Methods) generate neural progenitor cysts, polarised around a central lumen. These self-organise to generate a Shh-producing floorplate that patterns the

neural progenitors with ordering and proportions that mirror natural development (69) (Fig. 4L). To quantify in an unbiased manner the patterning consequences of GRN rewiring, we performed high-throughput imaging and automated 3D feature extraction using NTOs generated from our synGRN mESC lines (Materials and Methods; Fig. S12A). This revealed altered Olig2 patterning. The reduced FL-synCRE showed a near complete extinction of the Olig2 expressing domain, whereas the boosted FL-synCRE displayed an expanded domain of motor neuron progenitors (Fig. 4M-O). These patterns agree with the *in silico* predictions (Fig. 4G-H). Together, these results demonstrate that reordering of endogenous DNA fragments within a CRE, guided by quantitative design rules, serves as a tunable device for controlling developmental patterning. This finding establishes the programmable control of signal-dependent CRE activity for reshaping tissue pattern and redirecting cell fate decisions.

Discussion

Precise gene expression responses to external stimuli are a hallmark of cellular information processing. The complexity and robustness of cell-fate decisions that play out during development and homeostasis relies on the capacity of cells to measure subtle changes in signalling. This is typified in morphogen patterning (2, 30). A gradient of a single signalling molecule provides positional information to establish and organise distinct cell fates. CREs mediate the cellular decoding of spatial information by eliciting distinct gene expression programmes, determining when, where, and at which level, genes are turned on and off. Yet how the structure and composition of these sequences achieve precise signal decoding is poorly understood. Uncovering design principles of signal-dependent CRE activity has broad implications, from understanding how natural signal decoding is achieved, to providing insight into the evolutionary constraints on CREs, to applying composability rules to engineer synthetic CREs for programmable control of gene expression

Here, we introduce NeMECiS, a novel, high-throughput combinatorial screening strategy to explore systematically the design principles governing CRE function. By enabling the assessment of sequences longer than the typical ~200bp fragments assayed in many MPRA (41–43), NeMECiS reveals the combinatorial logic of CREs. It bridges the gap between high-throughput reporter assays and more extensive genomic engineering approaches and suggests that the field's historical focus on shorter fragments may have systematically underappreciated the modular composability rules of CREs. Our findings reveal that CRE activity can be decomposed into modular, interactional, and syntactical components, providing a predictive framework for engineering synthetic regulatory elements and offering new insights

into GRNs. We demonstrate that a CRE can be decomposed into a set of modular building-blocks, in which the activity of the collective is the product of its individual components, explaining why individual 200bp fragments of regulatory sequence display minimal activity, yet combine to make functional signal-decoding synCREs. This contrasts with the rigid conformation-dependent model proposed by enhanceosome models and suggests that many CREs operate under a more relaxed set of constraints.

The finding that CRE activity follows a multiplicative rather than additive rule provides mechanistic insight into how regulatory elements achieve signal sensitivity (49, 70). Individual 200bp fragments showed minimal activity, yet their combination generated thousands of functional signal-responsive elements. This multiplicative behaviour explains how weak individual contributions combine to produce strong overall activity, resembling the theoretical predictions of thermodynamic models of transcriptional regulation (50, 52–54).

Some pairs of modules interact synergistically, such that the activity of the pair is higher or lower than the product of its components. This means heterotypic fusions of signal-responsive modules can exceed the activities of their homotypic counterparts, and even that of the full length natural regulatory sequences from which these synCREs were constructed. We also find that interactions among pairs of modules can display distinct spacing-dependent syntax rules, meaning that the ordering of modules in a synCRE can alter its signal-responsiveness. These interactions are crucial for achieving the complex regulatory outcomes observed in GRNs.

The ability to identify and quantify these synergies across a large design space enables the systematic exploration of regulatory interactions that are difficult to predict from sequence information alone. We show how the learned composability rules can be used to rationalise the activity of an endogenous CRE and reprogramme it to increase or decrease its signal-dependence. Such rational design enables the rewiring of signal-decoding GRNs: we use the FL-synCREs to re-engineer signal-dependent control of the motor neuron specifying TF Olig2, producing predictable changes in expression and patterning.

Neural network based models of CRE accessibility and activity (20, 62, 71) have provided renewed hope for understanding how non-coding sequences enable their cellular functions. High-throughput constructionist methods such as NeMECiS can help test these models beyond the natural training set, allowing one to “hold out the genome” (24). However, the challenge in the use of these models is that their size and opacity prohibit the interpretable derivation of design principles. Our simple linear regression model achieves comparable and very similar predictive performance to the more sophisticated neural network models, while

offering interpretability. Whereas neural networks remain opaque black boxes, our results demonstrate that CRE activity can be explained through a small set of hierarchical, quantitative rules that provide clear mechanistic insights. This approach is intentionally coarse-grained to map the combinatorial space and hence capture the higher-order composability rules. The equivalent performance of our interpretable framework suggests that the opaque higher-order structures learned by neural network models may simply be approximating these fundamental rules. This has the potential to bridge the gap between black-box predictions and mechanistic understanding, offering a path toward rational design of synthetic CREs.

Synthetic regulatory sequences with programmable signal-responsiveness, cell type specificity and expression levels hold promise in reprogramming cell fates and targeting the delivery of gene therapy cargoes. The modular composability rules our study has uncovered suggest that practices from engineering can be applied to the design of regulatory sequences, allowing for independent optimisation of modules and their integration into bespoke synthetic sequences. This contrasts with the computationally and experimentally intractable task of a complete sequence space search that would be necessary in the absence of modularity. Practically, this will enable the construction of tailored synthetic regulatory sequences by combining the desired parts from a catalogue of modules, which can be adjusted or upgraded as this catalogue expands.

We demonstrate the power of this rule-based engineering approach by rewiring the signal-dependent expression of Olig2, allowing us to reprogramme the pattern of motor neuron progenitors. Moreover, the observation that the rearrangement of natural CREs results in tuneable changes in signal response suggests that evolutionary processes have optimised extant CREs to balance multiple constraints in addition to precisely titrated signal sensitivity to meet tissue patterning needs. Understanding these constraints could provide insight into both the evolution of gene regulation and the design of synthetic regulatory elements for specific applications.

Taken together, our study provides new insight into CRE function and its modular, interactional, and syntactical rules. By systematically exploring the design space of synCREs, we provide a robust framework for the rational engineering of synthetic regulatory elements. These findings have broad implications for synthetic biology and regenerative medicine, offering new avenues for the precise control of gene expression and cellular behaviour. As the field of synthetic genomics continues to progress, the principles uncovered here will serve as a foundation for the next generation of synthetic regulatory tools and therapeutic strategies.

Our approach provides a link between neural network models of CRE function and interpretable design principles, suggesting that a set of simple, quantitative rules can explain how the activity of a CRE depends on its constituent parts. This insight not only advances our understanding of natural regulatory sequences but also opens up new possibilities for engineering synthetic genomes with programmable functions.

Acknowledgements

We thank Margarida Cardoso Moreira, Zena Hadjivasiliou, Greg Findlay, Jonathan Chubb, Rory Maizels and Hilary Knowles for critically reading the manuscript; Steven Lim, Muhammad Saeed, and the Flow Science Technology Platform for flow cytometry support; Elly Tanaka for advice on the NTO system; Laura Cubitt and the Advanced Sequencing Science Technology Platform for sequencing support; Advanced Light Microscopy Sequencing Science Technology Platform for imaging support; Molly Strom, Ana Cunha and Hilary Knowles for molecular biology advice; Ruben Perez-Carrasco for advice on theory and statistical analysis; Nicholas M. Luscombe for initial discussions; Thomas Frith for advice on mESC maintenance.

Funding

This work was supported by the Francis Crick Institute, which receives its core funding from Cancer Research UK (CC001051), the UK Medical Research Council (CC001051) and the Wellcome Trust (CC001051); and by the Wellcome Trust (220379/D/20/Z). M.J.D. was supported by the Wellcome Trust Career Development Award (227326/Z/23/Z). J.C.S. was supported by a Boehringer Ingelheim Fonds and Francis Crick Institute PhD Fellowships. D.B (671-2020) and A.P (860-2019) were supported by EMBO ALTFs. G.L.M.B was supported by EMBO ALTF (792-2021) and UKRI (EP/X031225/1). H.T.S was supported by a Sir Henry Wellcome postdoctoral fellowship.

Author contributions

J.C.S, M.J.D and J.B conceived the project, and acquired funding. J.C.S, D.B, M.J.D and J.B interpreted the data, and wrote the manuscript. J.C.S and D.B performed experiments. J.C.S established theoretical results. J.C.S and A.P performed dynamical modelling simulations. J.C.S, A.P, T.Y, B.D, L.G conceived and performed statistical analyses. A.S generated the Olig2 knockout HM1-TetOn mESC line. S.B. assisted in molecular cloning. E.F assisted with flow-cytometry. G.L.M.B and O.C.K.I assisted in conceptualisation. H.T.S established the NTO protocol. H.T.S. and M.M assisted with its implementation and analysis.

Competing interests

The authors declare no competing or financial interests.

Data and materials availability

Code for sequencing processing and statistical analysis is provided at <https://github.com/jakesorel/NeMECiS>. Accession number for raw and processed barcode sequencing data reported in this paper is GSE291283.

References

1. H. L. Ashe, J. Briscoe, The interpretation of morphogen gradients. *Development* **133**, 385–394 (2006).
2. J. Briscoe, S. Small, Morphogen rules: design principles of gradient-mediated embryo pattern ing. *Development* **142**, 3996–4009 (2015).
3. S. Kim, J. Wysocka, Deciphering the multi-scale, quantitative *cis*-regulatory code. *Molecular Cell* **83**, 373–392 (2023).
4. C. W. Garvie, C. Wolberger, Recognition of Specific DNA Sequences. *Molecular Cell* **8**, 937–946 (2001).
5. F. Spitz, E. E. M. Furlong, Transcription factors: from enhancer binding to developmental control. *Nature Reviews Genetics* **13**, 613–626 (2012).
6. J. P. Junker, K. A. Peterson, Y. Nishi, J. Mao, A. P. McMahon, A. van Oudenaarden, A Predictive Model of Bifunctional Transcription Factor Signaling during Embryonic Tissue Patterning. *Developmental Cell* **31**, 448–460 (2014).
7. S. Kim, J. Wysocka, Deciphering the multi-scale, quantitative *cis*-regulatory code. *Molecular Cell* **83**, 373–392 (2023).
8. J. Jiang, M. Levine, Binding affinities and cooperative interactions with bHLH activators delimit threshold responses to the dorsal gradient morphogen. *Cell* **72**, 741–752 (1993).
9. G. Struhl, K. Struhl, P. M. Macdonald, The gradient morphogen *bicoid* is a concentration-dependent transcriptional activator. *Cell* **57**, 1259–1273 (1989).
10. D. N. Arnosti, M. M. Kulkarni, Transcriptional enhancers: Intelligent enhanceosomes or flexible billboards? *J Cell Biochem* **94**, 890–898 (2005).
11. G. A. Jindal, E. K. Farley, Enhancer grammar in development, evolution, and disease: dependencies and interplay. *Dev Cell* **56**, 575–587 (2021).
12. M. M. Kulkarni, D. N. Arnosti, Information display by transcriptional enhancers. *Development* **130**, 6569–6575 (2003).
13. D. P. Bazett-Jones, B. Leblanc, M. Herfort, T. Moss, Short-Range DNA Looping by the Xenopus HMG-Box Transcription Factor, xUBF. *Science* **264**, 1134–1137 (1994).
14. D. Thanos, T. Maniatis, Virus induction of human IFN beta gene expression requires the assembly of an enhanceosome. *Cell* **83**, 1091–1100 (1995).
15. D. Panne, T. Maniatis, S. C. Harrison, An Atomic Model of the Interferon- β Enhanceosome. *Cell* **129**, 1111–1123 (2007).
16. E. K. Farley, K. M. Olson, W. Zhang, A. J. Brandt, D. S. Rokhsar, M. S. Levine, Suboptimization of developmental enhancers. *Science* **350**, 325–328 (2015).
17. L. Luna-Zurita, C. U. Stirnimann, S. Glatt, B. L. Kaynak, S. Thomas, F. Baudin, M. A. H. Samee, D. He, E. M. Small, M. Mileikovsky, A. Nagy, A. K. Holloway, K. S. Pollard, C. W. Müller, B. G. Bruneau, Complex Interdependence Regulates Heterotypic Transcription Factor Distribution and Coordinates Cardiogenesis. *Cell* **164**, 999–1014 (2016).
18. S. Kim, E. Morgunova, S. Naqvi, S. Goovaerts, M. Bader, M. Koska, A. Popov, C. Luong, A. Pogson, T. Swigut, P. Claes, J. Taipale, J. Wysocka, DNA-guided transcription factor cooperativity shapes face and limb mesenchyme. *Cell* **187**, 692–711.e26 (2024).
19. G. Junion, M. Spivakov, C. Girardot, M. Braun, E. H. Gustafson, E. Birney, E. E. M. Furlong, A Transcription Factor Collective Defines Cardiac Cell Fate and Reflects Lineage History. *Cell* **148**, 473–486 (2012).

20. B. P. de Almeida, F. Reiter, M. Pagani, A. Stark, DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* **54**, 613–624 (2022).
21. I. I. Taskiran, K. I. Spanier, H. Dickmanken, N. Kempynck, A. Pančíková, E. C. Ekşi, G. Hulselmans, J. N. Ismail, K. Theunis, R. Vandepoel, V. Christiaens, D. Mauduit, S. Aerts, Cell-type-directed design of synthetic enhancers. *Nature* **626**, 212–220 (2024).
22. B. P. De Almeida, C. Schaub, M. Pagani, S. Secchia, E. E. M. Furlong, A. Stark, Targeted design of synthetic enhancers for selected tissues in the Drosophila embryo. *Nature* **626**, 207–211 (2024).
23. G. Novakovsky, O. Fornes, M. Saraswat, S. Mostafavi, W. W. Wasserman, ExplainNN: interpretable and transparent neural networks for genomics. *Genome Biology* **24**, 154 (2023).
24. C. G. de Boer, J. Taipale, Hold out the genome: a roadmap to solving the cis-regulatory code. *Nature* **625**, 41–50 (2024).
25. S. Pinglay, M. Bulajić, D. P. Rahe, E. Huang, R. Brosh, N. E. Mamrak, B. R. King, S. German, J. A. Cadley, L. Rieber, N. Easo, T. Lionnet, S. Mahony, M. T. Maurano, L. J. Holt, E. O. Mazzoni, J. D. Boeke, Synthetic regulatory reconstitution reveals principles of mammalian Hox cluster regulation. *Science* **377**, eabk2820 (2022).
26. J. Davies, Using synthetic biology to explore principles of development. *Development* **144**, 1146–1158 (2017).
27. G. M. Church, M. B. Elowitz, C. D. Smolke, C. A. Voigt, R. Weiss, Realizing the potential of synthetic biology. *Nat Rev Mol Cell Biol* **15**, 289–294 (2014).
28. S. J. Gray, S. B. Foti, J. W. Schwartz, L. Bachaboina, B. Taylor-Blake, J. Coleman, M. D. Ehlers, M. J. Zylka, T. J. McCown, R. J. Samulski, Optimizing promoters for recombinant adeno-associated virus-mediated gene expression in the peripheral and central nervous system using self-complementary vectors. *Hum Gene Ther* **22**, 1143–1153 (2011).
29. J. Jüttner, A. Szabo, B. Gross-Scherf, R. K. Morikawa, S. B. Rompani, P. Hantz, T. Szikra, F. Esposti, C. S. Cowan, A. Bharioke, C. P. Patino-Alvarez, Ö. Keles, A. Kusnyerik, T. Azoulay, D. Hartl, A. R. Krebs, D. Schübeler, R. I. Hajdu, A. Lukats, J. Nemeth, Z. Z. Nagy, K.-C. Wu, R.-H. Wu, L. Xiang, X.-L. Fang, Z.-B. Jin, D. Goldblum, P. W. Hasler, H. P. N. Scholl, J. Krol, B. Roska, Targeting neuronal and glial cell types with synthetic promoter AAVs in mice, non-human primates and humans. *Nat Neurosci* **22**, 1345–1356 (2019).
30. A. Sagner, J. Briscoe, Establishing neuronal diversity in the spinal cord: a time and a place. *Development* **146**, dev182154 (2019).
31. K. A. Peterson, Y. Nishi, W. Ma, A. Vedenko, L. Shokri, X. Zhang, M. McFarlane, J.-M. Baizabal, J. P. Junker, A. Van Oudenaarden, T. Mikkelsen, B. E. Bernstein, T. L. Bailey, M. L. Bulyk, W. H. Wong, A. P. McMahon, Neural-specific Sox2 input and differential Gli-binding affinity provide context and positional information in Shh-directed neural patterning. *Genes Dev.* **26**, 2802–2816 (2012).
32. T. Oosterveen, S. Kurdija, Z. Alekseenko, C. W. Uhde, M. Bergsland, M. Sandberg, E. Andersson, J. M. Dias, J. Muhr, J. Ericson, Mechanistic Differences in the Transcriptional Interpretation of Local and Long-Range Shh Morphogen Signaling. *Developmental Cell* **23**, 1006–1019 (2012).
33. K. Exelby, E. Herrera-Delgado, L. G. Perez, R. Perez-Carrasco, A. Sagner, V. Metzis, P. Sollich, J. Briscoe, Precision of tissue patterning is controlled by dynamical properties of gene regulatory networks. *Development* **148**, dev197566 (2021).
34. A. Sagner, Z. B. Gaber, J. Delile, J. H. Kong, D. L. Rouso, C. A. Pearson, S. E. Weicksel, M. Melchionda, S. N. M. Gharavy, J. Briscoe, B. G. Novitch, Olig2 and Hes regulatory dynamics during motor neuron differentiation revealed by single cell transcriptomics. *PLOS Biology* **16**, e2003127 (2018).
35. M. Gouti, A. Tsakiridis, F. J. Wymeersch, Y. Huang, J. Kleinjung, V. Wilson, J. Briscoe, In Vitro Generation of Neuromesodermal Progenitors Reveals Distinct Roles for Wnt Signalling in the Specification of Spinal Cord and Paraxial Mesoderm Identity. *PLOS Biology* **12**, e1001937 (2014).
36. M. J. Delás, C. M. Kalaitzis, T. Fawzi, M. Demuth, I. Zhang, H. T. Stuart, E. Costantini, K. Ivanovitch, E. M. Tanaka, J. Briscoe, Developmental cell fate choice in neural tube progenitors employs two distinct cis-regulatory strategies. *Dev Cell* **58**, 3-17.e8 (2023).
37. E. Z. Kvon, Y. Zhu, G. Kelman, C. S. Novak, I. Plajzer-Frick, M. Kato, T. H. Garvin, Q. Pham, A. N. Harrington, R. D. Hunter, J. Godoy, E. M. Meky, J. A. Akiyama, V. Afzal, S. Tran, F. Escande, B. Gilbert-Dussardier, N. Jean-Marçais, S. Hudaiberdiev, I. Ovcharenko, M. B. Dobbs, C. A. Gurnett, S. Manouvrier-Hanu, F. Petit, A. Visel, D. E. Dickel, L. A. Pennacchio, Comprehensive In Vivo Interrogation Reveals Phenotypic Impact of Human Enhancer Variants. *Cell* **180**, 1262-1271.e15 (2020).

38. T. Rito, A. R. G. Libby, M. Demuth, M.-C. Domart, J. Cornwall-Scoones, J. Briscoe, Timely TGF β signalling inhibition induces notochord. *Nature*, 1–10 (2024).
39. J. C. Klein, V. Agarwal, F. Inoue, A. Keith, B. Martin, M. Kircher, N. Ahituv, J. Shendure, A systematic evaluation of the design and context dependencies of massively parallel reporter assays. *Nat Methods* **17**, 1083–1091 (2020).
40. A. Melnikov, A. Murugan, X. Zhang, T. Tesileanu, L. Wang, P. Rogov, S. Feizi, A. Gnirke, C. G. Callan, J. B. Kinney, M. Kellis, E. S. Lander, T. S. Mikkelsen, Systematic dissection and optimization of inducible enhancers in human cells using a massively parallel reporter assay. *Nat Biotechnol* **30**, 271–277 (2012).
41. F. Inoue, M. Kircher, B. Martin, G. M. Cooper, D. M. Witten, M. T. McManus, N. Ahituv, J. Shendure, A systematic comparison reveals substantial differences in chromosomal versus episomal encoding of enhancer activity. *Genome Res* **27**, 38–52 (2017).
42. V. Agarwal, F. Inoue, M. Schubach, D. Penzar, B. K. Martin, P. M. Dash, P. Keukeleire, Z. Zhang, A. Sohota, J. Zhao, I. Georgakopoulos-Soares, W. S. Noble, G. G. Yardımcı, I. V. Kulakovskiy, M. Kircher, J. Shendure, N. Ahituv, Massively parallel characterization of transcriptional regulatory elements. *Nature*, 1–10 (2025).
43. S. J. Gosai, R. I. Castro, N. Fuentes, J. C. Butts, K. Mouri, M. Alasoadura, S. Kales, T. T. L. Nguyen, R. R. Noche, A. S. Rao, M. T. Joy, P. C. Sabeti, S. K. Reilly, R. Tewhey, Machine-guided design of cell-type-targeting cis-regulatory elements. *Nature* **634**, 1211–1220 (2024).
44. T. Oosterveen, S. Kurdija, Z. Alekseenko, C. W. Uhde, M. Bergsland, M. Sandberg, E. Andersson, J. M. Dias, J. Muhr, J. Ericson, Mechanistic Differences in the Transcriptional Interpretation of Local and Long-Range Shh Morphogen Signaling. *Developmental Cell* **23**, 1006–1019 (2012).
45. C. M. Vockley, A. M. D'Ippolito, I. C. McDowell, W. H. Majoros, A. Safi, L. Song, G. E. Crawford, T. E. Reddy, Direct GR Binding Sites Potentiate Clusters of TF Binding across the Human Genome. *Cell* **166**, 1269–1281.e19 (2016).
46. L. Giorgetti, T. Siggers, G. Tiana, G. Caprara, S. Notarbartolo, T. Corona, M. Pasparakis, P. Milani, M. L. Bulyk, G. Natoli, Noncooperative Interactions between Transcription Factors and Clustered DNA Binding Sites Enable Graded Transcriptional Responses to Environmental Inputs. *Molecular Cell* **37**, 418–428 (2010).
47. C.-C. Hui, S. Angers, Gli proteins in development and disease. *Annu Rev Cell Dev Biol* **27**, 513–537 (2011).
48. J. Jiang, C.-C. Hui, Hedgehog signaling in development and cancer. *Dev Cell* **15**, 801–812 (2008).
49. V. Loubiere, B. P. de Almeida, M. Pagani, A. Stark, Developmental and housekeeping transcriptional programs display distinct modes of enhancer-enhancer cooperativity in Drosophila. *Nat Commun* **15**, 8584 (2024).
50. J. Gertz, E. D. Siggia, B. A. Cohen, Analysis of combinatorial cis-regulation in synthetic and genomic promoters. *Nature* **457**, 215–218 (2009).
51. M. A. Shea, G. K. Ackers, The OR control system of bacteriophage lambda. A physical-chemical model for gene regulation. *J Mol Biol* **181**, 211–230 (1985).
52. M. S. Sherman, B. A. Cohen, Thermodynamic State Ensemble Models of cis-Regulation. *PLOS Computational Biology* **8**, e1002407 (2012).
53. N. E. Buchler, U. Gerland, T. Hwa, On schemes of combinatorial transcription logic. *Proceedings of the National Academy of Sciences* **100**, 5136–5141 (2003).
54. L. Bintu, N. E. Buchler, H. G. Garcia, U. Gerland, T. Hwa, J. Kondev, R. Phillips, Transcriptional regulation by the numbers: models. *Curr Opin Genet Dev* **15**, 116–124 (2005).
55. L. Bintu, N. E. Buchler, H. G. Garcia, U. Gerland, T. Hwa, J. Kondev, T. Kuhlman, R. Phillips, Transcriptional regulation by the numbers: applications. *Curr Opin Genet Dev* **15**, 125–135 (2005).
56. X. Lin, Y. Liu, S. Liu, X. Zhu, L. Wu, Y. Zhu, D. Zhao, X. Xu, A. Chemparathy, H. Wang, Y. Cao, M. Nakamura, J. N. Noordermeer, M. La Russa, W. H. Wong, K. Zhao, L. S. Qi, Nested epistasis enhancer networks for robust genome regulation. *Science* **377**, 1077–1085 (2022).
57. J. Choi, K. Lysakovskaia, G. Stik, C. Demel, J. Söding, T. V. Tian, T. Graf, P. Cramer, Evidence for additive and synergistic action of mammalian enhancers during cell fate determination. *eLife* **10**, e65381 (2021).
58. J. P. Bothma, H. G. Garcia, S. Ng, M. W. Perry, T. Gregor, M. Levine, Enhancer additivity and non-additivity are determined by enhancer strength in the Drosophila embryo. *Elife* **4**, e07956 (2015).
59. J. Vierstra, J. Lazar, R. Sandstrom, J. Halow, K. Lee, D. Bates, M. Diegel, D. Dunn, F. Neri, E. Haugen, E. Rynes, A. Reynolds, J. Nelson, A. Johnson, M. Frerker, M. Buckley, R. Kaul, W. Meuleman,

- J. A. Stamatoyannopoulos, Global reference mapping of human transcription factor footprints. *Nature* **583**, 729–736 (2020).
60. T. Jana Lang, S. Brodsky, W. Manadre, M. Vidavski, G. Valinsky, V. Mindel, G. Ilan, M. Carmi, F. Jonas, N. Barkai, Massively parallel binding assay (MPBA) reveals limited transcription factor binding cooperativity, challenging models of specificity. *Nucleic Acids Res* **52**, 12227–12243 (2024).
61. D. Quang, X. Xie, DanQ: a hybrid convolutional and recurrent deep neural network for quantifying the function of DNA sequences. *Nucleic Acids Research* **44**, e107 (2016).
62. Ž. Avsec, M. Weilert, A. Shrikumar, S. Krueger, A. Alexandari, K. Dalal, R. Fropf, C. McAnany, J. Gagneur, A. Kundaje, J. Zeitlinger, Base-resolution models of transcription-factor binding reveal soft motif syntax. *Nat Genet* **53**, 354–366 (2021).
63. E. D. Vaishnav, C. G. de Boer, J. Molinet, M. Yassour, L. Fan, X. Adiconis, D. A. Thompson, J. Z. Levin, F. A. Cubillos, A. Regev, The evolution, evolvability and engineering of gene regulatory DNA. *Nature* **603**, 455–463 (2022).
64. D. R. Kelley, J. Snoek, J. L. Rinn, Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks. *Genome Res* **26**, 990–999 (2016).
65. D. Quang, X. Xie, DanQ: a hybrid convolutional and recurrent deep neural network for quantifying the function of DNA sequences. *Nucleic Acids Research* **44**, e107 (2016).
66. M. Cohen, K. M. Page, R. Perez-Carrasco, C. P. Barnes, J. Briscoe, A theoretical framework for the regulation of Shh morphogen-controlled gene expression. *Development* **141**, 3868–3878 (2014).
67. A. Pezzotta, J. Briscoe, Optimal control of gene regulatory networks for morphogen-driven tissue patterning. *Cell Syst* **14**, 940-952.e11 (2023).
68. A. Meinhardt, D. Eberle, A. Tazaki, A. Ranga, M. Niesche, M. Wilsch-Bräuninger, A. Stec, G. Schackert, M. Lutolf, E. M. Tanaka, 3D Reconstitution of the Patterned Neural Tube from Embryonic Stem Cells. *Stem Cell Reports* **3**, 987–999 (2014).
69. T. Krammer, H. T. Stuart, E. Gromberg, K. Ishihara, D. Cislo, M. Melchionda, F. B. Perez, J. Wang, E. Costantini, S. Lehr, L. Arbanas, A. Hörmann, R. A. Neumüller, N. Elvassore, E. Siggia, J. Briscoe, A. Kicheva, E. M. Tanaka, Mouse neural tube organoids self-organize floorplate through BMP-mediated cluster competition. *Developmental Cell* **59**, 1940-1953.e10 (2024).
70. M. Martinez-Ara, F. Comoglio, B. Van Steensel, Large-scale analysis of the integration of enhancer-enhancer signals by promoters. *eLife* **12**, RP91994 (2024).
71. Ž. Avsec, V. Agarwal, D. Visentin, J. R. Ledsam, A. Grabska-Barwinska, K. R. Taylor, Y. Assael, J. Jumper, P. Kohli, D. R. Kelley, Effective gene expression prediction from sequence by integrating long-range interactions. *Nat Methods* **18**, 1196–1203 (2021).
72. I. Zhang, G. L. Boezio, J. Cornwall-Scoones, T. Frith, M. Jiang, M. Howell, R. Lovell-Badge, A. Sagner, J. Briscoe, M. J. Delás, The cis-regulatory logic integrating spatial and temporal patterning in the vertebrate neural tube. bioRxiv [Preprint] (2024). <https://doi.org/10.1101/2024.04.17.589864>.
73. I. Rauluseviciute, R. Riudavets-Puig, R. Blanc-Mathieu, J. A. Castro-Mondragon, K. Ferenc, V. Kumar, R. B. Lemma, J. Lucas, J. Chèneby, D. Baranasic, A. Khan, O. Fornes, S. Gundersen, M. Johansen, E. Hovig, B. Lenhard, A. Sandelin, W. W. Wasserman, F. Parcy, A. Mathelier, JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Res* **52**, D174–D182 (2024).
74. Y. Nishi, X. Zhang, J. Jeong, K. A. Peterson, A. Vedenko, M. L. Bulyk, W. A. Hide, A. P. McMahon, A direct fate exclusion mechanism by Sonic hedgehog-regulated transcriptional repressors. *Development* **142**, 3286–3293 (2015).
75. K. S. Pollard, M. J. Hubisz, K. R. Rosenbloom, A. Siepel, Detection of nonneutral substitution rates on mammalian phylogenies. *Genome Res* **20**, 110–121 (2010).

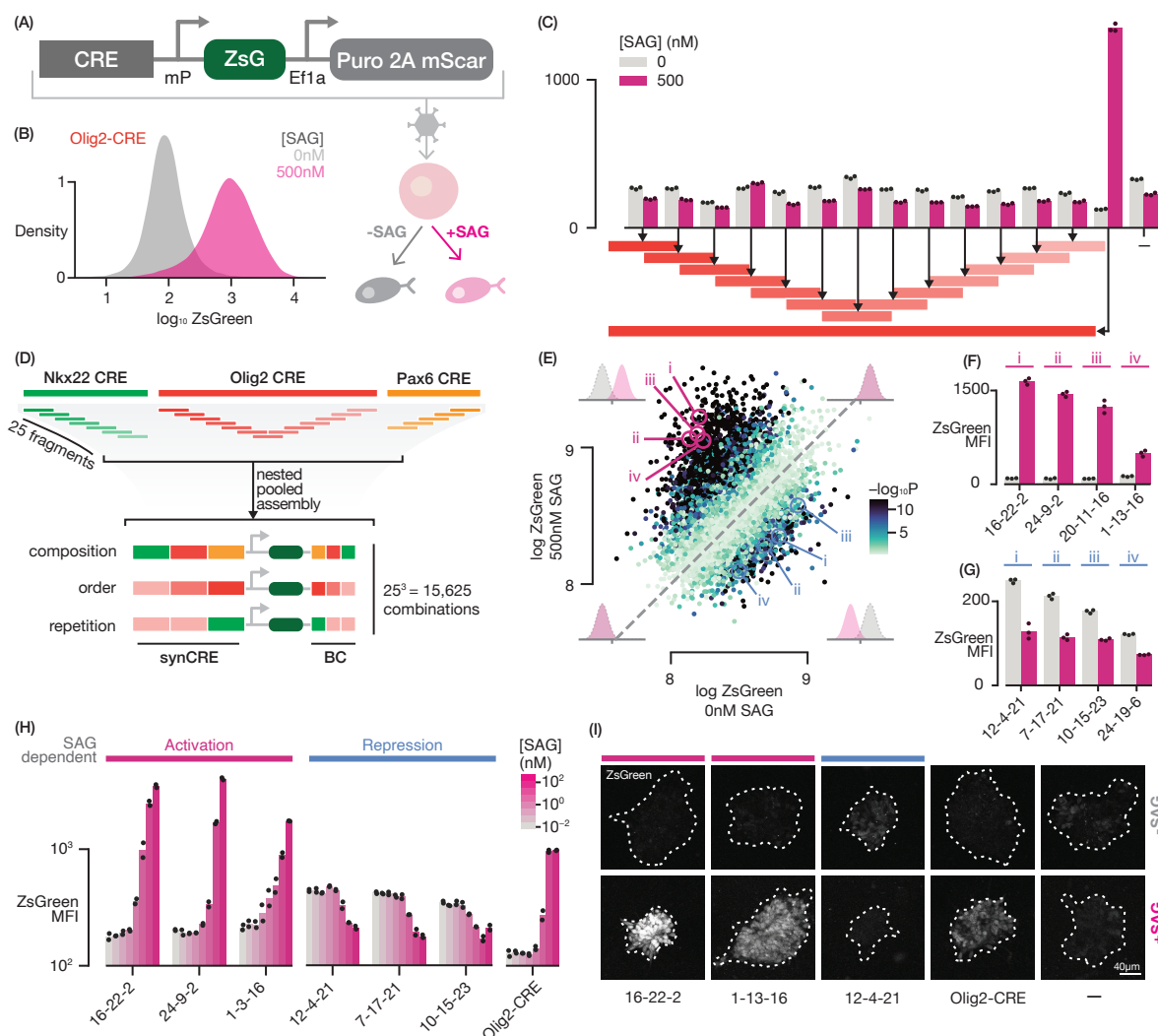


Figure 1. NeMECiS: A Combinatorial Screening Strategy Reveals Flexible Composability of Regulatory Elements.

(A) Schematic of lentiviral reporter construct, consisting of a CRE upstream of the *Shh* minimal promoter driving the green fluorescent protein ZsGreen, also containing the selectable markers mScarlet3 and Puromycin resistance. Lentivirus is generated from the cloned constructs, used to infect differentiating mESCs supplied with variable SAG concentrations, allowing for assaying of Hedgehog-dependent CRE activity.

(B) Live flow-cytometry of ZsGreen levels (kernel density estimate) in mScarlet3+ day-6 neural progenitor cells transduced with an Olig2-CRE reporter construct and treated with 0 or 500nM SAG.

(C) Using an equivalent live-flow cytometry quantification procedure, ZsGreen mean-fluorescence intensities (MFI) for day-6 neural progenitor cells cultured in 0 or 500nM SAG transduced with either the entire Olig2-CRE, or 200bp fragments of the Olig2-CRE (100bp offset), or promoter-only constructs without any CRE (“-”). Bars represent mean across biological replicates, and points represent individual replicate datapoints (N=3).

(D) Schematic of nested pooled sequential cloning strategy of NeMECiS, sequentially inserting one of 25 CRE fragments (with associated barcodes, BC) into a lentivirus backbone, yielding a pooled library of 15,625 combinations of CRE fragment triplet-fusions (synCREs), with barcode concatemers that reflect (in reverse) the composition and order of fragments by synCRE.

(E) Inferred log ZsGreen levels for the NeMECiS library in 0nM and 500nM SAG, coloured by the significance level of a non-zero difference between SAG concentrations ($-\log_{10}P$, z-test; $N=13,991$). Dashed-line represents equal inferred fluorescence in 0nM and 500nM SAG. Inset are diagrams of the expected fluorescence distributions for 0nM (grey) and 500nM (magenta) SAG in the four quadrants of the graph. Validation constructs in Fig. 1F-G are encircled and annotated.

(F) Independently constructed and tested validation constructs, assayed as in Fig. 1B-C, for synCREs predicted to show activation.

(G) As in Fig. 1F, but for repression. (H) Serial dilution of SAG concentration (8 concentrations at 5-fold dilution starting at 500nM), assayed as above.

(I) Representative images of day 6 neural progenitor colonies displaying ZsGreen levels for validation synCREs and controls, under 0nM and 500nM SAG. Magenta label represents “activation” synCREs and blue label represents “repression” synCREs. Colonies outlined (white dashed line) using a Sox2 immunofluorescence staining. Scale bar 40 μm .

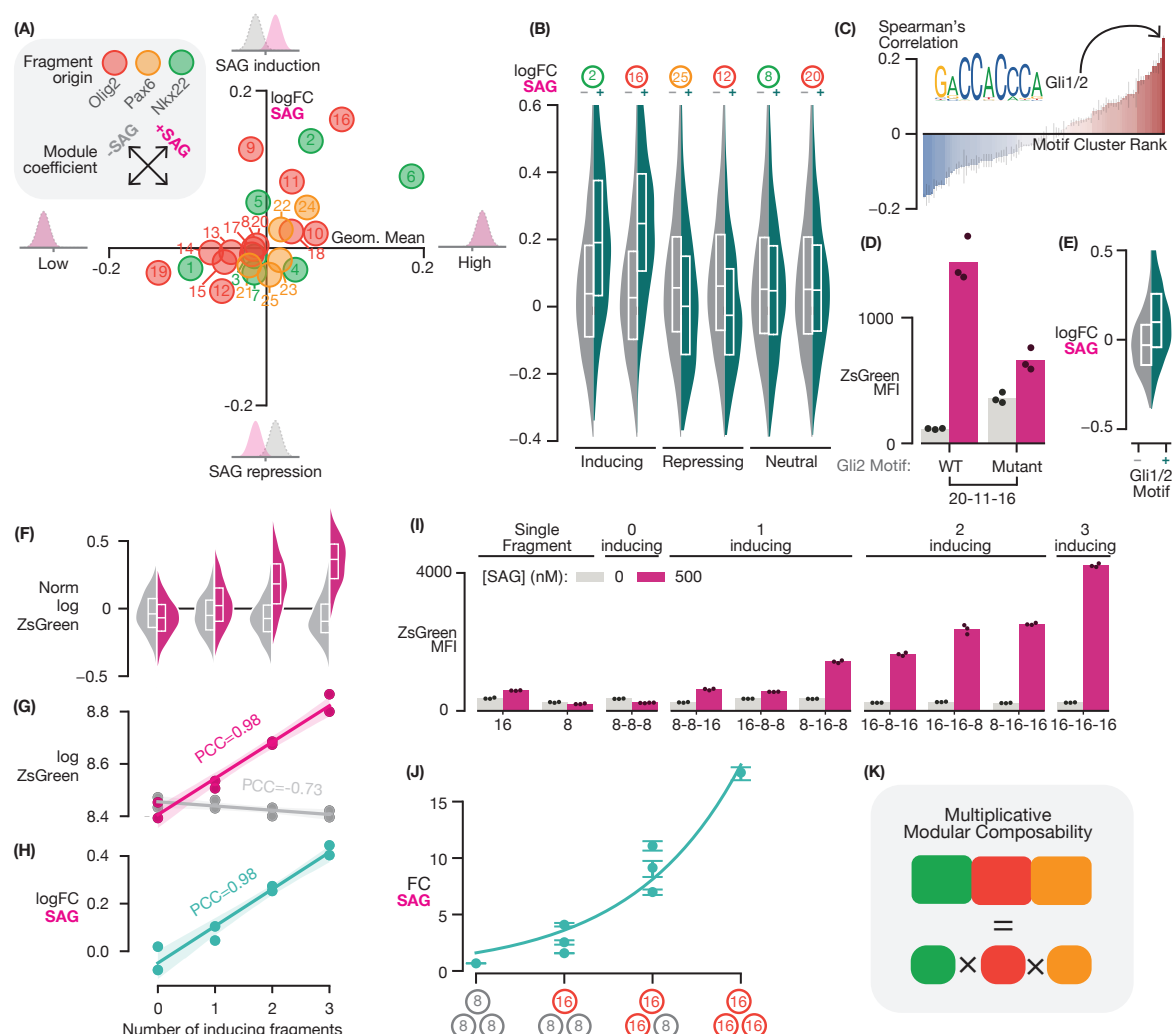


Figure 2. Modular Contributions of CRE Fragments to Activity Follow a Multiplicative Rule.

(A) Fragment contributions (module coefficient) from linear modelling of log ZsGreen ($\pm$ SAG) for each fragment in a synCRE. Each circle represents the contribution of a single fragment, coloured by their originating endogenous CRE. Each fragment has a weight for 0nM and 500nM SAG, which are used to compute logFC (logFC SAG (log fold change of ZsGreen = $w_{500} - w_0$) and their geom. (geometric) mean ($[w_0 + w_{500}]/2$). The diagonals of the graph represent the raw fragment weights (see key).

(B) Violin plots with associated quartiles for the logFC SAG of all synCREs without (grey) or with (teal) a given fragment, shown for two examples of each of the three classes of fragment activity. Fragments are denoted above the plot, coloured by their originating endogenous CREs.

(C) Spearman's correlation across synCREs of the presence of each DNA motif cluster with the corresponding logFC SAG. Performed separately for each replicate: bars represent mean across replicates; error bars indicate maximum/minimum. Annotated is the top hit motif cluster containing Gli1/2 and Zbtb7b/c, displaying the logo plot of the position weight matrix for Gli2 (JASPAR2024: MA0734.4 (73)).

(D) ZsGreen mean fluorescence intensity (MFI) in 0nM (grey) and 500nM (magenta) SAG for a wildtype (WT) strong inducing synCRE (f20-f11-f16) and a version where the (only) Gli2 motif was mutated (see Fig. S5D). Bars represent means across three biological replicates, and points represent individual replicates.

- (E) Violin plots with associated quartiles for the logFC SAG of all synCREs without (grey) or with (teal) a motif found in the cluster containing Gli1/2 and Zbtb7b/c.
- (F) Violin-plot and associated quartiles of log ZsGreen for 0nM (grey) or 500nM (magenta) SAG for the set of synCREs that lack repressing fragments, stratified by the number of inducing fragments. Values are normalised by subtracting the population mean across SAG concentrations (i.e. the intercept of the linear model).
- (G) The mean logZsGreen for the stratification detailed in Fig. 2F (PCC = pearson's correlation coefficient). Line represents replicate average, points represent individual replicates, shaded region represents 95% CI of a linear fit.
- (H) logFC SAG with the same stratification and depiction as Fig. 2F-G.
- (I) Individually built and assayed synCRE reporter constructs for the exhaustive set of ordered triplets that contain fragment 8 and 16, with corresponding single fragment reporter constructs. Data is plotted as in Fig. 2D. Number of inducing (i.e. fragment 16) fragments is annotated.
- (J) Fold change of mean fluorescence intensity of ZsGreen with respect to SAG (FC SAG) is plotted for the data in Fig. 2I. Each datapoint corresponds to a single synCRE, and the error bars represent the maximum/minimum FC SAG across all replicate-by-replicate comparisons. An exponential curve is fit to the data.
- (K) Schematic for the multiplicative modular composability model, where the activity of a synCRE is the product of the activities of each of its components.

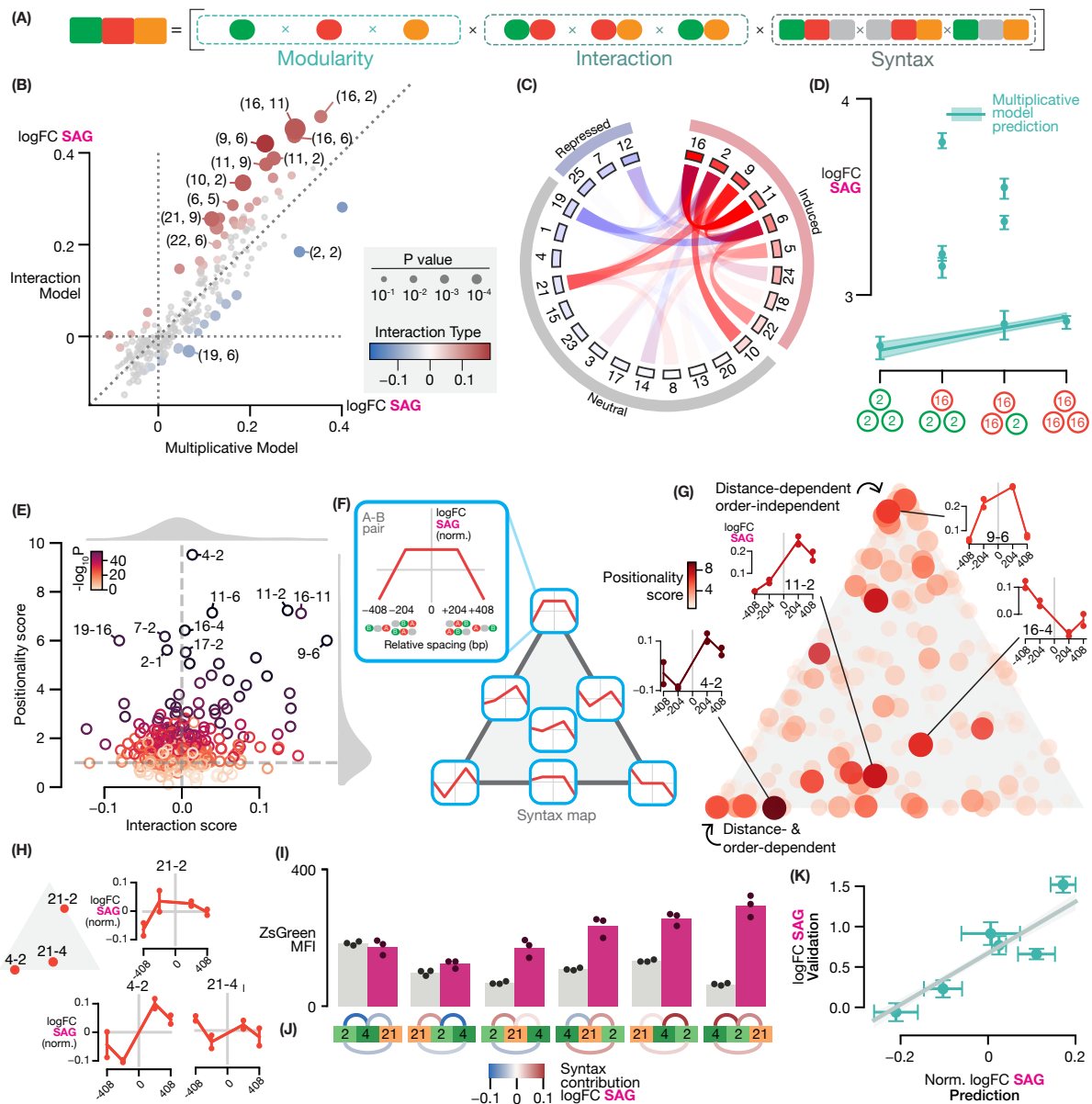


Figure 3. Pairwise Interactions and Syntax Rules Shape Regulatory Element Activity.

(A) Schematic of the extended linear model incorporating modularity, interaction and syntax layers. (B) Average contribution of each unique pair of fragments to the log fold change with respect to SAG (logFC SAG) in the multiplicative model and the interaction model. Dot size corresponds to the significance level of a z-test for non-zero interaction coefficients, and dot colour corresponds to the deviation of interaction coefficients from zero. Points above the dashed-line diagonal (red) indicate positive interactions (i.e. $\logFC\ SAG\ for\ fA \& fB > fA \times fB$) and points below (blue) indicate negative interactions (i.e. $\logFC\ SAG\ for\ fA \& fB < fA \times fB$). (C) Circos plot for the distribution of positive (red) and negative (blue) significant ($p < 0.01$) interactions between fragments. (D) Log fold change of ZsGreen mean fluorescence intensity with respect to SAG (logFC SAG) for individually built and assayed combinations of two activating fragments (f2 and f16), grouped by the number of each fragment. Datapoints represent mean across replicates for each synCRE, and error-bars represent the minimum and maximum value across exhaustive replicate comparison. Overlaid are the predicted activities from the multiplicative model for each of the combinations,

based on fitting to data considering combinations of f2 and f16 individually with a neutral fragment f8, where the error-bar is the 95% confidence interval.

(E) Distribution of interaction coefficient compared to positionality score (see Methods) for all heterologous fragment pairs. Top and right are kernel-density estimates of the marginal distributions. Points are coloured by significance of spacing dependent variance (Kruskal-Wallis H test for average logFC SAG across each possible spacing).

(F) Key for syntax map, built by orthonormal basis decomposition. Each position on the triangular manifold represents the relative contribution of each of three orthonormal bases to the total spacing-dependent variance of logFC SAG. Overlaid are diagrams of the logFC SAG (normalised to its mean) across all possible relative fragment spacings (see pull-out for an example of an ordered pair of fragments A & B).

(G) Orthonormal basis decomposition for all heterologous pairs of fragments, coloured by positionality score, with the shade and diameter of the points corresponding to the significance level in Fig. 3E. Inset are examples of spacing-dependent activity for different ordered fragment pairs.

(H) Case study of spacing-dependent activity for all pairings of a trimer (f2, f4, f21), where dots on the triangle correspond to the orthonormal basis decomposition for each pairing.

(I) Individually built and assayed exhaustively re-ordered synCREs comprising f2, f4, and f21 (MFI = mean fluorescence intensity; bars represent means across biological replicates; points represent individual replicates).

(J) Corresponding syntax contributions for each spaced-pair by synCRE (corresponding to the values in Fig. 3H).

(K) Comparison of predicted logFC SAG from linear modelling of NeMECiS to independently validated constructs for each of the six re-orderings of f2-f4-f21. Points represent means across replicates, and error bars represent maximum/minimum across replicates. Predicted logFC SAG is normalised by subtracting 'modular' and 'interactional' contributions.

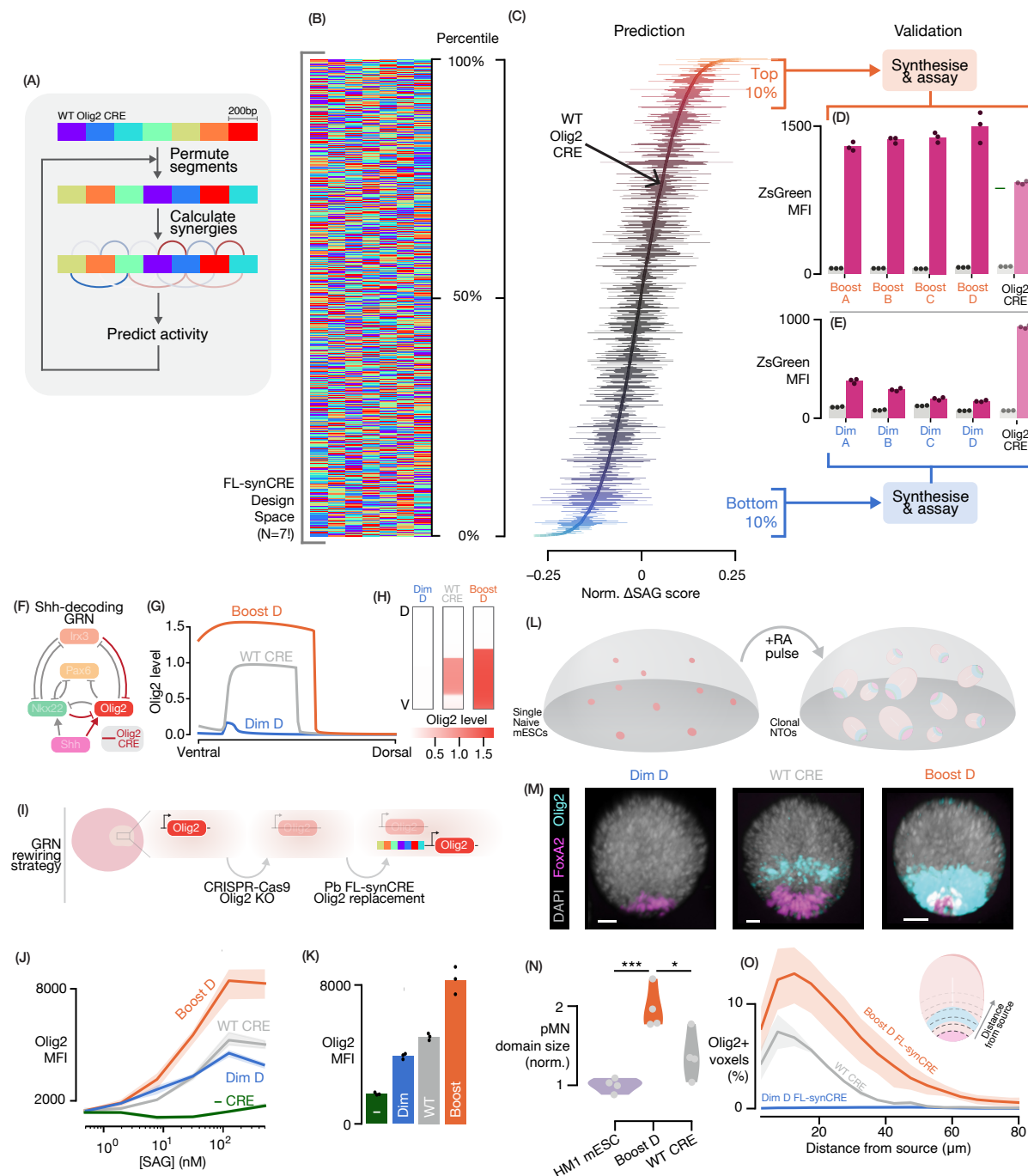


Figure 4. Predictable Engineering of CREs to Control Neural Patterning.

(A) Schematic representing iterative shuffling of 200bp fragments of the WT Olig2 CRE to generate FL-synCREs. Curved lines connecting fragments represent syntax contributions (see Fig. S11A). (B) Full FL-synCRE design space of permutations (5040 combinations), coloured as in Fig. 4A, ordered by predicted (normalised) log-fold change of ZsGreen level in response to 500nM SAG (logFC SAG). (C) Corresponding logFC SAG for each FL-synCRE, normalised to consider only positional-dependent effects, ordered by the average predicted logFC SAG across replicates. Horizontal bars represent predictions from each of the replicates of NeMECiS. Predicted logFC SAG of the WT ordering of fragments that make up the Olig2 CRE is annotated.

- (D) FL-synCREs from the top 10% of predicted logFC SAG were individually cloned and assayed for reporter activity under 0nM (grey) and 500nM (magenta) SAG. Bars represent means across biological replicates, and points represent individual replicates.
- (E) As in Fig. 4D but for the bottom 10%. Dim; Diminished.
- (F) Schematic of the ventral neural patterning gene regulatory network (GRN). Regulatory interactions corresponding to Olig2 CRE are coloured in red.
- (G) In silico prediction of Olig2 spatial expression pattern when the WT CRE is replaced with either Boost D or Dim D FL-synCREs.
- (H) Spatial distribution of Olig2 expression patterns from simulations in Fig. 4G.
- (I) Schematic of GRN rewiring strategy, involving: CRISPR-Cas9 mediated homozygous knockout of the Olig2 gene; re-introduction of a synthetic cassette containing a CRE, a minimal promoter and the Olig2 coding sequence.
- (J) Olig2 (AlexaFluor 488) mean fluorescence intensity (MFI) for populations of day 6 differentiating neural progenitors derived from FL-synCRE GRN re-wired mESCs, subjected to a logarithmic range of SAG concentrations. Solid lines represent mean across biological replicates, and shaded regions represent maxima/minima (N=3 replicates).
- (K) As in Fig. 4J, plotted for 500nM SAG. Bars represent means across replicates, and points represent individual biological replicates.
- (L) Schematic of clonal neural tube organoid (NTO) derivation. Single naïve mESCs are embedded in a Matrigel droplet and resulting NTOs analysed at day 6 of the differentiation.
- (M) Representative immunofluorescence images of FoxA2 and Olig2 expression in NTOs derived from FL-synCRE GRN re-wired mESCs. Scale bar = 25 μ m.
- (N) Quantification of pMN (motor neuron progenitor) domain size for wildtype HM1 mESCs, and mESCs subjected to GRN rewiring using the WT Olig2 CRE and the Boost D FL-synCRE. Dots are replicates.
- (O) Proportion of voxels that are Olig2 positive, binned by distance from the FoxA2+ pole (see Methods).

Predictable Engineering of Signal-Dependent Cis-Regulatory Elements

Materials and Methods

Contents

1	Cell culture	4
1.1	Mouse ES cell maintenance	4
1.2	Olig2 CRISPR knockout HM1 line generation	4
1.3	Lentivirus generation	4
1.4	Piggybac transgenesis	4
1.5	2D neural progenitor differentiation	5
1.5.1	Dual-colour differentiation for live flow cytometry	5
1.5.2	Neural progenitor colony differentiation imaging	5
1.5.3	Intracellular fixed flow-cytometry	5
1.6	Neural tube organoid differentiation	5
2	Flow cytometry	6
2.1	Live flow cytometry	6
2.2	Fixed flow cytometry	6
2.3	Flow cytometry analysis	7
3	Immunofluorescence (IF) staining	7
3.1	IF for 2D differentiations	7
3.2	IF and clearing of neural tube organoids	7
4	NeMECiS screen	7
4.1	NeMECiS mESC differentiation	7
4.2	NeMECiS flow sorting specifications	8
4.3	NeMECiS library preparation and sequencing	8
5	Molecular cloning	9
5.1	Cloning lentivirus backbones	9
5.2	Routine cloning, bacterial culture, plasmid DNA extraction and validation	9
5.3	FL-synCRE reporter constructs	9
5.4	FL-synCRE-Olig2 Piggybac plasmids	10
5.5	synCRE construction: pooled and individual	10
5.5.1	Overview	10
5.5.2	Preparation of synCRE backbone and assembly of the barcode concatemer	11
5.5.3	Allocation of CRE fragments	11
5.5.4	Preparation of 'even' and 'odd' subcloning vectors	11
5.5.5	Pooled sequential nested cloning	12
5.5.6	Arrayed preparation of individual synCRE reporter constructs	12
5.5.7	Barcode allocation	13
5.6	Gli1/2 binding site mutant reporter	13

6	NeMECiS sequencing processing	14
6.1	Alignment and barcode counts	14
6.2	Bayesian inference of fluorescence from barcode distributions	14
6.2.1	Outline of approach	14
6.2.2	Finite number effects	15
6.2.3	Calibrating bin gates	16
6.2.4	Implementation of Bayesian inference	16
7	Linear modelling framework for learning modular contributions and interactions	17
7.1	Overview	17
7.2	Remark about scale of data	18
7.3	Data filtering	18
7.4	Feature extraction	18
7.5	Fitting procedure, cross-validation and hyperparameter tuning	19
7.6	Model 1: Fragment contributions alone	19
7.6.1	Fragment class allocation	19
7.7	Model 2: Fragment contributions + interactions + spacing effects	19
7.7.1	Scoring interactions	19
7.7.2	Positionality score	19
7.7.3	Identifying syntax modes with k-means clustering	20
7.7.4	Orthogonal basis decomposition of syntax contributions	20
7.7.5	Inferring fragment order-dependent effects from spacing rules	20
7.7.6	FL-synCRE activity predictions	21
7.7.7	Modelling of individualised synCRE flow cytometry analysis	21
8	Comparison of deep learning approaches in predicting synCRE activity	21
8.1	Overview	21
8.2	Model Architecture	21
8.2.1	ExplaiNN	22
8.2.2	ConvNetDeep (DeepCNN)	22
8.2.3	DanQ	22
8.2.4	DeepSTARR	22
8.3	Data processing and model training	23
8.3.1	Data processing	23
8.3.2	Hyperparameter optimization	23
8.3.3	Model Training	23
8.3.4	Model evaluation	23
9	Motif analysis	24
9.1	Motif calling	24
9.2	Clustering of motifs into archetypes	24
9.3	Motif enrichment analysis	24
10	Dynamical modelling of neural tube patterning	24
10.1	Fitting FL-synCREs	25
11	Image analysis	26
11.1	Pre-processing and segmentation	26
11.2	Organoid geometry metrics	26
11.3	FoxA2+ floor-plate scoring	26
11.4	Organoid quality control filtering	27
11.5	Olig2 spatial expression analysis	27

12 Relating thermodynamic state ensemble models to synCRE activity	27
12.1 Transcriptional activation by a single site	27
12.2 Varying number of independently acting transcription factors	28
12.3 Breaking independence requirements	29
12.4 Defining regulatory modules	30
12.5 Linear modelling	31
13 Tables	32

1 Cell culture

1.1 Mouse ES cell maintenance

All experiments were carried out using the HM1 mouse embryonic stem cell (mESC) line [Magin et al., 1992], maintained at 37°C with 5% CO₂. For all experiments besides the generation of neural tube organoids (NTOs), mESCs were maintained in ESGRO Complete PLUS Clonal Grade Medium (Merck Millipore, Cat. No. SF001-500P) supplemented with 1% Penicillin/Streptomycin (Gibco, Cat. No. 15140122) on CellBIND tissue-culture plates (Corning, Cat No. 3335) coated with 0.15% gelatin (Gibco, Cat. no. G1393-100ML). mESCs were passaged every two days at 10^4 cells / cm^2 using StemPro Accutase (Gibco, Cat. No. A1110501) to dissociate cells. For generating NTOs, mESCs were maintained in 2i+LIF media at 3×10^3 cells / cm^2 : DMEM/F12(-/-) (Gibco, Cat. No. 21331-020) and Neurobasal (Gibco, Cat. No. 21103-049) at 1:1, supplemented with 1xN2 (Gibco Cat No. 17502001), 1xB27 (Gibco Cat. No. 17504001), 0.1% 50mM 2-mercaptoethanol (Gibco, Cat. No. 31350-010), 1% 200mM L-Glutamine (Gibco, Cat. No. 25030024), 1% Penicillin/Streptomycin (Gibco, Cat. No. 15140122), 500U/ml LIF (Chemicon, Cat. No. Int ESG1107), 1 μ M PD0325901 (Axon Medchem, Cat. No. 1408), 3 μ M CHIR99021 (Axon Medchem, Cat. No. 1386).

1.2 Olig2 CRISPR knockout HM1 line generation

A gRNA expression plasmid was built using the pX459v2 vector (Addgene #62988) and the following complementary 5' pre-phosphorylated oligonucleotides (CACCGTTTCTGCCCCGCCGAGCAA, AAAGTTGCTCCGGGCGGGCAGAAAC, Integrated DNA Technologies), following [Ran et al., 2013]. 4 μ g of a maxiprep (Qiagen, Cat. No. 12162) of the expression plasmid was electroporated into 2i+LIF cultured HM1-TetON mESCs [Serafimidis et al., 2008] with the AMAXA nucleofector kit (Lonza Cat no. VPH-1001) following manufacturer's instructions, and plated onto 0.1% gelatin coated 10cm tissue culture plates. After 24 hours, media was changed into 2i+LIF supplemented with 1.5 μ g/mL Puromycin Dihydrochloride (Gibco, Cat. No. 12122530). After a further 48 hours, media was changed to 2i+LIF. Once visible, clones were expanded and genotyped (F:atggactcggacgccagc; R:gacggtgacttgagcagc). An homozygous knockout clone was identified to carry a 23bp deletion (CGGAGCAAGGGGGGAAGCAGCAG), resulting in a 2bp frameshift after 24 amino acids, and a premature stop codon after 94 amino acids. Knockout was confirmed at the protein level by fixed intracellular flow-cytometry upon 6-day differentiation (Goat Olig2 unconjugated, 1:400; R&D, Cat. No. AF2418) (Fig. S11G).

1.3 Lentivirus generation

To generate lentivirus, HEK293T were plated at 1.5×10^6 per 60mm tissue culture plate, cultured in DMEM (Gibco, Cat. No. 21969-035), supplemented with 10% Fetal Bovine Serum (PAN-Biotech P30-3031), 1% 200mM L-Glutamine (Gibco, Cat. No. 25030024), 1% Penicillin/Streptomycin (Gibco, Cat. No. 15140122). The next day, the media was changed (3.5mL). 3rd generation lentivirus packaging plasmids (0.24 μ g CMV-Rev, 0.46 μ g pMDLg, 0.34 μ g VSV-G) and 2.28 μ g of the transfer plasmid were mixed with 360 μ L Opti-Mem (Gibco, Cat. No. 31985062), 11.9 μ L XtremeGene-HP (Roche, Cat. No. 6366236001), vortexed briefly and incubated at room-temperature for 15 minutes. This was added dropwise to the HEK293T cells. 16 hours later, the media was changed into N2B27 media (2mL): DMEM-F12 (Gibco, Cat. No. 21331-020) and Neurobasal (Gibco, Cat. No. 21103-049) (1:1), supplemented with 1xN2 (Gibco, Cat No. 17502001), 1xB27 (Gibco, Cat. No. 17504001), 2mM L-glutamine (Gibco, Cat. No. 25030024), 40 μ g/ml BSA (Sigma-Aldrich, Cat. No. A7979-50ML), and 0.1mM 2-mercaptoethanol. After 30 hours, the supernatant was collected with a syringe and filtered through a 0.45 μ m filter. Lentivirus preps were scaled up and down by culture plate area. To generate stable mESC lines, lentivirus supernatant was added to mESCs at the time of passaging at MOI<1; after 48hrs and from then onwards, cells were maintained with 1 μ g /mL Puromycin Dihydrochloride (Gibco, Cat. No. 12122530).

1.4 Piggybac transgenesis

To generate Piggybac transgenic mESC lines, 0.1 μ g of a mouse-codon optimised PiggyBac transposase plasmid [Cadinanos and Bradley, 2007] was vortexed in a 1.5mL Eppendorf microtube with 0.4 μ g of PiggyBac transfer plasmid, 25 μ L Opti-Mem (Gibco, Cat No. 31985062). In a separate microtube, 1 μ L of

Lipofectamine Stem Transfection Reagent (ThermoFischer, Cat. no. STEM00001) was vortexed with 25 μ L Opti-Mem, which were combined together and incubated for 10 minutes. mESCs were passaged in ES-GRO Complete PLUS Clonal Grade Medium (Merck Millipore, Cat. No. SF001-500P) onto 24-well tissue culture plates coated with 1% Vitronectin (Gibco, Cat. No. A31804), and the transfection mixture was added dropwise. After 48 hours, cells were maintained in 10 μ g/mL Blastidin S HCl (Gibco, Cat. No. A1113903), including during 2D and 3D differentiations.

1.5 2D neural progenitor differentiation

1.5.1 Dual-colour differentiation for live flow cytometry

10cm tissue culture plates were coated with 1% Vitronectin (Gibco, Cat. No. A31804) in Phosphate Buffer Saline (PBS) (Gibco, Cat. No. 14190-094) for 1 hour. mESCs were dissociated using StemPro Accutase (Gibco, Cat. No. A1110501) and centrifuged and resuspended in N2B27 (see above). Cells were seeded at 3×10^5 cells per plate (8mL) supplemented with 10ng/ml bFGF (R&D, Cat. No. 100-18B). Two days later, 12-well tissue culture plates were coated for 1 hour with a 1:100 dilution of GelTrex (Gibco, Cat. No. A1413202) in DMEM-F12(-/-) (Cat. No. 21331046). After 48hrs, cells were dissociated with StemPro Accutase (Gibco, Cat. No. A1110501), centrifuged and resuspended in N2B27 supplemented with 10ng/ml bFGF (R&D, Cat. No. 100-18B) and 5 μ M CHIR99021 (Axon Medchem, Cat. No. 1386) and plated at a 1x 10cm plate to 1x 12 well-plate ratio (0.95mL media per well), supplemented additionally with 50 μ L lentivirus supernatant. 20 hours later, media was changed into N2B27 supplemented with 100nM RA (Sigma, Cat. No. R2625) and variable concentrations of SAG (Merck, Cat. No. 566660-5mg). Media was changed daily. 72 hours later (day 6 of the differentiation), cells were collected for flow cytometry analysis (see below).

1.5.2 Neural progenitor colony differentiation imaging

The differentiation was set up in an identical manner to the above, but in 12-well tissue culture plates, plating 2.1×10^4 cells per well. After 48 hours, media was changed into 10ng/ml bFGF (R&D, Cat. No. 100-18B) and 5 μ M CHIR99021 (Axon Medchem, Cat. No. 1386). After 20 hours, media was changed into 100nM RA (Sigma, Cat. No. R2625) and variable concentrations of SAG (Merck, Cat. No. 566660-5mg). The next day, 24-well imaging plates (Miltényi Biotec, Cat. No. 130-098-263) were coated with a 1:50 dilution of Matrigel (Corning, Cat. No. 356231) in DMEM-F12 (-/-) (Gibco, Cat. No. 21331-020). After 24 hours of further culture, cells were dissociated with StemPro Accutase (Gibco, Cat. No. A1110501) and resuspended in fresh N2B27 with the same concentrations of supplements, and re-plated from a 1x 12-well well to 8x 24-well wells, which were pre-coated for 1 hour with with 1:100 GelTrex (Gibco, Cat. No. A1413202) in DMEM-F12 (Gibco, Cat. No. 21331046), to obtain sparse colonies. Media was changed daily. After a further 48 hours, samples were washed twice with Phosphate Buffer Saline (PBS) (Gibco, Cat. No. 14190-094) and fixed for 10 minutes with 4% methanol-free paraformaldehyde (ThermoScientific, Cat. No. 28908), followed by two subsequent washes in PBS (Gibco, Cat. No. 14190-094). Cells were then prepared for immunofluorescence staining (see below).

1.5.3 Intracellular fixed flow-cytometry

6-well tissue culture plates were coated for 1 hour with a 1:50 dilution of Matrigel (Corning, Cat. No. 356231) in DMEM-F12 (-/-) (Gibco, Cat. No. 21331-020). mESCs were dissociated and resuspended in N2B27 supplemented with bFGF as above and plated at 2×10^4 cells per well. Media was changed 48 hours later into bFGF and CHIR99021 as above. After 20 hours, media was changed into 100nM RA and variable SAG as above. Media was changed daily for another 72 hours, then cells were fixed for intracellular-stained flow cytometry (see below)

1.6 Neural tube organoid differentiation

Following the protocol for neural tube organoid generation [Krammer et al., 2024], mESCs maintained in 2i+LIF were dissociated with 200 μ L StemPro Accutase (Gibco, Cat. No. A1110501), and resuspended in 2mL 'Advanced N2B27' (Advanced DMEM/F12 (Gibco, Cat. No. 12634-010) and Neurobasal (Gibco, Cat. No. 21103-049) at 1:1, supplemented with 1xN2 (Gibco Cat., No. 17502001), 1xB27 (Gibco Cat. No. 17504001), 0.1% 50 mM 2-mercaptoethanol (Gibco, Cat. No. 31350-010), 1% 200mM L-Glutamine

(Gibco, Cat. No. 25030024), 1% Penicillin/Streptomycin (Gibco, Cat. No. 15140122), 1% non-essential amino acids (Gibco, Cat. No. 11140-035). Cells were aliquoted into a new 15mL Falcon, centrifuged, aspirated dry and gently resuspended in ice-cold Matrigel at a concentration of 10^2 cell / μ L (Corning, Cat. No. 354234). In small batches, 4 μ L of the mix was plated per well of an optically clear flat-bottom 96-well imaging plate (PerkinElmer Cat. No. 6055302) by drawing an inwards spiral in the centre of each well, ensuring a flat droplet of Matrigel that does not touch the sides. Plates were incubated for 3 minutes in a 37°C incubator, then supplemented with 100 μ L of Advanced N2B27. 48 hours later, 250nM all-trans Retinoic Acid (Merck, Cat. No. R2625) was supplemented into the media, and replaced with unsupplemented Advanced N2B27 after 18 hours. Media was changed daily. After 6 days from plating, 100 μ L of 4% paraformaldehyde (made from powder, as in [Krammer et al., 2024]) was added to each well on top of existing media and incubated at room temperature for 30 minutes. Wells were then washed twice with PBS.

2 Flow cytometry

2.1 Live flow cytometry

For live-cell flow cytometry collection, 1 μ L of LIVE/DEAD Fixable Dead Cell Stain Near-IR fluorescent reactive dye (ThermoScientific, Cat. No. L34976) was added to each 12-well tissue-culture plate well (1:1000) dilution and incubated at 37°C for 30 minutes. Wells were then washed twice with PBS, followed by dissociation with 100 μ L StemPro Accutase and resuspension in 120 μ L PBS+0.5% BSA (Sigma-Aldrich, Cat. No. A7979-50ML). Cells were transferred to a LoBind Microplate 96/V-PP 96-well plate (Eppendorf, Cat. No. 0030603303) and centrifuged at 500g for 5 minutes. Supernatant was removed by inversion of the plate. Cells were resuspended in 40 μ L of PBS + 2mM EDTA +0.5% BSA. Samples were run in high-throughput mode either on the BioRad ZE5 Cell Analyzer or the BD LSRFortessa Cell Analyzer, analysing the entire well for each. Samples that were infected with lentiviruses expressing ZsGreen or mScarlet3 singly and constitutively were run for compensation (without LIVE/DEAD staining), as were uninfected cells stained with LIVE/DEAD.

2.2 Fixed flow cytometry

For fixed intracellular-stained flow cytometry collection, following [Delas et al., 2023], LIVE/DEAD Fixable Dead Cell Stain Near-IR fluorescent reactive dye (ThermoScientific, Cat. No. L34976) was incubated at a 1:1000 dilution at 37°C, then washed, as above. Cells were then dissociated with StemPro Accutase and resuspended in PBS + 0.5% BSA as above. Cells were then transferred into 1.5mL DNA LoBind Eppendorf microfuge tubes (Eppendorf, Cat. No. Z666548), centrifuged at 500g for 5 minutes, supernatant removed by aspiration, and resuspended in 100 μ L of 4% methanol-free paraformaldehyde (ThermoScientific, Cat. No. 28908). After a 10 minute room-temperature incubation, 1mL of PBS was supplemented to each tube, followed by a 5 minute centrifugation at 2000g, and subsequent aspiration of supernatant. Cells were then resuspended in PBS + 0.5% BSA and stored at 4°C. For subsequent intracellular staining, 1×10^6 cells were transferred per well of a LoBind Microplate 96/V-PP 96-well plate (Eppendorf, Cat. No. 0030603303), and centrifuged for 5 minutes at 2000g, removing the supernatant by inversion. Cells were resuspended in a primary antibody mix, with a base of PBS + 0.5% BSA and 0.1% Triton-X100 (VWR Chemicals, Cat. No. 28817.295): Sox2-V450 (1:100; clone O30-678; BD Biosciences, Cat. No. 561610), Pax6-PerCPCy5.5 (1:100; clone 018-1330; BD Biosciences, Cat. No. 562388), Nkx6.1-AlexaFluor647 (1:100; clone R11-560; BD Biosciences, Cat. No. 563338), Goat Olig2 unconjugated (1:400; R&D, Cat. No. AF2418), Nkx2.2-PE (1:100; clone 74.5A5; BD Biosciences, Cat. No. 564730). The samples were incubated for 16 hours at 4°C, followed by two rounds of centrifugation, inversion and resuspension in PBS + 0.5% BSA and 0.1% Triton-X100. Cells were then resuspended in a secondary antibody mix with the same base: Donkey anti-goat AF488 (1:1000; Invitrogen, Cat. No. A11055). This was incubated for 40 minutes at room temperature, then washed twice as above. Cells were resuspended in 200 μ L of PBS + 0.5% BSA and run on the BD LSRFortessa Cell Analyzer, analysing 150 μ L of each sample. Single stain controls were additionally run, using UltraComp eBeads Plus Compensation Beads (Invitrogen, 01-3333-41) stained with the above conjugated antibodies.

2.3 Flow cytometry analysis

For all flow-cytometry analysis, live and fixed, FCS files were analysed using FlowJo (BD Biosciences), gating in the following order to obtain live singlets: FSC-A vs SSC-A; SSC-A vs SSC-H; FSC-A vs FSC-H; FSC-A vs LIVE/DEAD Near-IR. For dual-colour live flow cytometry analysis for ZsGreen reporter experiments, mScarlet3 positive cells were further subset, then mean fluorescence intensities in the ZsGreen channel were calculated. For fixed intracellular flow cytometry analysis, live singlets were further gated on Sox2 positivity to subset on neural progenitors.

3 Immunofluorescence (IF) staining

3.1 IF for 2D differentiations

Samples were pre-incubated with PBS + 0.5% BSA and 0.1% Triton-X100 (VWR Chemicals, Cat. No. 28817.295). Generically, primary antibody mixes were generated with the same base, added to wells at 150 μ L per 24-well well and incubated for 16 hours at 4°C. Wells were washed three times for 10 minutes each with PBS + 0.5% BSA and 0.1% Triton-X100, then supplied with the secondary antibody mix with the same base, and incubated at room temperature for 1 hour. Finally, wells were washed three times as above, then supplemented with PBS. For the single-colony re-plating experiment, we used the following primary (anti-Sox2 (1:1000, Invitrogen, Cat. No. 14-9811-80), goat anti-Olig2 (1:1000, R&D, Cat. No. AF2418), and mouse anti-Pax7 (1:500, DSHB)) and secondary (CS405 donkey anti-rat (Biotium, Cat. No. 20419); AF647 donkey anti-goat (Invitrogen, Cat. No. A21447); AF568 donkey anti-mouse (Invitrogen, Cat. No. A10037); all 1:1000) antibody mixes. 2D replated single-colonies were imaged with the an inverted Olympus FV3000 laser-scanning confocal microscope with an Olympus UPlanSApo 10x-0.40NA objective.

3.2 IF and clearing of neural tube organoids

For immunofluorescence staining of 3D neural tube organoids, samples were pre-incubated for 4 days at 4°C with PBS + 0.5% BSA and 0.1% Triton-X100 (VWR Chemicals, Cat. No. 28817.295). 50 μ L of the primary antibody mix (1:800 Olig2 chicken (Aves lab Cat. No OLIG2-0020); 1:400 FoxA2 goat (R&D Cat No. AF2400); 1:2000 Nkx6.1 rabbit (Novus Cat No. NBP1-49672)) was supplied to each well and incubated overnight at 4°C. Samples were washed once for 10 minutes each with PBS + 0.5% BSA and 0.1% Triton-X100 (VWR Chemicals, Cat. No. 28817.295) then incubated at room temperature for 6 hours, followed by a subsequent wash. 50 μ L of the Alexa Fluor conjugated secondary antibody mix (donkey anti-goat AF647 (Invitrogen Cat No. A-21447); donkey anti-chick AF568 (Jackson Cat No. 703-575-155); donkey anti-rabbit AF488 (Invitrogen Cat No. A-21206; all 1:1000) and 1:1000 DAPI (Cell Signalling 4083) was then supplied to each well and incubated overnight at 4°C. Samples were then washed three times in PBS + 0.5% BSA and 0.1% Triton-X100 for 10 minutes each. To clear samples, the buffer was aspirated and replaced with 100 μ L CUBIC-R+(N) [Matsumoto et al., 2019], incubated for 15 minutes, replaced with fresh CUBIC-R+(N), and repeated once more.

3D neural tube organoids were imaged with Olympus CSU-W1 SoRa Spinning Disk using a 10x air objective (10x UPLXAPO 0.4NA). Per well, a 3x3 tile-scan was performed per Z-stack, acquiring 72 Z-stacks at 3 μ m separation, allowing for the acquisition of the entire 3D volume of every organoid in each well.

4 NeMECiS screen

4.1 NeMECiS mESC differentiation

In duplicate, 10^7 mESCs were infected with the NeMECiS lentivirus library in ESGRO at MOI<1, maintained in CellBIND T175 flasks (Corning, Cat. No. 3292). After 48 hours, media was changed into ESGRO supplemented with 1 μ g/mL Puromycin Dihydrochloride (Gibco, Cat. No. 12122530). Cells were then expanded for one passage in ESGRO and Puromycin Dihydrochloride. 150mm tissue culture plates were coated with 1:100 Vitronectin (Gibco, Cat. No. A31804) diluted in DMEM-F12 (-/-) (Gibco, Cat. No. 21331-020) for 1 hour. Per replicate, a total of 6×10^7 cells were plated in N2B27 supplemented with bFGF as above. After

48 hours, media was changed into N2B27 supplemented with bFGF and CHIR99021 as above. After a further 20 hours, cells were dissociated with StemPro Accutase (Gibco, Cat. No. A1110501), centrifuged, and re-plated at the same density in N2B27 supplemented with 100nM RA (Sigma, Cat. No. R2625, additionally supplementing the media for half of the cells with 500nM SAG (Merck, Cat. No. 566660-5mg), onto 100mm tissue-culture plates coated with 1:100 GelTrex (Gibco, Cat. No. A1413202) diluted in DMEM-F12 (Gibco, Cat. No. 21331046). Media was changed daily. After 72 hours, cells were dissociated with StemPro Accutase, centrifuged and resuspended in 0.5% BSA (Sigma-Aldrich, Cat. No. A7979-50ML) in PBS (Gibco, Cat. No. 14190-094). 30% of each sample was immediately pelleted and frozen, and the remaining cells were flow sorted, as described below.

4.2 NeMECiS flow sorting specifications

24 hours before flow sorting, 10mL polypropylene tubes were incubated with 100% Fetal Bovine Serum. Prior to sorting, samples were supplemented with 10 ng/ml DAPI (Invitrogen, Cat. No. D1306). Samples were run on MoFlo XDP and BD InFlux. Sorted events were gated on live singlets (FSC-A vs SSC-A; SSC-A vs SSC-H; FSC-A vs DAPI). Four bins of ZsGreen intensity were established for each sample (two replicates, each for 0nM and 500nM SAG), demarcating the following percentile bins, following an optimal binning strategy found in [de Boer et al., 2020]: [0–15], [28.333–43.333], [56.666–71.666], [85–100]. FBS in polypropylene tubes was decanted and samples were sorted according to the above gating strategy. To assess sort quality and calibrate downstream analysis, 10 μ L of each sorted sample, as well as the unsorted pool, were extracted and run on the BD LSRFortessa Cell Analyzer, as described above. The final number of sorted cells is reported in Fig. S2E.

4.3 NeMECiS library preparation and sequencing

To extract genomic DNA, sorted cells for the NeMECiS screen were pelleted in 1.5mL DNA LoBind microfuge tubes (Eppendorf, Cat. No. Z666548) by centrifugation for 10 minutes at 500g. Genomic DNA was extracted for sorted and unsorted pellets using a Quick-DNA Miniprep Plus Kit according to the manufacturer's instructions (Zymo Research, Cat. No. D4069), dividing each sorted sample into six columns each. Each column was eluted in 75 μ L of the supplied elution buffer (10 mM Tris-HCl, pH 8.5, 0.1 mM EDTA) pre-warmed to 60°C, into 1.5mL DNA LoBind microfuge tubes, pooling the samples of the same origin to bring the total eluted gDNA to 450 μ L. Integrated barcode concatemers were amplified by PCR using NEBNext High-Fidelity 2X PCR Master Mix (New England Biolabs, Cat. No. M0541L) by combining on ice: 450 μ L gDNA, 500 μ L 2X master mix, 5 μ L each of 100 μ M Nextera-adaptor primers (indexing each sample, see below), and 40 μ L nuclease-free water. Amplification was carried out under the following program: 5 minutes at 72°C, 30 seconds at 98°C, followed by 17 cycles of 30 seconds at 98°C, 30 seconds at 63 °C and 60 seconds at 72°C, and subsequently 5 minutes at 72°C. At this step, all PCR products were pooled. The pooled amplicon DNA was then cleaned up using 1.8x v/v of AMPureXP beads (Beckman Coulter, Cat. No. A63882) and eluted in 50 μ L EB buffer (10 mM Tris-Cl, pH 8.5). Library concentration was determined using the KAPA Library Quantification Complete Kit Universal (Roche, Cat. No. 07960140001) according to manufacturer's instructions, using the QuantStudio 12K Flex Real-Time PCR System for qPCR (Applied Biosystems, Cat. No. 4471087). 50% of the amplified sequence was further amplified using NEBNext High-Fidelity 2X PCR Master Mix and P5 (AATGATACGGCGACCACCGA) and P7 primers (CAAGCAGAA-GACGGCATACGAGAT) using the same programme as above, then re-quantified as above to establish a concentration in excess of 1nM. Size spectra were examined with a Bioanalyzer using the DNA High Sensitivity DNA Kit (Agilent, Part No. 5067-4626) to confirm library quality. The library was then sequenced using the Illumina NextSeq 500 Sequencer on a 'High Output' run using the NextSeq 500/550 High Output Kit v2.5 (300 Cycles) (Illumina, Cat. No. 20024908) according to the manufacturer's instructions. Sequence demultiplexing was performed with the built-in software by supplying the corresponding indexing sequences for each sample.

5 Molecular cloning

5.1 Cloning lentivirus backbones

We based all lentivirus backbones on T3G-zsGreen-ultramiR-SFFV-rtTA (L3zUSR) [Delas et al., 2019]. Specifically, we generated a Golden Gate compatible lentivirus base vector – pLV-GG-WPRE – by cloning a bacterially expressible superfolder GFP flanked by Esp3I cut sites (derived from Addgene: 123943; [Fonseca et al., 2019]) between the CTS and WPRE sequences, replacing the existing contents of the plasmid with a Golden Gate landing-pad sequence (overhangs: 5'CTGA–AGCA 3'). This construct produces bacterial colonies that are visibly green under a blue-light box, whereas successful subsequent Golden Gate cloning with this vector produces non-green colonies.

We constructed reporter constructs through two alternative cloning strategies. We either inserted CREs into a backbone containing a minimal promoter, fluorescent protein, and one or more selectable marker; or we first inserted the CRE (or sequentially blocks thereof) into an 'empty' backbone, introducing the reporter and selectable cassettes in a second step (see synCRE construction).

For the first strategy, we built a series of base reporter vectors. We exclusively use the Sonic Hedgehog minimal promoter derived in [Kvon et al., 2020] (Addgene: 139098), optimised for high signal-to-noise in an in vivo mouse reporter context, to drive the expression of our reporter fluorescent protein ZsGreen, terminated by the SV40 polyA sequence. Upstream of the minimal promoter, we insert the same bacterially expressible superfolder GFP as above, this time flanked by PaqCI cut sites (overhangs: 5'CTGA–AGCA 3'), to act as a landing-pad for subsequent cloning. Downstream of the reporter cassette, we introduced one of two selectable cassettes, driven by an Ef1a promoter, and flanked by the 3' WPRE of the backbone: (1) Puromycin resistance gene (PuroR); (2) Puro-t2A-mScarlet3 (Addgene: 189753; [Gadella Jr et al., 2023]). These backbones are termed pLV-GG-CRErep-Puro and pLV-GG-CRErep-Puro2AmScar3 respectively. These backbones can be used in cloning reactions to insert any CRE with corresponding PaqCI golden-gate adaptors, where successful clonings can be screened by a lack of green fluorescence in bacterial colonies.

5.2 Routine cloning, bacterial culture, plasmid DNA extraction and validation

All cloning was performed using Golden Gate [Engler et al., 2008], using T4 Ligase (New England Biolabs, Cat. No. M0202L) and either Esp3I (Thermo Scientific, Cat. No. ER0451) or PaqCI (New England Biolabs, Cat. No. R0745L) as Type IIS restriction enzymes. In every case, 10 μ L reactions were prepared using 25fmol of each part, 1 μ L T4 Ligase Buffer, 0.1 μ L T4 Ligase, and 0.5 μ L of the Type IIS restriction enzyme ($\pm$ 'PaqCI Activator' in the case of PaqCI), thermocycling the reaction with the following protocol: for 25 cycles, 37 $^{\circ}$ C for 2 minutes, 16 $^{\circ}$ C for 5 minutes; then 37 $^{\circ}$ C for 30 mins; then 50 $^{\circ}$ C for 30 mins; then 80 $^{\circ}$ C for 10 minutes.

Besides the pooled sequential cloning used to build the NeMECiS plasmid library, all bacterial transformations were performed using MAX Efficiency Stbl2 Competent Cells (Invitrogen, Cat. No. 10268019), according to manufacturer's instructions. All bacteria were grown at 37 $^{\circ}$ C overnight in L-Broth, supplemented with the appropriate antibiotics (Carbenicillin (for Ampicillin resistance) at 100 μ g/mL (Merck Cat. No C1389-5G), Kanamycin at 50 μ g/mL (Sigma Aldrich Cat No. 60615), and Tetracycline at 15 μ g/mL (Sigma Aldrich Cat No. 87128). Minipreps were grown in 800 μ L L-Broth, using the fast protocol in the Zypmy Plasmid Miniprep Kit for extraction (Zymo Research, Cat. No. D4037). For larger DNA preps, midipreps were grown in 30mL of L-Broth, extracted using the QIAGEN Plasmid Plus Midi-Prep kit (Qia-gen, Cat. No. 12945). In both cases, plasmids were eluted in nuclease-free water. All plasmids used in the study were validated with whole-plasmid Nanopore sequencing, provided by Full Circle Labs Ltd (UK).

5.3 FL-synCRE reporter constructs

Based on our design-space survey of Olig2 CRE fragment re-orderings, we built sets of FL-synCREs that were predicted to have either boosted or diminished activity. Specifically, we synthesised these 1400bp FL-synCREs (without the 4bp spacer sequences inherent to the synCREs studied with NeMECiS) with appropriate Golden Gate adaptor sequences as Integrated DNA Technologies gBlocks, and cloned them into pLV-GG-CRErep-Puro2AmScar3.

5.4 FL-synCRE-Olig2 Piggybac plasmids

We established a base vector for experiments in which the transcription factor Olig2 is placed under the control of FL-synCREs by assembling using Esp3I-based Golden Gate: (1) a Piggybac backbone derived from [Benzinger and Briscoe, 2024]; (2) an expression cassette comprising the Sonic Hedgehog minimal promoter, 3×FLAG-FKBP12^{F36V}-GSGSGS-Olig2 (mouse CDS, synthesised as a gBlock), and an SV40 polyA; (3) upstream placing, a superfolder GFP, PaqCI-flanked Golden-Gate landing pad; (4) downstream placing an [Ef1a]-[Blasticidin]-[SV40 polyA] resistance cassette:

[Landing Pad] [Shh mP] [FKBP-Olig2] [polyA] [Ef1a] [BSD] [polyA]

FL-synCREs were cloned into this backbone using the same synthesised sequences as above.

5.5 synCRE construction: pooled and individual

5.5.1 Overview

synCREs are assembled by sequential cloning of CRE fragments into a lentivirus backbone. For trimer synCREs, this is achieved by three sequential cloning reactions, inserting into the backbone a single fragment and a single barcode per step. This sequential nested cloning approach builds up CRE fragments and barcodes one by one from the outside inwards. Once inserted, fragments and barcodes remain physically linked via the plasmid backbone, meaning the composition and order of fragments can be read out (in reverse) by the composition and order of their cognate barcodes. Additionally, at each step, a distinct secondary antibiotic resistance (in addition to the backbone's existing ampicillin resistance) is introduced, allowing successful cloning events to be selected for during bacterial culture. In a final step, the landing pads and associated secondary antibiotic resistances are replaced by the reporter construct.

This is achieved by alternate usage of pairs of Type IIS (Golden Gate) restriction enzymes and secondary bacterial antibiotic resistance genes. Specifically, each part is designated either an 'odd' structure:

CACCTGCatataACCA[CRE fragment]CTGatGAGACG[KanR]CGTCTCaAGCA[8bp Barcode]GATTatGCAGGTG
or an 'even' structure

CGTCTCaCTGA[CRE fragment]ACCAatataGCAGGT[TetR]CACCTGCatataGATT[8bp Barcode]AGCAatGAGACG
with the following key:

- Esp3I recognition sequence
- Esp3I associated 4bp overhang
- PaqCI recognition sequence
- PaqCI associated 4bp overhang
- Spacer sequence
- [Feature]

Here, parts with an 'odd' structure can be inserted into plasmid with a compatible PaqCI ('odd') landing pad. 'Odd' parts – and consequently their cloned products – contain an internal Esp3I ('even') landing pad. In the 'even' structure, the reverse is true, where the part can be inserted into a compatible Esp3I ('even') landing pad, and contains within it a PaqCI landing pad.

Consequently, inserting an 'odd' part into an 'odd' backbone yields the following reaction

[LV backbone]ACCAatataGCAGGT...CACCTGCatataGATT[LV backbone]

+ 'odd' part

⇒ [LVbb]ACCA[Frag]CTGatGAGACG[KR]CGTCTCaAGCA[BC]GATT[LVbb]

This has centrally an 'even' landing pad (CTGatGAGACG[KR]CGTCTCaAGCA) into which an 'even' part can be inserted in a subsequent step

⇒ [LVbb]ACCA[Frag]CTGA[Frag]ACCAatataGCAGGT[TR]CACCTGCatataGATT[BC]AGCA[BC]GATT[LVbb]

This recovers the initial structure with an 'odd' landing pad (ACCAatataGCAGGT[TR]CACCTGCatataGATT), allowing for repeated cycling, for one more time in our case, but indefinitely in principle.

⇒ [LVbb]ACCA[Frag]CTGA[Frag]ACCA[Frag]CTGatGAGACG[KR]CGTCTCaAGCA[BC]GATT[BC]AGCA[BC]GATT[LVbb]

(Note here we use the following shorthand notation: Frag = CRE fragment; KR = Kanamycin Resistance; TR = Tetracycline Resistance; BC = 8bp Barcode; LVbb = Lentivirus backbone).

In a final step, the landing pads and associated secondary antibiotic resistances are replaced by the reporter construct, using a *terminal* plasmid, which has the following structure:

CGTCTCtCTGA [ShhmP-ZsGreen-polyA-Ef1a-Puro+t2AmScarlet3] [Nextera Read 1] **AGCAaGAGACG** where we insert the Nextera Read 1 recognition sequence (TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG)) directly into the plasmid. The terminal plasmid containing Puro alone is termed *pNeMECiS-term-Puro*, and the plasmid additionally containing mScarlet3 is termed *pNeMECiS-term-Puro2AmScar*.

Note that this terminal plasmid is equally compatible for insertion into plasmids with one or three fragments/barcodes, allowing us to build single-fragment controls.

5.5.2 Preparation of synCRE backbone and assembly of the barcode concatemer

To perform the above nested sequential cloning, we established an 'odd' compatible backbone, containing the necessary parts for barcode extraction later. Specifically, using *pLV-GG-WPRE* as a base vector, we cloned in the following insert

ACCAat**atGCAGGTG** [lacZ cassette] **CACCTGC**at**atGATT** [min-pA] [Nextera Read 2 RC]

Here, Nextera Read 2 RC is the reverse complement of the Nextera Read 2 recognition sequence (RC = CTGTCTCTTATACATCTCCGAGCCACGAGAC), and min-pA is a 56bp truncated variant of the beta-globin polyadenylation signal (GATTAATAAAGGAAATTTATTTCATTGCAATAGTGTGTGGAATTTGTGTCTCTCA). We term this backbone *pLV-eNextpA-WPRE*. Note that this plasmid contains an 'odd' landing pad (**ACCA**at**atGCAGGTG** [...] **CACCTGC**at**atGATT**).

Upon sequential nested cloning, and introduction of the terminal sequence, one yields the following 163bp sequence built into the plasmid:

[Nextera Read 1] **AGCA** [BC] **GATT** [BC] **AGCA** [BC] **GATT** [min-pA] [Nextera Read 2]

This can be directly amplified from the plasmid, or upon lentivirally mediated integration, using Nextera-compatible primers with structures

5' - AATGATACGGCGACCAACGAGATCTACAC [seqBC] TCGTCGGCAGCGTCAGATGTG-3'

5' - CAAGCAGAAGACGGCATACGAGAT [seqBC] GTCTCGTGGGCTCGGAGATGT-3'

Giving a total Illumina sequencing compatible amplicon with the following structure

5' - [P5] [seqBC] [Nextera Rd1] **AGCA** [BC] **GATT** [BC] **AGCA** [BC] **GATT** [min-pA] [Nextera Rd2] [seqBC] [P7] -3'

5.5.3 Allocation of CRE fragments

200bp CRE fragments were derived from validated CREs for three well-established Sonic Hedgehog dependent TFs: Olig2, Nkx22, Pax6 [Oosterveen et al., 2012, Peterson et al., 2012]. Their full sequences are provided in Table 1. Note here that we use a combination of the sequences for the Olig2 CRE defined in [Oosterveen et al., 2012] and [Peterson et al., 2012], which is 1305bp in length.

We generated a 200bp fragmentation starting from the 5' end of each of the above sequences, with the final fragment overlapping and extending beyond the natural sequence's 3' extent. We then generated a 200bp fragmentation with a 100bp 3' offset. Sequences are provided in Table 2

5.5.4 Preparation of 'even' and 'odd' subcloning vectors

Nested cloning proceeded by sequentially inserting 'odd' parts into 'odd' backbones to generate 'even' plasmids, followed by the insertion of 'even' parts to return the plasmid to 'odd'. Practically, this was achieved by Golden Gate reactions between two types of plasmid: a lentivirus backbone and a (pool of) subcloning vectors. Each subcloning vector has an overall structure:

... [Ori] [Esp3I/PaqCI] [Frag] [PaqCI/Esp3I] [TetR/KanR] [PaqCI/Esp3I] [BC] [Esp3I/PaqCI] ...

These plasmids were cloned as follows. Integrated DNA Technologies eBlocks were synthesised the following structure, allocating the connection between 8bp barcodes ([BC]) and 200bp CRE fragments ([Frag]):

'Odd':

atta**CACCTGC**at**atACCA** [Frag] **CTGA**at**atGCAGGTG** [Spacer Odd] **CACCTGC**at**atAGCA** [BC] **GATT**at**atGCAGGTG**ttaa

'Even':

atta**CGTCTCtCTGA** [Frag] **ACCAaGAGACG** [Spacer Even] **CGTCTCtGATT** [BC] **AGCAaGAGACG**ttaa

Where [Spacer Odd] = CGTCGGGCTCGACATCGGCAAGGT

and [Spacer Even] = CGTCGGGCTCGACATCGGCAAGGTGCGACTGAATTCATGC.

NB: these spacer sequences act as fillers and are removed in the assembly of the subcloning vectors.

Next, a high-copy-number ColE1/pMB1/pBR322/pUC origin of replication ([Ori]) was amplified by PCR from Addgene: 123943 [Fonseca et al., 2019], adding on ‘odd’ or ‘even’ adaptors, introducing the outer Type IIS restriction sites into the plasmid that are subsequently used for sequential, nested cloning (note the inverted architecture of recognition cut-site pairs at the amplicon termini):

Odd: 5’-ttttGATTatGAGGTTG [Ori] CACCTGcatatACCAaaaa

Even: 5’-ttttAGCATGAGACG [Ori] CGTCTCaCTGAAAA-3’

Cognate resistance cassettes were then amplified by PCR: [KanR] (Kanamycin resistance gene, amplified from Addgene: 123943 [Fonseca et al., 2019]) for ‘odd’ subcloning vectors; and [TetR] (Tetracycline resistance gene, amplified from Addgene: 92394 [Williams et al., 2018]) for ‘even’ subcloning vectors.

Even: 5’-ttttCGTCTCaCCAatGAGGTTG [TetR] CACCTGcatatGATTaGAGACGaaaa-3’

Odd: 5’-ttttCACCTGcatatCTGatGAGACG [KanR] CGTCTCaAGCAatGAGGTTGaaaa -3’

where the multicoloured sequences NNNN denote dual potential usage by Esp3I and PaqCI. PCRs were performed using Phusion Flash High-Fidelity PCR Master Mix (Thermo Scientific, Cat No. F548S) by manufacturer’s instructions, and purified using the QIAquick PCR Purification Kit (QIAGEN, Cat No. 28104).

PCRs and eBlocks were then normalised to 25fmol/ μ L and cloned to generate plasmids using Golden Gate (‘odd’ with PaqCI; and ‘even’ with Esp3I), using a modified reaction protocol (10 cycles at 37°C for 2 minutes, 16°C for 5 minutes; 30 minutes at 16°C), mixed at a 1:1 ratio with a T4 ligase mix (per 10 μ L: 0.5 μ L T4 ligase, 1 μ L T4 ligase buffer, New England Biolabs, Cat No. M0202L), and incubated for 16 hours at 4°C. Cloning products were then transformed in NEB-5 α Competent E. coli (New England Biolabs, Cat No. C2987H), miniprep (Zyppy Plasmid Miniprep Kit; Zymo Research, Cat No. D4037), verified by Sanger Sequencing, and midiprep (QIAGEN Plasmid Plus Midi-Prep kit, Cat. No. 12945).

Additionally, 1mL live cultures (L-Broth) for each set of ‘odd’/‘even’ subcloning vectors (25 each) were serially cultured twice in 96 deep well plates (Axygen, Cat No. P-2ML-SQ-C-S), pooled in an equivolometric ratio and midiprep (QIAGEN Plasmid Plus Midi-Prep kit, Cat. No. 12945) together to generate equimolar pools.

5.5.5 Pooled sequential nested cloning

Firstly, pLV-eNextpA-WPRE (containing an ‘odd’ landing pad) and the ‘odd’ subcloning plasmid pool were individually secondarily cleaned up using DNA-Concentrator 5 (Zymo Research, Cat No. D4014) according to manufacturer’s instructions, eluting in nuclease-free water and normalising (using Qubit quantification of DNA concentration using the broad-range kit, Invitrogen, Cat No. Q32850) to 25fmol/ μ L. 24 $\times$ 10 μ L Golden Gate reactions with PaqCI were run (50 cycles), and immediately cleaned up using DNA-Concentrator 5, eluting in 50 μ L water. 2 μ L of this elution was transformed with Endura Electrocompetent Cells (Biosearch Technologies, Cat No. 60242-1) according to manufacturer’s instructions, then cultured directly at 28°C (170rpm) in 1L L-Broth supplemented with Carbenicillin and Kanamycin until the OD₆₀₀ reached 1.0 (~ 14 hours). 100mL of the culture was then maxiprep using the QIAGEN Plasmid Plus Maxi Kit (Cat No. 12965).

Next, an equivalent procedure was performed by cleaning up the *round 1* cloned pool (containing an ‘even’ landing pad) with the ‘even’ subcloning plasmid pool with DNA-Concentrator 5, normalising, and running a Golden Gate reaction with the same specifications, this time with Esp3I. An identical transformation, culture and DNA extraction procedure was then performed, this time using Tetracycline in place of Kanamycin, to establish the *round 2 cloned pool*. After this, the same protocol was performed as in round 1, but using *round 2 cloned pool* in place of pLV-eNextpA-WPRE. Finally, the same procedure was performed as in round 2, but combining *round 3 cloned pool* with pNeMECiS-term-Puro, this time culturing the transformation with Carbenicillin alone, and performing and pooling 4 $\times$ maxipreps.

5.5.6 Arrayed preparation of individual synCRE reporter constructs

synCRE reporter constructs were established in an analogous manner to the above, but in an arrayed fashion, allowing for the construction of desired fragment combinations and orderings. In this, clean-up steps before Golden-Gate reaction preparation and transformation were omitted, and MAX Efficiency Stbl2 Competent Cells (Invitrogen, Cat No. 10268019) were used. Here, 1 μ L of the reaction was combined with 10 μ L competent cells, heat-shocked and recovered according to manufacturer’s instructions, and grown at 37°C for 16 hours in 800 μ L L-Broth, split across two wells of 1.1mL 96 deep well plates (Axygen, Cat No. 736-0339), purifying minipreps using the Zyppy Plasmid Miniprep Kit for extraction (Zymo Research,

Cat No. D4037). In the final step, *pNeMECiS-term-Puro2AmScar3* was used to establish dual-colour reporter plasmids. Single fragment reporter constructs were established in an identical manner, cloning in *pNeMECiS-term-Puro2AmScar3* in the first step rather than the third. Matched full-length natural CRE and CRE-less, minimal-promoter only (spacer sequence = CGAGCGTCTCT) reporters were cloned in a similar manner.

5.5.7 Barcode allocation

NeMECiS relies on sequencing barcode concatemers to retrieve compositional information. Consequently, it is crucial that the barcodes used are maximally informative, given potential misassignment. We built a Markov chain Monte Carlo Metropolis Hastings algorithm to scan sequence space for optimal combinations of 25 barcodes. The algorithm is summarised below:

1. Generate 25 random sequences of 8bp. Add the overhangs for cloning to each side. Re-generate if any of the random sequences has: (i) >2 repeated bases; or (ii) a GC content below 40% or above 60%.
2. Calculate the sum H of Hamming distances among each pair of the 25 sequences (=300 pairs).
3. Randomly select one of the 25 sequences and replace the central 8bp with another random sequence that follows the two conditions above.
4. Calculate H_{new} of the new set of sequences.
5. If $H - H_{new} < 0$, accept the swap. Else, accept the swap if $p < \exp((H_{new} - H)/T)$, where p is a random number between [0,1]. ($T=0.01$). Otherwise, reject the swap.
6. Repeat 2-5 for another 2989 iterations.
7. Repeat 2-5 for another 10 iterations, logging the sequence combinations.
8. Repeat 1-7 500 times to establish an ensemble of sequence combinations.
9. Calculate E for each logged sequence:
 - $E = H + H_{shift} + H_{delete}$
 - Where H_{shift} is the sum of the minimum of the Hamming distances among pairs of sequences where sequences are shifted 1bp 5' or 3'.
 - And H_{delete} is the sum of the minimum Hamming distances among pairs of sequences where each of the base-pairs of one of the sequences is deleted.
10. Determine the combination of 25 barcodes with the maximum E .

The resulting barcodes are given in Table 3

5.6 Gli1/2 binding site mutant reporter

Using the motif enrichment analysis described below, we identified a putative Gli1/2 binding site in a strongly signal responsive synCRE f20-f11-f16:

...GTGAAATCTTGGGTGGTATGGGCACCG...

Key: surrounding sequence, expanded motif, MOODS/JASPAR2024 called motif

Adding 3bp to each side of the called motif to ensure complete disruption, we computationally scrambled the base-pairs to generate the sequence:

...GTGAAGGTTGACCGTGGGTATCACCG...

We then synthesised both the WT and Gli-mutated variants of f20-f11-f16 as Integrated DNA Technologies gBlocks, and cloned them into *pLV-GG-CRErep-Puro2AmScar3*.

6 NeMECiS sequencing processing

6.1 Alignment and barcode counts

A white-list of all 15,625 possible barcode concatemers was generated using a custom script. These were appended with left and right adaptor sequences to establish `amplicons.fa`, containing sequences each of length 163bp, labelled with the corresponding composition and ordering of CRE fragments. These have a generic form:

TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGAGCANNNNNNNGATTNNNNNNNAGCANNNNNNNGATT
 GATTAATAAAGGAAATTTATTTCAATAGTGTGTTGGAATTTGTGTCTCTCACTGTCTCTTATACACATCTCCGAGCCACGAGAC
 Key: **Nextera Rd1** Golden Gate overhang **8bp-BC sequence** **polyA terminator sequence** **Nextera Rd2**

Bowtie 2 was used for barcode calling. Index files were generated using the command:

```
bowtie2-build amplicons.fa amplicons
```

Global alignment of barcode concatemers for each *fastq* file from the NextSeq 500 demultiplexed output was achieved by running the following command, trimming alignment to exclude Nextera adaptors:

```
bowtie2 --threads 32 --trim-to 3:96 -x amplicons -1 fastq_file_read1.fastq -2 fastq_file_read2.fastq
```

From the alignments recorded in the *.sam* file output, a custom python script `process_sam_files.py` was used to generate a count-matrix for each barcode in each sample. Here, reads by barcode were counted that had a perfect match (CIGAR output of 96M), and successfully aligned to a member of the white-list.

6.2 Bayesian inference of fluorescence from barcode distributions

The output of NeMECiS is the frequencies of each of the 15,625 synCRE barcodes in the entire population (background distribution) and the frequencies of each barcode after flow-sorting. We converted these frequencies into estimated fluorescence intensities, with associated uncertainties in estimates, by a Bayesian inference approach. This allows for quantitative comparison and batch-normalisation across conditions, by placing all values on a common scale, and further aggregates the multiple measurements per barcode into a single summary statistic.

6.2.1 Outline of approach

The basic principle of the approach is that reporter fluorescence for each synCRE, and corresponding barcode i , will follow a distribution in intensity (y) as measured by flow cytometry ($P(y|i)$). We assume that this is a log-normal distribution for simplicity, or rather is normally distributed in log transformed intensities ($x = \log y; P(x|i) \sim \mathcal{N}(\mu_i, \sigma_i^2)$). Our task is to learn the parameters $\{\mu_i, \sigma_i^2\}$ given the measured data.

Measurements are performed in pool. This means that all fluorescence distributions measured during quality-control flow cytometry are a mixture of the barcode-specific distributions, weighted by their relative frequencies in the pool π_i . Therefore, the global distribution of (log) fluorescence in the pool (for a specific condition/replicate) is given by a Gaussian Mixture Model.

$$P(x) = \sum_i \pi_i \mathcal{N}(\mu_i, \sigma_i^2) \quad (1)$$

To deconvolve the barcode-specific components of this global distribution, we can compare the frequencies of barcodes in the unsorted pool to the frequencies in each of the sorted bins $b_j \in \{1, 2, 3, 4\}$. The flow sorter is assumed to transfer cells into bin b_j if the log ZsGreen fluorescence level lies between the limits of the sort gate $[b_j^L, b_j^U]$. This assumes that sorting is 'perfect' i.e. that the sort bin only contains cells with log fluorescence values within these limits. Thus the probability of sorting into gate b_j given a particular log fluorescence x is given by:

$$P(b_j|x) = \frac{1}{b_j^U - b_j^L} \text{ for } b_j^L \leq x \leq b_j^U \quad (2)$$

Flow-sorting is performed in a manner that is indiscriminate to synCREs (once conditioned on fluorescence) i.e. $P(b_j|x, i) = P(b_j|x)$. Consequently, we can calculate the probability of being sorted into a bin b_j for a particular synCRE i through marginalisation to obtain $P(b_j|i)$:

$$P(b_j|i) = \int_{-\infty}^{\infty} P(b_j|x)P(x|i)dx \quad (3)$$

$$= \phi\left(\frac{b_j^U - \mu_i}{\sigma_i}\right) - \phi\left(\frac{b_j^L - \mu_i}{\sigma_i}\right) \quad (4)$$

where $\phi(z)$ is the cumulative distribution function of the normal distribution with parameters $\mathcal{N}(0, 1)$. To then calculate the proportion of a synCRE i in a bin b_j , we apply Bayes's rule:

$$P(i|b_j) = \frac{P(i)P(b_j|i)}{P(b_j)} \quad (5)$$

Note that $P(b_j)$ represents the probability of cells displaying fluorescence values between the bin-limits of b_j across the entire unsorted pool. Consequently, $P(b_j)$ can be determined empirically from the flow-cytometry distribution of log ZsGreen fluorescences in the unsorted pool $P(x)$

$$P(b_j) = \int_{-\infty}^{\infty} P(b_j|x)P(x)dx \quad (6)$$

$$= \int_{b_j^L}^{b_j^U} P(x)dx \quad (7)$$

Therefore, defining π_i as the density of each synCRE in the unsorted pool, we have:

$$P(i|b_j) = \frac{\pi_i[\phi(\frac{b_j^U - \mu_i}{\sigma_i}) - \phi(\frac{b_j^L - \mu_i}{\sigma_i})]}{\int_{b_j^L}^{b_j^U} P(x)dx} \quad (8)$$

6.2.2 Finite number effects

The above describes the probability a given cell is sorted into a particular bin, given it contains a particular barcode. As we sorted a finite, while large, number of cells, we additionally consider and integrate the implications of these finite number effects on the confidence of our measurements.

There are two additional potential sources of noise: (a) the relative frequencies of cells containing each barcode found in cells actually sorted in each bin; (b) the relative frequencies of sequencing reads given the frequencies of the barcodes in each bin. Both, strictly speaking, follow multinomial distributions, containing the frequencies/probabilities for each of the 15,625 barcodes. In practice, such scenarios are well approximated by binomial distributions (given the covariance among any pair of different barcodes in this multinomial distribution will be vanishingly small).

$P(i|b_j)$ is the probability that a cell contains barcode i given it is allocated to bin b_j . If $N_{sort,j}$ cells are sorted into bin b_j , $N_{cell}(i|b_j)$ is the number of those cells that contain barcode i . Specifically, this follows: $N_{cell}(i|b_j) \sim \text{Binom}(N_{sort,j}, P(i|b_j))$. Correspondingly, the probability that a cell contains barcode i among a finite number of sorted cells is given by: $P(i|b_j, SORTED) = N_{cell}(i|b_j)/N_{sort,j}$. For computational tractability, this distribution is approximated by a normal distribution

$$P(i|b_j, SORTED) \sim \mathcal{N}\left(P(i|b_j), \frac{P(i|b_j)(1 - P(i|b_j))}{N_{sort,j}}\right) \quad (9)$$

From this, the absolute read counts C by barcode is:

$$C(i|b_j) \sim \text{Binom}(N_{read,j}, P(i|b_j, SORTED)) \quad (10)$$

For the unsorted pool, one can likewise write:

$$C_{pool}(i) \sim \text{Binom}(N_{read}^{pool}, \pi_i) \quad (11)$$

6.2.3 Calibrating bin gates

To achieve the scale of the flow-sort required for this screen, each sample was run on independent flow sorting machines. Consequently, we have to calibrate the bin gates among conditions to ensure that all estimated fluorescence values are on a common scale and without systematic errors. This was done using the small fraction of sorted/unsorted cells we kept out for quality control, run on a common flow cytometer.

Firstly, the raw fluorescence distributions of cells (gating on single, live cells) were log transformed and independently fit to Gaussian Mixture Models ($n_{\text{component}} = 3$ using the *sklearn* package), providing a continuous approximation of the empirical fluorescence probability distributions. Let us call these $P_{\text{emp}}(x|b_j)$ for each gate and $P_{\text{emp}}(x)$ for the global, unsorted pool. We then fit the following function to each $P_{\text{emp}}(x|b_j)$ to obtain the bin specifications:

$$P_{\text{fit}}(x|b_j) = P_{\text{emp}}(x) \cdot \frac{\alpha_j}{2} [\tanh(m_j(x - b_j^L)) + \tanh(m_j(b_j^U - x))] \quad (12)$$

which describes the convolution between the global fluorescence distribution of the pool and a step-like function that goes between 0 and α_j , smoothed by the parameter m_j , which accounts for experimental error in the measurement. Parameters were fit using *scipy.optimize*'s implementation of the Newton-CG minimisation method, using *JAXlib*'s autodifferentiation capacities to calculate Jacobians and Hessians numerically. We adopted the following cost function on which to perform this optimisation with respect to the free parameters α_j, b_j^L, b_j^U :

$$\begin{aligned} \mathcal{E} = & \sum_r \sum_j \left[\int_{-\infty}^{\infty} (P_{\text{fit}}(x|b_j, r) - P_{\text{emp}}(x|b_j, r))^2 \right] \\ & + \beta \sum_r \sum_j \left[(\tilde{b}_{j,r}^U - b_{j,r}^U)^2 + (\tilde{b}_{j,r}^L - b_{j,r}^L)^2 \right] \\ & + \gamma \sum_i \left[(\Delta_r \hat{\mu}_i)^2 \left(\sum_j C_{i,j,r} / 4 \right) \right] \end{aligned} \quad (13)$$

where r defines the replicate number $r \in \{1, 2\}$. Here, the latter two sets of terms are used to guide fit values towards the theoretical values. The first term, scaled by the hyperparameter β , compares the fit upper and lower bin limits $b_{j,r}^U, b_{j,r}^L$ to the values expected from the global distribution if they were placed at exactly their desired percentile positions (see experimental methods). For example, for $b_j = 2$, $\int_{\tilde{b}_{j,r}^L}^{\tilde{b}_{j,r}^U} P_{\text{emp}}(x|r) = 0.15$ and $\int_0^{\tilde{b}_{j,r}^L} P_{\text{emp}}(x|r) = 0.15 + 0.1333 = 0.2833$. The second term, scaled by the hyperparameter γ compares the approximate mean fluorescence levels of each of a set of barcodes $\mu_{i,r}$ across replicates, weighted by their frequency in the data. This can be viewed as guiding batch correction. Specifically:

$$\hat{\mu}_{i,r} = \frac{\sum_j G_{j,r} C_{i,j,r}}{\sum_j C_{i,j,r}} \quad (14)$$

where $C_{i,j,r}$ is the read-count matrix by barcode, gate and replicate, and $G_{j,r}$ is the mean empirical log fluorescence level for a given gate b_j and replicate r . The value $\Delta_r \hat{\mu}_i$ is then calculated as $\Delta_r \hat{\mu}_i = \hat{\mu}_{i,1} - \hat{\mu}_{i,2}$. Together this defined the the inferred bin-positions in subsequent calculations (Fig. S2F shaded regions).

6.2.4 Implementation of Bayesian inference

Provided with the read-count matrix $C_{i,j,r,s}$ across barcodes, bins, replicates, and SAG concentrations, we used *pySTAN* [Stan Development Team, 2024] to infer $\{\mu_{i,r,s}, \sigma_{i,r,s}, \pi_{i,r,s}\}$. This is done in a per-barcode/replicate/SAG concentration fashion, given the binomial approximation to the multinomial problem described above, allowing for parallel implementation.

We specify the following priors for these fit parameters:

$$\mu_{i,r,s} \sim \mathcal{N}(6, 1) \quad (15)$$

$$\sigma_{i,r,s} \sim \psi inv\text{-}\chi^2(1, 0.1) \quad (16)$$

$$\pi_{i,r,s} \sim \text{Beta}(1, 1) \quad (17)$$

(where scaled inverse χ^2 distribution is denoted $\psi inv\text{-}\chi^2$)

The *Stan* script used in the fitting, takes the following measurements as input:

- Read-counts by gate for a particular barcode, replicate and SAG concentration ($C_{i,j,r,s}$)
- Number of cells sorted by gate ($N_{sort,j,r,s}$)
- Number of reads by gate ($N_{read,j,r,s}$)
- Number of reads for the unsorted, pooled population ($N_{read,r,s}^{pool}$)
- Upper and lower bounds of each gate, as inferred from the above
- Normalisation constant in the calculation of $P(i|b_j)$, i.e. $P(b_{j,r,s}) = \int_{b_{j,r,s}^L}^{b_{j,r,s}^U} P_{emp}(x|r, s)$. This is calculated empirically once the gate boundaries are inferred using the above.

Using the approach described above, we model using binomial functions the number of reads expected in the pool and in each of the bins, and use the No-U-Turn-Sampler (NUTS) sampler to generate posterior distributions and maximum-likelihood estimates with the following specifications ($n_chains=8$, $iter_warmup=500$, $iter_sampling=500$). We additionally specify the initial condition around which to sample as: $\mu_i \sim \mathcal{N}(\hat{\mu}, 0.5)$; $\sigma_i \sim [0.8, 1.5]$; $\pi_i = C_{i,r,s}^{pool}(i) / \sum_j C_{j,r,s}^{pool}$. From this, we gain estimates of the expected value of each parameter ($\mathbb{E}(\theta)$), and its associated sample standard deviation ($s(\theta)$).

7 Linear modelling framework for learning modular contributions and interactions

7.1 Overview

The overarching strategy of our linear modelling framework is to describe synCRE structure as a set of categorical features, and fit a joint model for CRE activity with and without SAG.

Let the vector $\mathbf{y}$ denote the expected activities of each synCRE $\times$ SAG combination. Let $\mathbf{X}$ be a $N_{observation} \times N_{feature}$ count-matrix of features. Here, $N_{feature} = 2n_{feature}$ where, depending on if the observation was without or with SAG, either the first (approximately) half or second half (respectively) of the entries in a given column are set to zero. From this, we find the optimal parameters of the following ridge-regression loss function:

$$\mathcal{E} = \sum_i^{N_{observation}} [\hat{w}_i(y_i - \beta^T \mathbf{X}_i + c)^2] + \lambda \|\beta\|^2 \quad (18)$$

Here, β is a $(2n_{feature})$ coefficient vector of regressed values, with the first and second halves corresponding to the coefficients without and with SAG respectively, and c is the intercept. We additionally include a ridge regularisation factor λ which performs two functions: (a) to guard against over-fitting; (b) to encourage a parsimonious fit (i.e. involving the fewest number of features) in cases of parametric degeneracy. We additionally include a weighting parameter $\hat{w}_i$ which weights the fit towards observations with lower uncertainty. $\mathbf{y}$ and $\hat{\mathbf{w}}$ are taken directly from the Bayesian inference of log fluorescence levels, as described above, where $y_i = \mathbb{E}(\mu_i)$ and $w_i = 1/s(\mu_i)^2$; $\hat{w}_i = (N_{observation} w_i) / \sum_{j=1}^{N_{observation}} w_j$.

7.2 Remark about scale of data

We choose to fit the ridge regression on log-transformed data. As discussed in 12, motivated by thermodynamic models of CRE activity, we anticipate that contributions from each feature (modular contributions and interactions) should multiply to give the overall activity of a synCRE. Consequently, if fluorescence $f = \prod_j f_j$ over f_j feature-decomposed contributions, then $\log(f) = \sum_j \log(f_j)$, hence allowing for a functional form amenable to linear regression modelling.

7.3 Data filtering

For all models, the following filtering steps were performed on the data. Firstly, we excluded fragment 18 in position 2 as inspection of fragment sequences revealed mutations, and fragment 17 in position 3, due to significantly fewer reads observed for this fragment-position pair. Secondly, we removed all data-points where $s(\mu_i) > 0.1$, deemed to be too large an uncertainty for further analysis. This left 28520 and 28468 data-points for replicate 1 and 2 respectively (out of a total of $25^3 \cdot 2 = 31250$).

7.4 Feature extraction

To extract the features that comprise $\mathbf{X}$ in all of the models from the synCRE fragment composition, the following operations are performed. Let z_i be a triple of fragment indices ($z_i = (z_i^1, z_i^2, z_i^3)$; $z \in \{1, 2, \dots, 25\}$) in a 5' to 3' direction from furthest to closest to the minimal promoter. For example, for a synCRE 12 – 14 – 16, $z_i^1 = 12, z_i^2 = 14, z_i^3 = 16$. Let $\mathbf{Z}_i$ be the one-hot encoding of z_i ($\mathbf{Z}_i = (e_i^1, e_i^2, e_i^3)$), where e_i^j for a given synCRE i and position j is given by: $e_i^j[z_i^j] = 1$ and is 0 otherwise.

The fragment count matrix $\mathbf{X}_F^S$ for a given SAG is given by:

$$\mathbf{X}_F^S|_i = \left[\sum_{j=1}^3 e_i^j \right] \quad (19)$$

Note that

$$\mathbf{X}_F^S|_i \cdot \mathbf{1} \equiv 3 \quad (20)$$

The fragment pair count matrix $\mathbf{X}_P^S$ for a given pair of fragments $\{u, v\}$ for a given synCRE i is given by

$$\mathbf{X}_P^S|_{i,u,v} = e_i^1[u]e_i^2[v] + e_i^2[u]e_i^3[v] + e_i^3[u]e_i^1[v] \quad (21)$$

Note that, summing over pairs in $\mathbf{X}_P^S$ for a given synCRE:

$$\sum_u \sum_{v \geq u} [\mathbf{X}_P^S|_{i,u,v}] \equiv 3 \quad (22)$$

The fragment spaced pair count matrix $\mathbf{X}_{SP}^S$ for a given pair of fragments $\{u, v\}$ offset by a distance d (e.g. $1 - x - 2 \implies (u = 1, v = 2, d = 2)$ or $(v = 1, u = 2, d = -2)$ where x is a wildcard), is given by the following set of expressions:

$$\mathbf{X}_{SP}^S|_{i,u,v,d} = \begin{cases} e_i^3[u]e_i^1[v]; & \text{if } d = -2 \\ e_i^3[u]e_i^2[v] + e_i^2[u]e_i^1[v]; & \text{if } d = -1 \\ e_i^2[u]e_i^3[v] + e_i^1[u]e_i^2[v]; & \text{if } d = 1 \\ e_i^1[u]e_i^3[v]; & \text{if } d = 2 \end{cases} \quad (23)$$

Note that the sum of $\mathbf{X}_{SP}^S|_{i,u,v,d}$ over non-degenerate spaced pairs of fragments is always 3.

$\mathbf{X}_P^S$ and $\mathbf{X}_{SP}^S$ are constructed such that descriptions of (spaced) pairs is not degenerate. Thus there are 325 columns in $\mathbf{X}_P^S$ and 1250 columns in $\mathbf{X}_{SP}^S$. The overall count matrices are then constructed, for a given feature class c , as the concatenation $\mathbf{X}_c = \{\mathbf{X}_c^-, \mathbf{X}_c^+\}$, where the rows are all data-points for a given replicate (across SAG levels).

7.5 Fitting procedure, cross-validation and hyperparameter tuning

Models were fit using the *sklearn.linear_model.Ridge* package, specifying w_i with the *sample_weight* optional field. Models were fit separately (including independent hyperparameter tuning) on each of the two replicates.

To identify the optimal value of the regularisation parameter λ , we performed hyperparameter scans over the following range $\lambda = 10^q$ (q is evenly sampled between $-2, +2$ inclusive in 32 regularly spaced steps). Specifically, we identify a 5-fold split of the data that preserves feature abundance. To do this, we randomly partition $N_{\text{observation}}$ into 5 bins, holding out each of the 5 bins b individually in turn ($\mathbf{X}_{-b}$), and calculated $\sum_b (\langle \mathbf{X}_{-b} \rangle_j - \langle \mathbf{X} \rangle_j)^2$. Iterating 100 times, we take the minimum of this cost function to find an optimal split.

We then trained the model on each of the 4/5 subsets, for each specified value of λ , and calculate the explained variance in each of the withheld bins, and take a mean. This produces a graph with a maximum, identifying an optimal value of λ (λ^*) the balances over and underfitting.

Next, we used λ^* to fit the full model, using all $N_{\text{observation}}$. For sensitivity analysis, we fit the model on a bootstrapped resampling of the data ($N_{\text{boot}} = 50$).

7.6 Model 1: Fragment contributions alone

We start with a model considering fragment contributions alone. Hence here $N_{\text{feature}} = 2 \cdot 25 = 50$, corresponding to each of the 25 fragments, with and without SAG. When fit, the values of β approximately correspond to the contributions of each fragment with respect to the population average, thus defining modular effects beyond this average.

If we define $\beta = \{\beta^-, \beta^+\}$ as the decomposition of feature coefficients without and with SAG respectively, we can determine the SAG-dependent and SAG-independent effects as $\Delta\beta = \beta^+ - \beta^- = \text{'log FC } \Delta\text{SAG'}$, and $\langle \beta \rangle = \frac{1}{2}(\beta^+ + \beta^-)$, respectively.

7.6.1 Fragment class allocation

Inspection of the distribution of fragment coefficients across $\{\langle \beta \rangle, \Delta\beta\}$ prompted us to score fragments into three categories based on their response to SAG: increased, repressed, neutral. To do this, we used the bootstrapped distributions of the coefficients and calculated $\bar{\beta}_{\text{boot}}$ and $s(\beta_{\text{boot}})$. We performed a two tailed Z-test, with $z_{\text{crit}} = 4.265$ to assign neutrality, then for fragments significantly different from neutrality, we define increasing fragments as those for which $\bar{\beta}_{\text{boot}} > 0$. We performed the test across replicates by using $\bar{\beta}_{\text{boot}} = \frac{1}{2}(\bar{\beta}_{\text{boot}}^{r=1} + \bar{\beta}_{\text{boot}}^{r=2})$, and $s(\beta_{\text{boot}}) = \sqrt{s(\beta_{\text{boot}}^{r=1})^2 + s(\beta_{\text{boot}}^{r=2})^2}$. We also use the Z-test to calculate logP-values of fragment contribution.

7.7 Model 2: Fragment contributions + interactions + spacing effects

7.7.1 Scoring interactions

Upon fitting a model with interactions and spacing effects, we first analysed the role of interactions. To do this, we define an interaction parameter $\alpha_{i,j} = \beta_P|_{\{i,j\}} + \sum_{d \in \{-2,2\}} \beta_{SP}|_{\{i,j,d\}} + 2 \sum_{d \in \{-1,1\}} \beta_{SP}|_{\{i,j,d\}}$, summing over both position and spacing dependent effects, and weighting the latter with respect to the frequency it appears in the data. From this, bootstrapped means and sample standard deviations could be calculated, and Z-tests could be performed as above.

For the comparison between the multiplicative model (incorporating fragment effects alone) and the interaction model, we calculated the overall SAG-dependent activity of a pair $\Delta y_{i,j} = \Delta\beta_i + \Delta\beta_j + \Delta\alpha_{i,j}$ and compared it to $\Delta y_{i,j} = \Delta\beta_i + \Delta\beta_j$. When $\alpha_{i,j} \neq 0$, points on the graph that plot these models against each other deviate from the diagonal.

7.7.2 Positionality score

We devised a positionality score to rank the importance of spacing on the interactions between fragments. Define $M_{u,v,d,r,q}$ as an array of shape $(25 \times 25 \times 6 \times 2 \times N_{\text{boot}})$, holding the values of $\Delta\beta_{SP}$ for a given pair of fragments $\{u, v\}$, at a given spacing $d \in \{-2, -1, -1, 1, 1, 2\}$, for a given replicate $r \in \{1, 2\}$, and a given bootstrap sample $q \in \{1, 2, \dots, N_{\text{boot}}\}$. Note here that $d = \pm 1$ is duplicated to reflect the doubled

frequencies of these features in the data. Let $s_q(M)$ be the standard deviation of the array M along the q dimension, and likewise for s_d . We can therefore define $\rho_{u,v}$, the positionality score for the pair $\{u, v\}$, as:

$$\rho_{u,v} = \frac{6 \sum_r \sum_q s_d(M_{u,v,d,r,q})^2}{N_{boot} \sum_r \sum_d s_q(M_{u,v,d,r,q})^2} \quad (24)$$

This is a measurement of the ratio of the variance attributed to spacing compared with the variance attributed to uncertainty in inference (via bootstrapping). When $\rho_{u,v} \gg 1$, spacing-dependent effects are strong. We additionally perform significance testing using the Kruskal-Wallis H-test, over each of the four possible spacings $d \in \{-2, -1, 1, 2\}$.

7.7.3 Identifying syntax modes with k-means clustering

Having identified cases of strong spacing-dependent effects, we wanted to map the ‘modes’ of these spacing rules onto a continuum. Specifically, our fitted model generates a discrete function $\alpha_{u,v}(d)$ as a function of spacing d (or for simplicity the vector $\alpha_{u,v}$ of size 4).

To perform spatial clustering, we constructed a feature vector from the above. Specifically, we define:

$$\mathbf{Q}_{u,v} = \left\{ \sum_{d \in D} \alpha_{u,v} : D \subseteq \{-408, -204, 204, 408\}, 1 \leq |I| \leq 4 \right\} \quad (25)$$

Filtering by fragment-pairs where the positionality score exceeds 2, we perform k-means clustering on $\mathbf{Q}_{u,v}$ on all unique pairs of fragments, with $n_{cluster} = 3$, using *sklearn.cluster.KMeans*. Given pairs can be considered in either order, we choose u and v such that $\alpha_{u,v}(d)$ is on average increasing in d .

7.7.4 Orthogonal basis decomposition of syntax contributions

For a continuum description, we constructed a set of four orthonormal bases to describe $\alpha_{u,v}(d)$.

$$\hat{\mathbf{v}}_0 = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) \quad (26)$$

$$\hat{\mathbf{v}}_1 = \left(0, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \right) \quad (27)$$

$$\hat{\mathbf{v}}_2 = \left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right) \quad (28)$$

$$\hat{\mathbf{v}}_3 = \left(\frac{\sqrt{2}}{2}, 0, 0, -\frac{\sqrt{2}}{2} \right) \quad (29)$$

from which one can note immediately that $\hat{\mathbf{v}}_0$ specifies the average interaction over spacing, and the other three vectors describe the grammatical modes. The coefficients of this transformed vector space ($m_{u,v,k} = \hat{\mathbf{v}}_k \cdot \alpha_{u,v}$) thus define the contributions of each of these basis vectors to the overall shape of $\alpha_{u,v}(d)$. Noting that $Var_d(\alpha_{u,v}(d)) = \sum_{k=1}^3 m_{u,v,k}^2$, one can therefore define the percentage of variance owing to each of the remaining three components $\omega_{u,v,k} = m_{u,v,k}^2 / \sum_{l=1}^3 m_{u,v,l}^2$. This therefore allowed us to plot the spacing-dependent rules of each pair of fragments on a ternary plot, describing the relative contribution of spacing-dependent variance with respect to each of the three remaining modes ($\hat{\mathbf{v}}_1, \hat{\mathbf{v}}_2, \hat{\mathbf{v}}_3$). To perform the calculations, we let $\alpha_{u,v}(d) = \frac{1}{2}(\beta_{SP}|_{u,v,d}^{r=1} + \beta_{SP}|_{u,v,d}^{r=2})$.

7.7.5 Inferring fragment order-dependent effects from spacing rules

Given the above analysis, we sought to determine whether the spacing rules inferred from NeMECiS are sufficient to predict changes in SAG-dependent synCRE activity upon permuting the order of its comprising fragments. In this model, permutation of fragments leaves the contributions owing to individual fragments (β_F) and owing to spacing-independent synergies (β_P) unchanged, meaning β_{SP} can be solely used to infer order-dependent effects. Put another way, this experimental test is determining whether pairwise contributions are sufficient to explain the activity of a synCRE as a whole.

For a given triple of fragment indices $\{u, v, w\}$, for which exhaustive permutation sets (size 6) were generated and tested experimentally ($\mathcal{P}_{\{u,v,w\}} \mapsto \{(u, v, w), (u, w, v), \dots\}$) we calculated the normalised activity of each ordering as:

$$\Delta y_{(u,v,w)} = \alpha_{u,v}(d=1) + \alpha_{v,w}(d=1) + \alpha_{u,w}(d=2) \quad (30)$$

$$\Delta \hat{y}_{(u,v,w)} = \Delta y_{(u,v,w)} - \langle \Delta y_{(p,q,r)} \rangle_{(p,q,r) \in \mathcal{P}_{\{u,v,w\}}} \quad (31)$$

This was performed on coefficients from each replicate individually (defining the lower and upper error-bars), as well as their average.

7.7.6 FL-synCRE activity predictions

Extending the above, we next asked whether our model could extrapolate to larger permutation sets, testing if the spacing-dependent rules learned from NeMECiS could generalise to arbitrarily lengthed sequences. To do this, we took $7 \times$ contiguous fragments comprising the Olig2 CRE (fragments 8, 9, ..., 14) and calculated the SAG-dependent activities upon exhaustive permutation ($N_{\text{Design Space}} = 7! = 5040$). Defining $\mathbf{f} = \{f_1, f_2, \dots, f_7\}$ as a particular ordering of fragments 8–14, we could calculate the normalised activities upon shuffling (including the natural ordering where $\mathbf{f} = (8, 9, \dots, 14)$) as:

$$\Delta y_{\mathbf{f}} = \sum_{i=1}^6 \alpha_{f_i, f_{i+1}}(d=1) + \sum_{i=1}^5 \alpha_{f_i, f_{i+2}}(d=2) \quad (32)$$

This was performed on the average of coefficients across replicates, as well as each replicate in turn to determine reproducibility.

7.7.7 Modelling of individualised synCRE flow cytometry analysis

For the individualised tests of the multiplicative model, considering the exhaustive construction of triplets of fragments taken with replacement from a pair of fragments $\{i, j\}$, we fit the function $\Delta y_{N_i, N_j} = \exp(\alpha + N_i \beta_i + N_j \beta_j)$ (where $N_i + N_j = 3$) using `scipy.optimize.curve_fit`. For analyses comparing the number of inducing fragments against $\log \Delta y_{N_i, N_j}$, we use the linear regression function `scipy.stats.linregress`.

For the individualised tests for interactions between fragments 2 and 16, we fit an expanded integrative model on all triplets containing $\{2, 8\}$ and $\{16, 8\}$ (where 8 is a neutral fragment and 2 and 16 are inducing fragments): $\Delta y_{N_2, N_8, N_{16}} = \exp(\alpha + N_2 \beta_2 + N_8 \beta_8 + N_{16} \beta_{16})$ (where $N_2 + N_8 + N_{16} = 3$). To determine sensitivity to experimental error, we performed a bootstrap analysis, re-fitting the above model to the data sampled evenly by synCRE structure, but sampled with replacement for 500SAG-0SAG comparisons ($N_{boot} = 1000$), plotting the interval $\Delta y_{pred} \pm \sigma_{\Delta y_{pred}}$.

8 Comparison of deep learning approaches in predicting synCRE activity

8.1 Overview

We evaluated the predictive performance of four convolutional neural network (CNN) based deep learning models – ExplainNN [Novakovsky et al., 2023], DeepCNN [Kelley et al., 2016], DanQ [Quang and Xie, 2016], and DeepSTARR [de Almeida et al., 2022] – for predicting the activity of cis-regulatory elements (CREs) with and without SAG treatment. These models have been widely used in genomic applications, including transcription factor binding prediction, chromatin accessibility inference, and enhancer activity characterization [Novakovsky et al., 2023, Kelley et al., 2016, Quang and Xie, 2016, de Almeida et al., 2022].

8.2 Model Architecture

Each model received as input a one-hot-encoded representation of synCRE sequences and was trained to predict the corresponding CRE activity levels with and without SAG treatment. The architectures were adapted from the original publications [Novakovsky et al., 2023] with modifications tailored to the task as follows.

8.2.1 ExplaiNN

ExplaiNN was implemented as described in [Novakovsky et al., 2023] and consisted of units with a single convolutional layer followed by fully connected layers. The architecture included:

- a convolutional layer with one 19×4 filter, batch normalization, an exponential activation function, and max pooling (kernel size = 7, stride = 7);
- a fully connected layer with 100 nodes, batch normalization, ReLU activation, and 30% dropout;
- a second fully connected layer with a single node, batch normalization, and ReLU activation;
- a final linear output layer predicting CRE activity with and without SAG treatment.

8.2.2 ConvNetDeep (DeepCNN)

ConvNetDeep, referred to as DeepCNN in [Novakovsky et al., 2023] and originally presented in [Kelley et al., 2016], was adapted by modifying filter sizes and the final output layer. In detail, this implementation included

- three convolutional layers with 100 filters each, using kernel sizes of 19×4 , 11×1 , and 7×1 , respectively, each followed by batch normalization, ReLU activation, and max pooling (kernel size = 3, stride = 3);
- two fully connected layers with 1,000 nodes, batch normalization, ReLU activation, and 30% dropout;
- a final fully connected layer generating two outputs.

8.2.3 DanQ

DanQ [Quang and Xie, 2016] integrates convolutional and recurrent layers to capture dependencies. The original DanQ architecture was altered to incorporate a fully connected output layer producing two outputs [Novakovsky et al., 2023]. The architecture included:

- a first convolutional layer with 320 filters (26×4), ReLU activation, 20% dropout, and max pooling (kernel size = 13, stride = 13);
- two bidirectional long short-term memory (biLSTM) layers with a hidden state size of 320 and 50% dropout
- a fully connected layer with 925 nodes and ReLU activation;
- a final fully connected output layer generating two outputs.

8.2.4 DeepSTARR

DeepSTARR [de Almeida et al., 2022, Novakovsky et al., 2023], was implemented as follows:

- four convolutional layers with 256, 60, 60, and 120 filters, respectively, each using batch normalization, ReLU activation, and max pooling;
- two fully connected layers with 256 nodes each, batch normalization, ReLU activation, and 40% dropout;
- a final fully connected layer generating two outputs.

All models were implemented in PyTorch [Paszke et al., 2019].

8.3 Data processing and model training

8.3.1 Data processing

We first filtered the data following the criteria outlined in Section 3.3. Raw fluorescence measurements for CRE sequences were then preprocessed by averaging replicate values to obtain a single activity measurement per sequence. This resulted in 14,301 data-points of synCRE sequences paired with their corresponding CRE activity values with and without SAG. Data were split into training, validation, and test sets in a 7:1.5:1.5 ratio.

Each dataset was loaded using the PyTorch DataLoader function. Training data were shuffled during each epoch to enhance generalization, whereas validation and test sets remained fixed to ensure consistency.

8.3.2 Hyperparameter optimization

Hyperparameters were optimized independently for each model.

- **ExplaiNN.** The number of convolutional units (num_cnns), batch size (bs), and learning rate (lr) were explored over the following ranges: num_cnns $\in \{10, 20, 30, \dots, 250\}$, bs $\in \{96, 168, 322\}$ and lr $\in \{5 \times 10^{-5}, 1 \times 10^{-4}, 5 \times 10^{-4}\}$
- **Others.** For ConvNetDeep, DanQ, and DeepSTARR we optimized the batch size (bs) and learning rate (lr) over: bs $\in \{96, 168, 322\}$ and lr $\in \{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$.

Each model was trained five times per hyperparameter setting with different random seeds.

8.3.3 Model Training

All models were trained for up to 200 epochs with early stopping to prevent overfitting. An early stop was triggered if the validation loss (loss calculated for the validation set) did not improve for 15 consecutive epochs.

The validation loss was computed using mean squared error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the true label, $\hat{y}_i$ is the predicted label ($i \in \{1, 2\}$ in this case), and n is the number of data points in the validation set. The total loss was the sum of MSE values for CRE activity with and without SAG:

$$\text{Total Loss} = \text{MSE}_{\text{with_SAG}} + \text{MSE}_{\text{without_SAG}}$$

8.3.4 Model evaluation

Performance was assessed using Pearson correlation between predicted and true activity values on the test set:

$$\text{pearson}_{\text{corr}} = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}}$$

where y_i and $\hat{y}_i$ still denote the true and predicted labels, $\bar{y}$ is the mean of y_i , and $\bar{\hat{y}}$ is the mean of $\hat{y}_i$. The epoch with the highest Pearson correlation was selected as the best-performing model.

9 Motif analysis

9.1 Motif calling

Pulling all vertebrate non-redundant .pfm motif files from <https://jaspar.elixir.no/downloads/> [Rauluseviciute et al., 2024], we ran MOODS ([Korhonen et al., 2009], <https://github.com/jhkorhonen/MOODS>) to enrich for motifs in each of the twenty-five 200bp fragment sequences using the following command:

```
python MOODS/python/scripts/moods-dna.py -m reference/JASPAR2024_CORE_PROCESSED/*.pfm  
-s fasta/fragment_i.fa -p 0.0001 > alignment/fragment_i.csv
```

This returns .csv files with the following columns: [Fragment, Motif, Start Position, Orientation, Match Score, Matched Sequence].

9.2 Clustering of motifs into archetypes

Given substantial sequence similarity among sets of motif sequences, we followed [Vierstra et al., 2020] by clustering motifs based on sequence similarity into ‘archetypes’. Specifically, we pulled pre-computed pairwise similarity scores from JASPAR2024 and performed agglomerative clustering (`sklearn.cluster.AgglomerativeClustering`, `n_clusters=150`). This provided a grouping of motifs by similarity, which were used for subsequent feature-enrichment analysis.

9.3 Motif enrichment analysis

We next scored whether particular motif clusters showed a correlation with SAG-dependent activity. Specifically, we took the MOODS-enriched motifs from each fragment, and subsetting by `Match Score > 6.5`. We then built a series of sparse, boolean matrices: $M_{motif,arch}$, a motif to archetype association built on the above clustering; and $M_{frag,motif}$, a fragment to motif matrix; and $M_{synCRE,frag}$, a boolean matrix determining the presence of a synCRE. From this, the below matrix product describes the archetype-count matrix (where $B(M)$ describes a boolean transformation).

$$B(M_{synCRE,frag} M_{frag,motif} M_{motif,arch}) = M_{synCRE,arch} \quad (33)$$

We then calculate the Spearman’s correlation coefficient across synCREs between the presence of a given archetype and Δy (change in predicted fluorescence with respect to SAG) for each replicate separately, filtering synCRE data-points by quality as in our linear modelling framework. We subsequently filtered for archetypes for which at least one of the corresponding genes is expressed (TPM > 20 for any day 4, 5, 6 sample from [Delas et al., 2023]). Our strongest enriched archetype was cluster 90, containing the motifs [Gli1, Gli2, Zbtb7b, Zbtb7c].

10 Dynamical modelling of neural tube patterning

We use the previously established and parameterised model of neural progenitor patterning [Cohen et al., 2014, Pezzotta and Briscoe, 2023, Exelby et al., 2021] to explore the effects of changes in expression amplitude in response to a gradient of Shh. This describes the gene-regulatory network underpinning Shh morphogen decoding during neural progenitor patterning as a set of ordinary differential equations for the level of four transcription factors $S = (P, O, N, I)$, corresponding to Pax6, Olig2, Nkx22, and Irx3 respectively. The Shh gradient is integrated into the model by graded activation of a fixed amount of Gli (G) protein ($G = A + R = 1$), which was either in an activated (A) or repressive (R) form.

$$\frac{dS}{dt} = f_S = (f_P, f_O, f_N, f_I) \quad (34)$$

$$H^+(y) = \frac{y}{1+y} \quad (35)$$

$$A = e^{-0.15x} \quad (36)$$

$$R = 1 - A \quad (37)$$

$$f_P = \alpha_{Pax} H^+ \left(\frac{K_{Pax,Pol} c_{Pol}}{(1 + K_{Pax,Oli} O)^2 (1 + K_{Pax,Nkx} N)^2} \right) - \beta_{Pax} P \quad (38)$$

$$f_O = \alpha_{Oli} H^+ \left(\left[\frac{K_{Oli,Pol} c_{Pol}}{(1 + K_{Oli,Nkx} N)^2 (1 + K_{Oli,Irx} I)^2} \right] \left[\frac{1 + f_A K_{Oli,Gli} A}{1 + K_{Oli,Gli} (A + R)} \right] \right) - \beta_{Oli} O \quad (39)$$

$$f_N = \alpha_{Nkx} H^+ \left(\left[\frac{K_{Nkx,Pol} c_{Pol}}{(1 + K_{Nkx,Pax} P)^2 (1 + K_{Nkx,Oli} O)^2 (1 + K_{Nkx,Irx} I)^2} \right] \left[\frac{1 + f_A K_{Nkx,Gli} A}{1 + K_{Nkx,Gli} (A + R)} \right] \right) - \beta_{Nkx} N \quad (40)$$

$$f_I = \alpha_{Irx} H^+ \left(\frac{K_{Irx,Pol} c_{Pol}}{(1 + K_{Irx,Oli} O)^2 (1 + K_{Irx,Nkx} N)^2} \right) - \beta_{Irx} I \quad (41)$$

In all cases, the model was simulated using *scipy.integrate.solve_ivp* subject to an initial condition of $S = (0.1, 0, 0, 0.1)$.

Varying the parameter α_{Olig2} for a fixed range ($x \times 10^q$, where q is linearly evenly spaced between X and Y), we measured the predicted distribution of O (Olig2) as a function of position, identifying shifts in the patterning.

10.1 Fitting FL-synCREs

To predict the changes in Olig2 spatiotemporal expression upon GRN rewiring, we calibrated the model. In vitro differentiation under SAG of neural progenitors yields multiple cell fates that shift in proportions as the concentration of SAG changes. Thus to calibrate the in vivo model to the in vitro differentiation, we used fate proportions of p3, pMN and p2+ cells at day 6 500nM SAG determined in [Delas et al., 2023] (p3=20%; pMN=45%; p2+ = 35%). By binarising cell fate in the in silico model as a function of dorsoventral position y , we specify $pMN^+(y) = O(y) > \max_y(O(y))/2$, $p3^+(y) = N(y) > \max_y(N(y))/2$ and $p2^+(y) = \neg(pMN^+(y) + p3^+(y))$. We then suggest that 500nM SAG in vitro differentiations maps to a Gaussian kernel of spatial coordinates in the in vivo model:

$$K(y) = \exp\left(\frac{-(y - y_0)^2}{\sigma_y^2}\right) \quad (42)$$

From this, we can determine the values of y_0 and σ_y for which the following cost-function is minimised using a grid-search.

$$C = \left(\frac{\int_0^1 pMN^+(y) K(y) dy}{\int_0^1 p2^+(y) K(y) dy} - \frac{P(pMN)}{P(p2+)} \right)^2 + \left(\frac{\int_0^1 p3^+(y) K(y) dy}{\int_0^1 p2^+(y) K(y) dy} - \frac{P(p3)}{P(p2+)} \right)^2 \quad (43)$$

Then using the measured ZsGreen fluorescence values, we calculated the fold changes with respect to the WT Olig2 CRE for the boosted and diminished FL-synCREs. We then calculated the kernel-averaged value of Olig2 expression for 500nM SAG as a function of α_{Oli} , normalised to the WT value:

$$\tilde{O}(\alpha_{Oli}) = \int_0^1 O(y|\alpha_{Oli}) K(y) dy \quad (44)$$

$$\hat{o}(\alpha_{Oli}) = \frac{\tilde{O}(\alpha_{Oli})}{\tilde{O}(\alpha_{Oli} = \text{WT value})} \quad (45)$$

From this, we can compute α_{Oli} for the FL-synCREs as the value that satisfies:

$$\hat{o}(\alpha_{Oli}^{FL-synCRE}) = \frac{ZsGreen_{FL-synCRE}}{ZsGreen_{WT}} \quad (46)$$

To determine uncertainty, error was propagated for $ZsGreen_{FL-synCRE}$, $ZsGreen_{WT}$.

11 Image analysis

11.1 Pre-processing and segmentation

We exported a $4 \times x, y$ downsampled version of our 3D images inherent to the pyramidal .vsi files from Olympus Spinning Disk microscopy to .ome.tiff using the Bio-Formats [Linkert et al., 2010] converter in Fiji-ImageJ [Schindelin et al., 2012], ensuring that physical spatial dimensions are preserved.

We then repurposed the cell-volume segmentation software *giani* [Barry et al., 2022] to segment entire organoids from the 3D imaging volumes, using the following command:

```
srunch java -Xmx750G -jar path-to-giani.jar path-to-joblist.txt path-to-properties.xml
index
```

This produces a directory of Nucleus_masks, with .tiff files for each z-slice (where here nuclei imply organoids). We use these masks to compile a series of .tiff files for each uniquely identified organoid, taking the bounding-box of a given organoid index, and concatenating into it the original microscopy data. This now allows for parallelisation of subsequent analysis for each organoid.

From here, we perform a set of operations to clean the mask by organoid:

- Perform `scipy.ndimage.binary_erosion` on the mask with `iterations=2`
- Rescale the image and mask such that each voxel has dimension ($1\mu m \times 1\mu m \times 1\mu m$) using `scipy.ndimage.zoom` with `order=0`.

11.2 Organoid geometry metrics

From our cleaned, isometric organoid mask (M), we calculated a series of geometric metrics:

- Volume ($V_{mask} = \sum_{\{x,y,z\}} M_{i,j,k}$) using `mask_iso.sum()` [units = μm^3]
- Convexity ($\mathcal{C} = V_{hull}/V_{mask} - 1$), where we calculate the convex hull of the mask, and compare its volume to the mask volume. The ratio V_{hull}/V_{mask} will tend to 1 when the mask is perfectly convex (i.e. all boundary pixels lie on their convex hull).

11.3 FoxA2+ floor-plate scoring

We next scored each organoid for their floor-plate composition. As an overview, we generated a mask for where the FoxA2 channel exceeds a critical intensity, and sequentially processed it by merging clusters and filling holes, such that we can group sets of FoxA2 voxels as being members of distinct floor-plates.

Specifically, we performed the following operations on the isometrised FoxA2 channel:

- Define M^{FA2} , the mask of FoxA2+ voxels within its isometrised image, as $M_{i,j}^{FA2} = \{1 \text{ if } F_{i,j} > F_{thresh}; \text{ else } 0\}$, where $F_{thresh} = 1000$.
- Perform the operation `process_3d_mask`. This identifies the connected components with `scipy.ndimage.label`, and from the corresponding labelled mask, performs `scipy.ndimage.remove_small_objects` with `min_size=200`.
- Smooth the mask by performing `scipy.ndimage.gaussian_filter`, then binarise with the threshold 0.1.
- Perform `process_3d_mask` a second time, this time with `min_size=3000`.
- We update our FoxA2+ voxel classification (removing the spurious positive pixels removed in the above smoothing) by taking the intersection between $M \otimes M^{FA2} \otimes M_{processed}^{FA2} = M_{final}^{FA2}$

- Use `scipy.ndimage.label` to label the product of the above. This label mask is used to classify voxels in M_{final}^{FA2} , and the corresponding number of connected components is taken as the number of floor-plates.

11.4 Organoid quality control filtering

We perform analysis only on organoids with regular and equivalent geometries, and with precisely one floor-plate, for valid subsequent comparisons. Specifically, we require:

- Number of floor plates = 1
- Volume constraints: $10^6/3 < V_{mask} < 3 \times 10^3$
- Convexity constraints: $\mathcal{C} < 0.5$

11.5 Olig2 spatial expression analysis

We then determine the euclidian distance from each voxel within the processed isometrised organoid mask to its nearest FoxA2+ voxel in M_{final}^{FA2} using `scipy.ndimage.distance_transform_edt` on the inverse of M_{final}^{FA2} . We then bin these distances to a linear range of bins ($\{0, 5, 10, \dots\} \mu m$), converting exact distances to bin mid-points (i.e. $\{2.5, 7.5, \dots\} \mu m$), excluding voxels that lie outside of M , the processed isometrised organoid mask.

We then perform spatial analysis on the distribution of Olig2 expression. Specifically, we binarise the (isometrised) Olig2 channel with a threshold 3000. From this, we can compare the fraction of Olig2+ voxels at each spatial bin position in an organoid-by-organoid fashion ($p_j^{Olig2}(x)$, where x is a set of distances (bin mid-points) from the nearest FoxA2+ voxel, for organid j). To understand the average behaviour by genotype, we take the mean of this spatial profile (i.e. effectively removing an otherwise inherent weighting based on variations in organoid volume; $p_{ensemble}^{Olig2}(x) = \frac{1}{N_{organoid}} \sum_{j=1}^{N_{organoid}} p_j^{Olig2}$). We report these values in a replicate-by-replicate fashion, excluding replicates for which $N_{organoid} \leq 20$.

For domain-size calculations, we calculate a binary spatial profile for positions where p_j^{Olig2} exceeds a threshold percentage ($b_j^{Olig2}(x) = (p_j^{Olig2}(x) > 0.1)$). This is then processed such that $b_j^{Olig2}(x > x_0) = 0$ where x_0 is the lowest value of x for which $b_j^{Olig2} = 0$, so as to consider only called Olig2 domains near the floor-plate source, removing rare instances where iso-lines of Olig2 expression exceed the threshold far from the source (likely due to imaging noise). We then take a per-replicate ensemble mean as a function of x : $\langle b_j^{Olig2}(x) \rangle_j$. Next, we calculate $d^{Olig} = \sum_{x \in \text{bin mid-points}} [x \cdot \langle b_j^{Olig2}(x) \rangle_j] / \sum_{x \in \text{bin mid-points}} \langle b_j^{Olig2}(x) \rangle_j$ to determine the domain size by organoid. This is then normalised by the wild-type domain size averaged across replicates $d_{cond.,rep.}^{Olig2} = d_{cond.,rep.}^{Olig2} / \langle d_{WT,rep.}^{Olig2} \rangle_{rep.}$.

12 Relating thermodynamic state ensemble models to synCRE activity

12.1 Transcriptional activation by a single site

We started from a basic model of gene regulation, which offers a baseline framework for understanding the composability rules of regulatory sequences. This approach draws heavily from prior work [Bintu et al., 2005, Buchler et al., 2003, Veitia, 2003]. We consider a regulatory element containing a single binding site for a transcriptional activator A and a binding site for RNA polymerase II (RNAPII) P at the transcription start site (TSS). Additionally, we assume that A and P can physically interact when bound simultaneously. The system can occupy four distinct states: neither molecule bound, only A bound, only P bound, or both bound. Under the assumption of thermodynamic equilibrium, the partition function describing this system can be written as:

$$Z = 1 + Ae^{-E_a/kT} + Pe^{-E_p/kT} + APe^{-(E_a+E_p+E_{ap})/kT} \quad (47)$$

where T is the temperature and k is the Boltzmann constant. Here E_{xy} relates to a binding/unbinding process by:



$$k_{xy}^+/k_{xy}^- = K_{xy} = e^{-E_{xy}/kT} \quad (49)$$

$$(50)$$

From this, the probability that RNAPII is bound can be calculated by:

$$p_{bound} = \frac{Pe^{-E_p/kT} + APe^{-(E_a+E_p+E_{ap})/kT}}{Z} \quad (51)$$

$$= K_p P (1 + K_a A e^{-E_{ap}/kT}) / (1 + K_a A + K_p P + K_a K_p A P e^{-E_{ap}/kT}) \quad (52)$$

$$= \frac{1}{1 + \frac{1}{K_p P} \cdot \frac{1 + K_a A}{1 + K_a A e^{-E_{ap}/kT}}} \quad (53)$$

which can be written more simply as

$$p_{bound} = \frac{1}{1 + \frac{\alpha}{F_A}} \quad (54)$$

$$\alpha = \frac{1}{K_p P} \quad (55)$$

$$F_A = \frac{1 + f_a K_a A}{1 + K_a A} \quad (56)$$

$$f = e^{-E_{ap}/kT} \quad (57)$$

$$(58)$$

This reparameterisation provides an intuitive interpretation [Bintu et al., 2005], where F_A quantifies the regulatory effect of A on polymerase binding. When $F_A < 1$, the polymerase binds less frequently than in the absence of the TFBS, indicating that A acts as a repressor. In contrast, if $F_A > 1$, the polymerase is more likely to bind, implying that A functions as an activator. To express this in terms of the transcription rate r , we assume that polymerase recruitment is the rate-limiting step in transcription initiation, such that:

$$r = \beta_m \cdot p_{bound} \quad (59)$$

Consequently, assuming the system is at steady state, the mean level of the corresponding protein is a read-out for p_{bound}

$$\frac{d[mRNA]}{dt} = r - \lambda_m [mRNA] \quad (60)$$

$$\frac{d[Protein]}{dt} = \beta_p [mRNA] - \lambda_p [Protein] \quad (61)$$

$$\implies [Protein] = \frac{\beta_p \beta_m}{\lambda_m \lambda_p} p_{bound} = c \cdot p_{bound} \quad (62)$$

12.2 Varying number of independently acting transcription factors

What happens when there are two different transcription factors A and B with two corresponding TFBSs? Suppose A and B are not directly interacting, such that binding is not cooperative. If we further assume that A and B independently interact with RNAPII, this yields a partition function:

$$Z_{AB} = 1 + K_a A + K_b B + K_p P + K_a K_b AB + K_a K_p f_a AP + K_b K_p f_b BP + K_a K_b K_p f_a f_b ABP \quad (63)$$

implying:

$$F_{AB} = \frac{(1 + f_a K_a A)(1 + f_b K_b B)}{(1 + K_a A)(1 + K_b B)} \quad (64)$$

$$p_{bound}^{AB} = \frac{1}{1 + \frac{\alpha}{F_{AB}}} \quad (65)$$

Comparing a CRE with either of the two TFBSs to the two-site CRE, one can see that individual “contributions” (i.e. F s) multiply:

$$F_{AB} = F_A \cdot F_B \quad (66)$$

Therefore, the function of a cis-regulatory element made up of a set S of non-interacting TFBSs is expected to follow a logistic curve based on their individual effects.

$$p_{bound}^{\{S\}} = \frac{1}{1 + \exp(-(\log 1/\alpha + \sum_{s \in \{S\}} \log F_s))} \quad (67)$$

Additionally, when $F_s \ll \alpha$, meaning RNAPII loading to the TSS is far from saturation, we can approximate p_{bound} as the product of the individual contributions of each TFBS. As a result, the activity of a CRE consisting of different TFBSs should be directly proportional to the product of the activities of each TFBS independently.

$$p_{bound}^{\{S\}} \approx \frac{1}{\alpha} \prod_{s \in \{S\}} F_s \propto \prod_{s \in \{S\}} p_{bound}^s \quad (68)$$

When multiple identical TFBSs are concatenated, with F_s held constant, the activity will increase exponentially with the number of binding sites, following an exponential distribution.

$$p_{bound}(N) \approx \frac{1}{\alpha} F^N \propto \exp(N \cdot \log F) \quad (69)$$

This outcome also suggests that combining two CREs, each containing distinct sets of TFBSs S_1 and S_2 , should adhere to a multiplicative rule, a principle that can be generalized to any N-way fusion.

$$p_{bound}^{\{S_1, S_2\}} \propto p_{bound}^{\{S_1\}} p_{bound}^{\{S_2\}} \quad (70)$$

Together, the activity of a regulatory sequence consisting of independently acting TFBSs (those that independently bind and recruit RNAPII) adheres to a logistic composability rule based on the separate contributions of its TFBSs. In sub-saturating conditions, this behavior simplifies to a multiplicative rule. Consequently, when TFBSs act independently, the activity of any fusion of sequence fragments—regardless of the number or type of TFBSs—can be approximated by multiplying the activities of each individual fragment.

12.3 Breaking independence requirements

The previous section supports a multiplicative (with saturation) composability rule based on the contributions of TFBSs. However, it assumes that TFs act independently, both in terms of binding to the CRE and recruiting the polymerase. To relax these assumptions, we introduce additional energies that account for cooperative binding and post-binding synergies. For a CRE containing two TFBSs, this can be generically represented by the partition function.

$$Z_{AB} = 1 + K_a A + K_b B + K_p P + c_{ab} K_a K_b AB + f_a K_a K_p AP + f_b K_b K_p BP + c_{ab} f_a f_b K_a K_b K_p ABP \quad (71)$$

Here, c_{ab} quantifies how much the likelihood of both TFs binding together deviates from the expected probability if each TF were to bind independently, assuming the polymerase is not bound. If $c_{ab} \neq 1$, this signifies that the binding of the two TFs is cooperative.

$$c_{ab} = \frac{P(A \cap B | \neg P)}{P(A \cap \neg B | \neg P) P(B \cap \neg A | \neg P)} \quad (72)$$

f_{ab} reflects the difference in polymerase recruitment probability when both TFs are present, as compared to their individual recruitment abilities. When $f_{ab} \neq 1$, it indicates that the recruitment process is cooperative. that the TFs engage in post-binding synergy.

$$f_{ab} = \frac{1}{c_{ab}} \frac{P(A \cap B|P)}{P(A \cap \neg B|P)P(B \cap \neg A|P)} \quad (73)$$

We can then determine F_{AB} :

$$F_{AB} = \frac{1 + f_a K_a A + f_b K_b B + f_{ab} c_{ab} f_a f_b K_a K_b AB}{1 + K_a A + K_b B + c_{ab} K_a K_b} \quad (74)$$

This expression is no longer the product of F_A and F_B .

$$\frac{F_{AB}}{F_A F_B} = \frac{1 + (f_{ab} c_{ab} - 1) \left(\frac{f_a f_b}{F_A F_B} \right) \left(\frac{K_a K_b AB}{(1 + K_a A)(1 + K_b B)} \right)}{1 + (c_{ab} - 1) \left(\frac{K_a K_b AB}{(1 + K_a A)(1 + K_b B)} \right)} \quad (75)$$

The regulatory complexity in principle grows exponentially with the addition of more TFBSs. For instance, the partition function for a sequence with three TFBSs will include cooperative binding and post-binding synergy terms for each of the two possible interacting TFBS pairs ($c_{ac}, c_{bc}, f_{ac}, f_{bc}$), as well as higher-order terms (c_{abc}, f_{abc}).

12.4 Defining regulatory modules

One expects that, at least for larger regulatory elements, not all pairs of TFBSs engage in synergistic interactions (pre- or post-binding). Therefore, certain combinations of TFBSs may combine under the multiplicative rule. As a result, some TFBS combinations will follow the multiplicative rule, while others will show deviations from it. This observation leads to the concept of a regulatory module. In this view, a regulatory module is a group of TFBSs where no synergy exists between any TFBS in the module and any TFBS from another.

Consider a graph where the TFBSs in a CRE are represented as nodes, and the synergies (pre- or post-binding) between pairs or higher-order combinations are represented as edges. For instance, there would be an edge between nodes A and B if any of the following terms differ from 1: $f_{ab}, c_{ab}, f_{abc}, c_{abc}, f_{abcd}, c_{abcd}, \dots$. Modules are therefore defined as the connected components of the graph.

Define the partition function of the TFBSs comprising a module as Z_m . A CRE is then defined by a set of modules. A generic description of the partition function for a CRE with an assortment of potentially interacting TFBSs for TFs $T = \{A_1, A_2, \dots, A_n\}$ is given by

$$Z = 1 + K_p P + \sum_{\{A_1, A_2, \dots, A_n\}} K_i A_i (1 + f_i K_p P) + \sum_{S \subseteq \{A_1, A_2, \dots, A_n\}, |S| > 1} c_S \prod_{i \in S} K_i A_i \left(1 + f_S K_p P \prod_{i \in S} f_i \right) \quad (76)$$

$$= \left[1 + \sum_{A_i \in T} K_i A_i + \sum_{S \subseteq T, |S| > 1} c_S \prod_{i \in S} K_i A_i \right] + K_p P \left[1 + \sum_{A_i \in T} f_i K_i A_i + \sum_{S \subseteq T, |S| > 1} c_S f_S \prod_{i \in S} K_i A_i f_i \right] \quad (77)$$

$$\Rightarrow F_T = \frac{1 + \sum_{A_i \in T} f_i K_i A_i + \sum_{S \subseteq T, |S| > 1} c_S f_S \prod_{i \in S} K_i A_i f_i}{1 + \sum_{A_i \in T} K_i A_i + \sum_{S \subseteq T, |S| > 1} c_S \prod_{i \in S} K_i A_i} \quad (78)$$

If we can divide TFBSs into sets of modules ($M = \{M_1, M_2, M_3\}$, $M_1 = \{A_1, A_2, \dots, A_{m-1}\}$, $M_2 = \{A_m, A_{m+1}, \dots, A_{p-1}\}$, ...), one can show that

$$Z = \prod_j \left[1 + \sum_{A_i \in M_j} K_i A_i + \sum_{S \subseteq M_j, |S| > 1} c_S \prod_{i \in S} K_i A_i \right] + K_p P \prod_j \left[1 + \sum_{A_i \in M_j} f_i K_i A_i + \sum_{S \subseteq M_j, |S| > 1} c_S f_S \prod_{i \in S} K_i A_i f_i \right] \quad (79)$$

This therefore implies that

$$F_{\{M_1, M_2, \dots\}} = \prod_j F_{M_j} \quad (80)$$

Importantly, this demonstrates that we can decompose the activity of a CRE into the product of the activity of regulatory modules. As with the individual, non-interacting, TFBS examples, in the sub-saturating limit, this implies that the activity of a CRE will be linearly proportional to the product of the activities of each individual module.

Crucially, this shows that the activity of a CRE can be broken down into the product of the activities of its regulatory modules. Similar to the case with individual, non-interacting TFBSs, in the sub-saturating limit, this means that CRE activity will be directly proportional to the product of the activities of each module.

$$p_{bound}(\{M_1, M_2, \dots\}) \propto \prod_j p_{bound}(M_j) \quad (81)$$

12.5 Linear modelling

Due to the combinatorial design of synCREs in the NeMECiS dataset, it is possible to directly connect experimental measurements to the thermodynamic model of modularity. Assuming the thermodynamic model accurately represents transcriptional regulation, and that ZsGreen mRNA and protein levels reach steady state—with ZsGreen protein levels being linearly proportional to fluorescence—it becomes feasible to predict the SAG-dependent fluorescence of a synCRE (y^S ; $S \in -, +$). This fluorescence can be described in terms of the SAG-dependent regulatory contribution of a synCRE composed of fragments A , B , and C at a given SAG concentration (F_{ABC}^S).

$$y^S \propto \frac{1}{1 + \alpha/F_{ABC}^S} \quad (82)$$

In this context, F_{ABC}^S represents a scalar value rather than the function. Specifically, we consider the overall function $F_{ABC}(\mathbf{x})$, which is, in principle, a function of many components in the cell's state-space, $\mathbf{x}$, with measurements focusing on evaluating the overall function $F_{ABC}(\mathbf{x}(S))$ at two specific SAG concentrations: $F_{ABC}^- = F_{ABC}(\mathbf{x}(0))$ and $F_{ABC}^+ = F_{ABC}(\mathbf{x}(500))$.

We first test the hypothesis that all fragments in the NeMECiS dataset behave as independent modules. Consequently, one can calculate the SAG-dependent regulatory contributions of each individual fragment, such as $F^S A$, $F^S B$, F_C^S , Therefore:

$$F_{ABC}^S = F_A^S \cdot F_B^S \cdot F_C^S \quad (83)$$

Therefore, for a given SAG concentration, let each fragment contribute a regulatory value F_i^S (50 parameters in total). A synCRE can be represented by a triplet of indices $\mathbf{X} = (X_1, X_2, X_3)$. For instance, the synCRE labeled "4-5-6" would correspond to $X_1 = 4$, $X_2 = 5$, and $X_3 = 6$. Consequently, we can express the ZsGreen fluorescence of the synCRE as:

$$\log(y_{\mathbf{X}}^S) \approx c + \sum_{i=1}^3 \log(F_{X_i}^S) \quad (84)$$

Here, c is a proportionality constant. As a result, the log-transformed ZsGreen levels of synCREs measured by NeMECiS ($\log(y_{\mathbf{X}}^S)$) can be linked to the by-fragment weights ($\beta_i = \log(F_{X_i}^S)$) through linear regression, where the intercept parameter (c) represents the intrinsic transcription rate of the promoter.

It is important to note that this represents an under-determined system with 51 parameters and 50 variables. Intuitively, since synCREs always consist of three fragments, it is not feasible to determine an absolute baseline. However, differences in β are still meaningful. Thus, it is reasonable to choose a value for c that reflects the average activity of synCREs across the dataset, with the values of β capturing the deviations from this baseline. A common approach to address this is through regularization. Specifically, we chose to fit the model using Ridge regression, minimizing a cost function of the form:

$$C = \|\mathbf{Z}\boldsymbol{\beta} + y_0 - \mathbf{Y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2 \quad (85)$$

13 Tables

CRE	Sequence	Genomic Coordinates (mm10)
Nkx22	TTGGTTCTATTCATTTTTCCCTCTAACTGTCCTTGTTGGGTTGCCTTAG TCCAGGCGAAGCAGAACGCTCAGGCATCGAGTCTCCAAGCTCACAGCTG CTTGGGCCGCGGCGGCGCCCCACCACCCCGGGTTTCTAGAGCCTTGTG ACAGCGCCTCACTCCGAGCTGCTCAGCGGCTTTCTGCAACTCTCTGCGC TCGCCCCCAGTATGTGACGTGGGTGACAATGGCCCAGGTTGGAGCGAGCC CCACGTCGGCGCGTCTGGGTGGTCGGACCCGGGCAAACACAAATACAAA CCGATTGCTAAGCTGCGGACAATGAGGGAAATGTAGACAAATGTCCCGCT CCTGTTGGAAGCCTTTGTCCAGGCCCGGTTTTTGCATTTATTTCACTGGC GAAATAATACATGATTGACGCTCTCTTTCAATGTGTCCTAACTGTTTGA ATAAATCTAAGGTTGTCCCTAGTTGTCTATGGCATTCAACCCTTTTCAATG GACTATCTCTTCATTCATTTCTGAATCCGTCCACAATATTAAGGAAACCC CTTCATGTCGACGTTCCCCATTCTCTCTCTGCTTTTCTTTCTCCT TTCTTCTCCTTTTCCCCCTTGCAAGAATTAGTATCTGGTTTGAAACGTG TTCTCAGACTCATTACTTTCTGTCTTCCCCCTTCTGTTTTTCCAGC TTCTTTCTCAGCGTTAGAGCCTGTTGCCCTCTCCCTCCCTCTTTTCAGGC	chr2 147187985-147188733
Olig2	TGATTGCCAACTGCCTCCAACCCAACCTTGTAAAGCCGAGCCCTCATCCCT ACCCACTCCCGGTGTGTGTCAGATGGAACACCTAGGTGGCCACGGGGACCT CTGACCTCTATATCCTCTTCTCTTCTCCCTCCTTTGCTACTTTTCTACTG GATAAAAGGAGAGAGTGAGAGATAATTAACAAAAAACATGGCCCCGGGAC AATGAAGCAACTGGCCTTGCGCGGAAGCAACGATCCTGGTTTTCTAGGT AGAGTTTCTCCCATCAATCTTTCCTTTAACCTCCCTGTTCTGGAAGCAA TAGAAACACCACCCCTCCCTGAGCAAATGCTTTCTTTGACTGGAAC AAAAAGGGGGCCCGCAAAGACGGAGGTGAAATCTGGGTGGTATGGGCAC CGCACAATGGCCCCGCTGTTCTGCGCCCTGCTTGTGTTTTACAACAGGGG AGGGGCAGGCGGAATGGTCCGATGGTGGAGACAATCCCCCTGATTGAGG CTACAAATGCATCTTCTATTCCACACGGAGCTGAGCAGAAAGGATGGGGG TGACAAAGAGCATGGGCGGGGAGAGGGAAAAACAAATGTTTTGAGTTGAA AAAAAAATCTCTCATATCCTACACATCCTCAGAAGAGCTTCTATGGAGA AGGCCTTCGGAGAGTCCCAGCCCACAACCTCAAGGGCTTGTCTGAACTCT GATTTATTGATGAAGCTTAAGCGGCTCGCTAAGAAAGGCCTGGGGGTGTC TTTGCTTGAAGATAAAGTACAATAGGCCACAAGGGCCAAGATCTCTCGG GATGCTCTCAGGTCCTGCCTCTCTTGGCCCTCTCCTCCCTGCAAAATGCC AGCAGATGCTGAAAAAAAAAAAAACCCATCGGTGGTGTGGCTGGGAGTGC TGGGGACAAGCTGGGCCACTTGAGGTCTCCTTAAGAGGGTATTATGGCCA GGGAAAAATTTGCGCTCTAAGGATGGCACACTCCATTGATAATGGCTC TCATCTGCCTCAGATAATCGCCTCCCTCCCGGCTGTCAGGGGTGCAGCCA CTGCCAATTCACAGCGCCCTCCGAGAAAGTACCCTTGTCTGTGATGACCA AGATGGGGACATTGTGTTTACCTACTTGAGCAGAGGAGAAGGTGACCGTG AGGGCAGCCTGCATTGTAAATTACAATTAACAGAAACAGACAGTTTCCT GCTCTGCCCTGGGACCCCAACCAATAAATTATGGGTGGACATTAGGGGAG AGCCCAGGAAAGGTTGGGTCTGGGGAGGATCCCCCATCCCATAGCCTA CCGAC	chr16 91192436-91193740

Pax6	CTAAATAGCACCGCGGCGCCCGCTCTCCGGACAGTGATTAATGATAGCAG TGCAGAGGGGTAAACACACTTCGCTGAAAAAGTCTGTTGACTGAGCTTCTT GTAACACAATGTGGCCCGCTGCACGCCTCGAGAGAATCCTTTTGTGTCC GTGCTTATTGTGGCCTCAAAATCTGCCACGAAAGTTTGCCAACGCTCC TGCCCCAGGAGTTAATAGTTTCCCTTACTCGCGGGGCATTGTGTGGTGC TGAAAAGCAGCCCTCGCTATTCAAGTGTGGTGGTCATCTCAATAGATC TCCAAGGGCCCATATGGTGGCCAGTGCCGATGAATCCGCCTGTTAAATG GGGGAGAAAGTTGGGGTTTTAAACATTTCAAAGTTCCTGAAAAGATCCC ACTAGATCCTGTCACAATCCCTGAACGCTTTGAAGGCGCGGCCTATTGT CTCCTGGTTATAAATGATATTCTGGCCAAGTCGATTCCCACA	chr2 105689933-105690425
------	--	-----------------------------

Table 1: Natural CRE sequences

Fragment	Sequence	Origin
1	TTGGTTCTATTCATTTTTCCCTCTAACTGTCCTTGTTGGGTTGCCTTAG TCCAGGCGAAGCAGAACGCTCAGGCATCGAGTCCTCAAGCTCACAGCTG CTTGGGCGCGGGCGGCGCCCAACCACCCCGGGTTTCTAGAGCCTTGTG ACAGCGCCTCACTCCGAGCTGCTCAGCGGCTTTCTGCAACTCTCTGCCC	Nkx22 chr2 147187985-147188184
2	TCGCCCCAGTATGTGACGTGGGTGACAATGGCCCAGGTTGGAGCGAGCC CCACGTCGGGCGCGTCCTGGGTGGTTCGGACCCGGGCAAACACAAATACAAA CCGATTGCTAAGCTGCGGACAATGAGGGAAATGTAGACAAATGTCCCGCT CCTGTTGGAAGCCTTTGTCCAGGCGCGGTTTTTGCATTTATTTCACTGGC	Nkx22 chr2 147188185-147188384
3	GAAATAATACATGATTGACGCTCTCTTTCAATGTGTCCTAACTGTTTGA ATAAATCTAAGTTGTCCCTAGTTGTTCATGGCATTCAACCCTTTTCAATG GACTATCTCTTCATTCAATTTCTGAATCCGTCCACAATATTAAGGAAACCC CTTCATGTCGACGTTCCCCATTCTCTCTCTCTGCTTTTCTTTCTCCT	Nkx22 chr2 147188385-147188584
4	TTCTTCTCCTTTTTCCCTTGCAAGAATTAGTATCTGGTTTGAAACGTG TTCTCAGACTCATTACTTTCTGTCTTCCCCCTTCTGTTTTTCCAGC TTCTTTCTCAGCGTTAGAGCCTGTTGCCTCTCCCTCCCTCTTTTCAGGCT GGGGCTCTCTGCCCCGCGAGGTGGGAGGGGGGAAAGTCAGCTTTCAGAA	Nkx22 chr2 147188585-147188784
5	CTTGGGCGCGGGCGGCGCCCCACCACCCCGGGTTTCTAGAGCCTTGTG ACAGCGCCTCACTCCGAGCTGCTCAGCGGCTTTCTGCAACTCTCTGCCC TCGCCCCAGTATGTGACGTGGGTGACAATGGCCCAGGTTGGAGCGAGCC CCACGTCGGGCGCGTCCTGGGTGGTTCGGACCCGGGCAAACACAAATACAAA	Nkx22 chr2 147188085-147188284
6	CCGATTGCTAAGCTGCGGACAATGAGGGAAATGTAGACAAATGTCCCGCT CCTGTTGGAAGCCTTTGTCCAGGCGCGGTTTTTGCATTTATTTCACTGGC GAAATAATACATGATTGACGCTCTCTTTCAATGTGTCCTAACTGTTTGA ATAAATCTAAGTTGTCCCTAGTTGTTCATGGCATTCAACCCTTTTCAATG	Nkx22 chr2 147188285-147188484
7	GACTATCTCTTCATTCAATTTCTGAATCCGTCCACAATATTAAGGAAACCC CTTCATGTCGACGTTCCCCATTCTCTCTCTCTGCTTTTCTTTTCTCCT TTCTTCTCCTTTTTCCCTTGCAAGAATTAGTATCTGGTTTGAAACGTG TTCTCAGACTCATTACTTTCTGTCTTCCCCCTTCTGTTTTTCCAGC	Nkx22 chr2 147188485-147188684
8	TGATTGCCAACTGCCTCCAACCCAACCTTGTAAAGCCGAGCCCTCATCCCT ACCCACTCCCGGTGTGTGATGGAACACCTAGGTGGCCACGGGGACCT CTGACCTCTATATCCTCTTCTCTTCTCCCTCCTTTGCTACTTTTCTACTG GATAAAAGGAGAGAGTGAGAGATAATTAACAAAAAACATGCCCCGGGAC	Olig2 chr16 91192436-91192635
9	AATGAAGCAACTGGCCTTGGCCGGCAAGCAACGATCCTGGTTTTCTAGGT AGAGTTTCTCCCATCAATCTTTCCCTTTAACCTCCCTGTTCTGTGAAGCAA TAGAAACACCACCCCTCCCTGAGCAAATGCTTTTCTTTGACTGGAAGAAC AAAAAGGGGGCCCGGCAAAGACGGAGGTGAAATCTGGGTGGTATGGGCAC	Olig2 chr16 91192636-91192835
10	CGCACAATGGCCCCGCTGTTCTGGCCCTGCTTGTGTTTTACAACAGGGG AGGGGCAGGCGGAATGGTCCGATGGTGGAGACAATCCCCCTGATTGAGG CTACAAATGCATCTTCTATTCCACACGGAGCTGAGCAGAAAGGATGGGGG TGACAAAGAGCATGGGCGGGGAGAGGGAAAAACAAATGTTTTCACTTGAA	Olig2 chr16 91192836-91193035

11	AAAAAAATCTCTCATATCCTACACATCCTCAGAAGAGCTTCTATGGAGA AGGCCTTCGGAGAGTCCCAGCCCACAACCTCAAGGGCTTTGTCTGAACTCT GATTTATTGATGAAGCTTAAGCGGCTCGCTAAGAAAGGCCTGGGGGTGTC TTTGTCTTGAAGATAAAGTACAATAGGCCACAAGGCCAAGATCTCTCGG	Olig2 chr16 91193036-91193235
12	GATGCTCTCAGGTCCTGCCTCTCTCTTGCCTCTCCTCCCTGCAAAATGCC AGCAGATGCTGAAAAAAAAAAAAACCCATCGGTGGTGTGGCTGGGAGTGC TGGGGACAAGCTGGGCCACTTGAGGTCTCCTTAAGAGGGTATTATGGCCA GGGAAAAATTTTGCCTCTAAGGATGGCACACTCCATTTGATAATGGCTC	Olig2 chr16 91193236-91193435
13	TCATCTGCCTCAGATAATCGCCTCCCTCCCGGCTGTCAGGGGTGCAGCCA CTGCCAATTCACAGCGCCCTCCGAGAAAGTACCCTTGTCTGTGATGACCA AGATGGGGACATTGTGTTTACCTACTTGAGCAGAGGAGAAGGTGACCGTG AGGGCAGCCTGCATTGTAAATTACAATTAACAGAAACAGACAGTTTCCT	Olig2 chr16 91193436-91193635
14	GCTCTGCCCTGGGACCCCCACCAATAAATTATGGGTGGACATTAGGGGAG AGCCCAGGAAAGGTTGGGTCTGGGGAGGATCCCCCATCCCATAGCCTA CCGACAGGTCTTGATATAGGGATAGGGCTACTTGGGAGTCAAGGTAGAC TGGCTGGTTGACCACACACACTGGGATCCTCAGGAGGTTCCCCCACT	Olig2 chr16 91193636-91193835
15	CTGACCTCTATATCCTCTTCTCTTCTCCCTCCTTTGCTACTTTTCTACTG GATAAAAGGAGAGAGTGAGAGATAATTAACAAAAACATGGCCCCGGGAC AATGAAGCAACTGGCCTTGGCCGGCAAGCAACGATCCTGGTTTTCTAGGT AGAGTTTCTCCCATCAATCTTTCTTTAACCTCCCTGTTCTGTGAAGCAA	Olig2 chr16 91192536-91192735
16	TAGAAACACCACCCCTCCCTGAGCAAAATGCTTTCTTTGACTGGAAC AAAAAGGGGGCCCGGCAAGACGGAGGTGAAATCTGGGTGGTATGGGCAC CGCACAATGGCCCCGCTGTTCTGGCCCTGCTTGTGTTTTACAACAGGGG AGGGGCAGGCGGAATGGTCCGATGGTGGAGACAATCCCCCTGATTGAGG	Olig2 chr16 91192736-91192935
17	CTACAAATGCATCTTCTATTCCACACGGAGCTGAGCAGAAAGGATGGGGG TGACAAAGAGCATGGGCGGGGAGAGGAAAAACAAATGTTTTCACTTGAA AAAAAAATCTCTCATATCCTACACATCCTCAGAAGAGCTTCTATGGAGA AGGCCTTCGGAGAGTCCCAGCCCACAACCTCAAGGGCTTTGTCTGAACTCT	Olig2 chr16 91192936-91193135
18	GATTTATTGATGAAGCTTAAGCGGCTCGCTAAGAAAGGCCTGGGGGTGTC TTTGTCTTGAAGATAAAGTACAATAGGCCACAAGGCCAAGATCTCTCGG GATGCTCTCAGGTCCTGCCTCTCTCTTGCCTCTCCTCCCTGCAAAATGCC AGCAGATGCTGAAAAAAAAAAAAACCCATCGGTGGTGTGGCTGGGAGTGC	Olig2 chr16 91193136-91193335
19	TGGGGACAAGCTGGGCCACTTGAGGTCTCCTTAAGAGGGTATTATGGCCA GGGAAAAATTTTGCCTCTAAGGATGGCACACTCCATTTGATAATGGCTC TCATCTGCCTCAGATAATCGCCTCCCTCCCGGCTGTCAGGGGTGCAGCCA CTGCCAATTCACAGCGCCCTCCGAGAAAGTACCCTTGTCTGTGATGACCA	Olig2 chr16 91193336-91193535
20	AGATGGGGACATTGTGTTTACCTACTTGAGCAGAGGAGAAGGTGACCGTG AGGGCAGCCTGCATTGTAAATTACAATTAACAGAAACAGACAGTTTCCT GCTCTGCCCTGGGACCCCCACCAATAAATTATGGGTGGACATTAGGGGAG AGCCCAGGAAAGGTTGGGTCTGGGGAGGATCCCCCATCCCATAGCCTA	Olig2 chr16 91193536-91193735
21	CTAAATAGCACC CGCGCCCGCTCTCCGGACAGTGATTAATGATAGCAG TGCAGAGGGGTTAACACACTTCGCTGAAAAGTCTGTTGACTGAGCTTCTT GTAACACAATGTGGCCCGCTGCACGCCTCGAGAGAATCCTTTTGTGTCC GTGCTTATTGTGCCTCAAAATTCTGCCACGAAAGTTTGCCAACGCTCC	Olig2 chr16 105689933-105690132
22	TGCCCCAGGAGTTAATAGTTTCCCTTACTCGCGGGGCATTGTGTGGTGC TGAAAAGCAGCCCTCGCTATTCAAGTGTGGTGGTCATCTCAATAGATC TCCAAGGGCCCATATGGTGGCCAGTGCCGATGAATCCGCCTGTTTAAATG GGGGAGAAAGTTGGGGTTTTAAAAACATTCAAAGTTCCTGAAAAGATCCC	Pax6 chr2 105690133-105690332
23	ACTAGATCCTGTCACAATTCCTGAACGCTTTGAAGCGCGGCCTATTGT CTCCTGGTTATAAATGATATTCTGGCCAAGTCGATTCCACAGAGGCCT GGCGCCCCCTCCCCCAACTAGCCAAGTTTCTAGGATCCGGAGAGAACA GTTCTTGTGGTATCCCGTGGGGGCCACAGGCCTCAGGACCCAAGACAGC	Pax6 chr2 105690333-105690532

24	GTAACACAATGTGGCCCGCTGCACGCCTCGAGAGAATCCTTTTGTGTCC GTGCTTATTGTGGCCTCAAAATTCGCCACGAAAGTTTGCCAACGCTCC TGCCCCAGGAGTTTAATAGTTTCCCTTACTCGCGGGGCATTGTGTGGTGC TGAAGAGCAGCCCTCGCTATTCAAGTGTGGTGGTCATCTCAATAGATC	Pax6 chr2 105690033-105690232
25	TCCAAGGGCCCATATGGTGGCCAGTGCCGATGAATCCGCCTGTTTAAATG GGGAGAAAAGTTGGGGTTTTAAACATTCAAAGTTCCTGAAAAGATCCC ACTAGATCCTGTCACAATCCCTGAACGCTTTGAAGGCGCGGCCTATTGT CTCCTGGTTATAAATGATATTCTGGCCAAGTCGATTCCACAGAGGCCT	Pax6 chr2 105690233-105690432

Table 2: CRE Fragments

Fragment	'Odd' Barcodes (5' -3-)	'Even' Barcodes (5' -3-)
1	CACTGGAT	GACAACAC
2	CTCCTCTA	CCTGATCA
3	CTAAGGTG	CACTTGGC
4	TCGCGATA	GCCATTCT
5	GGCCTATT	GTACGAGT
6	TGTGTGCA	GATGAACG
7	GGATATGC	CGAGTCTA
8	ACTCTCTC	GTCAGTGT
9	TATCCAGC	CTACCATG
10	GAGTATCC	GAGTCTAC
11	CCTTCAAG	ATGTGCTG
12	TACGCCGT	GTTGCTAC
13	AGAACACG	AGGCTGTT
14	GCTAGTAG	CCGACGTA
15	AGCAACTC	AGCTCACT
16	CCAACCTGA	AGAGACAC
17	ATCGACCA	ACGAACAC
18	AGAGTGGT	AGAGCGGA
19	TGGATGCT	AACCGAAG
20	GAACACAC	GCTTGTCG
21	TCGTGAGT	CTCCGTG
22	GTGTAGCG	ACAGACCA
23	CATGCTCA	GAGTGCGT
24	ATGGCGAG	CCTCTAGT
25	GTGCGTAC	CGAGTGGC

Table 3: Allocated Barcodes by Fragment

References

- [Barry et al., 2022] Barry, D. J., Gerri, C., Bell, D. M., D'Antuono, R., and Niakan, K. K. (2022). Giani—open-source software for automated analysis of 3d microscopy images. *Journal of cell science*, 135(10):jcs259511.
- [Benzinger and Briscoe, 2024] Benzing, D. and Briscoe, J. (2024). Illuminating morphogen and patterning dynamics with optogenetic control of morphogen production. *bioRxiv*, pages 2024–06.
- [Bintu et al., 2005] Bintu, L., Buchler, N. E., Garcia, H. G., Gerland, U., Hwa, T., Kondev, J., and Phillips, R. (2005). Transcriptional regulation by the numbers: models. *Current opinion in genetics & development*, 15(2):116–124.
- [Buchler et al., 2003] Buchler, N. E., Gerland, U., and Hwa, T. (2003). On schemes of combinatorial transcription logic. *Proceedings of the National Academy of Sciences*, 100(9):5136–5141.

- [Cadinanos and Bradley, 2007] Cadinanos, J. and Bradley, A. (2007). Generation of an inducible and optimized piggybac transposon system. *Nucleic acids research*, 35(12):e87.
- [Cohen et al., 2014] Cohen, M., Page, K. M., Perez-Carrasco, R., Barnes, C. P., and Briscoe, J. (2014). A theoretical framework for the regulation of shh morphogen-controlled gene expression. *Development*, 141(20):3868–3878.
- [de Almeida et al., 2022] de Almeida, B. P., Reiter, F., Pagani, M., and Stark, A. (2022). Deepstarr predicts enhancer activity from dna sequence and enables the de novo design of synthetic enhancers. *Nature Genetics*, 54(5).
- [de Boer et al., 2020] de Boer, C. G., Ray, J. P., Hacohen, N., and Regev, A. (2020). Maude: inferring expression changes in sorting-based crispr screens. *Genome Biology*, 21(1):134.
- [Delas et al., 2019] Delas, M. J., Jackson, B. T., Kovacevic, T., Vangelisti, S., Maravilla, E. M., Wild, S. A., Stork, E. M., Erard, N., Knott, S. R., and Hannon, G. J. (2019). Incrna spehd regulates hematopoietic stem and progenitor cells and is required for multilineage differentiation. *Cell Reports*, 27(3):719–729.
- [Delas et al., 2023] Delas, M. J., Kalaitzis, C. M., Fawzi, T., Demuth, M., Zhang, I., Stuart, H. T., Costantini, E., Ivanovitch, K., Tanaka, E. M., and Briscoe, J. (2023). Developmental cell fate choice in neural tube progenitors employs two distinct cis-regulatory strategies. *Developmental cell*, 58(1):3–17.
- [Engler et al., 2008] Engler, C., Kandzia, R., and Marillonnet, S. (2008). A one pot, one step, precision cloning method with high throughput capability. *PloS one*, 3(11):e3647.
- [Exelby et al., 2021] Exelby, K., Herrera-Delgado, E., Perez, L. G., Perez-Carrasco, R., Sagner, A., Metzis, V., Sollich, P., and Briscoe, J. (2021). Precision of tissue patterning is controlled by dynamical properties of gene regulatory networks. *Development*, 148(4):dev197566.
- [Fonseca et al., 2019] Fonseca, J. P., Bonny, A. R., Kumar, G. R., Ng, A. H., Town, J., Wu, Q. C., Aslankoohi, E., Chen, S. Y., Dods, G., Harrigan, P., et al. (2019). A toolkit for rapid modular construction of biological circuits in mammalian cells. *ACS synthetic biology*, 8(11):2593–2606.
- [Gadella Jr et al., 2023] Gadella Jr, T. W., van Weeren, L., Stouthamer, J., Hink, M. A., Wolters, A. H., Giepmans, B. N., Aumonier, S., Dupuy, J., and Royant, A. (2023). mscarlet3: a brilliant and fast-maturing red fluorescent protein. *Nature methods*, 20(4):541–545.
- [Kelley et al., 2016] Kelley, D. R., Snoek, J., and Rinn, J. L. (2016). Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks. *Genome Research*, 26(7).
- [Korhonen et al., 2009] Korhonen, J., Martinmäki, P., Pizzi, C., Rastas, P., and Ukkonen, E. (2009). Moods: fast search for position weight matrix matches in dna sequences. *Bioinformatics*, 25(23):3181–3182.
- [Krammer et al., 2024] Krammer, T., Stuart, H. T., Gromberg, E., Ishihara, K., Cislo, D., Melchionda, M., Perez, F. B., Wang, J., Costantini, E., Lehr, S., et al. (2024). Mouse neural tube organoids self-organize floorplate through bmp-mediated cluster competition. *Developmental Cell*.
- [Kvon et al., 2020] Kvon, E. Z., Zhu, Y., Kelman, G., Novak, C. S., Plajzer-Frick, I., Kato, M., Garvin, T. H., Pham, Q., Harrington, A. N., Hunter, R. D., et al. (2020). Comprehensive in vivo interrogation reveals phenotypic impact of human enhancer variants. *Cell*, 180(6):1262–1271.
- [Linkert et al., 2010] Linkert, M., Rueden, C. T., Allan, C., Burel, J.-M., Moore, W., Patterson, A., Loranger, B., Moore, J., Neves, C., MacDonald, D., et al. (2010). Metadata matters: access to image data in the real world. *Journal of Cell Biology*, 189(5):777–782.
- [Magin et al., 1992] Magin, T. M., McWhir, J., and Melton, D. W. (1992). A new mouse embryonic stem cell line with good germ line contribution and gene targeting frequency. *Nucleic acids research*, 20(14):3795.
- [Matsumoto et al., 2019] Matsumoto, K., Mitani, T. T., Horiguchi, S. A., Kaneshiro, J., Murakami, T. C., Mano, T., Fujishima, H., Konno, A., Watanabe, T. M., Hirai, H., et al. (2019). Advanced cubic tissue clearing for whole-organ cell profiling. *Nature protocols*, 14(12):3506–3537.

- [Novakovsky et al., 2023] Novakovsky, G., Fornes, O., Saraswat, M., Mostafavi, S., and Wasserman, W. W. (2023). Explainn: interpretable and transparent neural networks for genomics. *Genome Biology*, 24:154.
- [Oosterveen et al., 2012] Oosterveen, T., Kurdija, S., Alekseenko, Z., Uhde, C. W., Bergsland, M., Sandberg, M., Andersson, E., Dias, J. M., Muhr, J., and Ericson, J. (2012). Mechanistic differences in the transcriptional interpretation of local and long-range shh morphogen signaling. *Developmental cell*, 23(5):1006–1019.
- [Paszke et al., 2019] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32.
- [Peterson et al., 2012] Peterson, K. A., Nishi, Y., Ma, W., Vedenko, A., Shokri, L., Zhang, X., McFarlane, M., Baizabal, J.-M., Junker, J. P., van Oudenaarden, A., et al. (2012). Neural-specific sox2 input and differential gli-binding affinity provide context and positional information in shh-directed neural patterning. *Genes & development*, 26(24):2802–2816.
- [Pezzotta and Briscoe, 2023] Pezzotta, A. and Briscoe, J. (2023). Optimal control of gene regulatory networks for morphogen-driven tissue patterning. *Cell Systems*, 14(11):940–952.
- [Quang and Xie, 2016] Quang, D. and Xie, X. (2016). Danq: a hybrid convolutional and recurrent deep neural network for quantifying the function of dna sequences. *Nucleic Acids Research*, 44(11).
- [Ran et al., 2013] Ran, F. A., Hsu, P. D., Wright, J., Agarwala, V., Scott, D. A., and Zhang, F. (2013). Genome engineering using the crispr-cas9 system. *Nature protocols*, 8(11):2281–2308.
- [Rauluseviciute et al., 2024] Rauluseviciute, I., Riudavets-Puig, R., Blanc-Mathieu, R., Castro-Mondragon, J. A., Ferenc, K., Kumar, V., Lemma, R. B., Lucas, J., Chèneby, J., Baranasic, D., et al. (2024). Jasp2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 52(D1):D174–D182.
- [Schindelin et al., 2012] Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S., Rueden, C., Saalfeld, S., Schmid, B., et al. (2012). Fiji: an open-source platform for biological-image analysis. *Nature methods*, 9(7):676–682.
- [Serafimidis et al., 2008] Serafimidis, I., Rakatzi, I., Episkopou, V., Gouti, M., and Gavalas, A. (2008). Novel effectors of directed and ngn3-mediated differentiation of mouse embryonic stem cells into endocrine pancreas progenitors. *Stem cells*, 26(1):3–16.
- [Stan Development Team, 2024] Stan Development Team (2024). *Stan Modeling Language User's Guide and Reference Manual*.
- [Veitia, 2003] Veitia, R. A. (2003). A sigmoidal transcriptional response: cooperativity, synergy and dosage effects. *Biological Reviews*, 78(1):149–170.
- [Vierstra et al., 2020] Vierstra, J., Lazar, J., Sandstrom, R., Halow, J., Lee, K., Bates, D., Diegel, M., Dunn, D., Neri, F., Haugen, E., et al. (2020). Global reference mapping of human transcription factor footprints. *Nature*, 583(7818):729–736.
- [Williams et al., 2018] Williams, R. M., Senanayake, U., Artibani, M., Taylor, G., Wells, D., Ahmed, A. A., and Sauka-Spengler, T. (2018). Genome and epigenome engineering crispr toolkit for in vivo modulation of cis-regulatory interactions and gene expression in the chicken embryo. *Development*, 145(4).